

Aeroelastic Flight-Testing Performed in Accordance with
Parametric Flutter Margin

A Thesis

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Abstract

The Parametric Flutter Margin (PFM) method is utilized to identify the aeroelastic characteristics of an Unmanned Aerial Vehicle's wing in a series of Ground Vibration Tests (GVTs) and initial flight-testing study. A self-contained electro-mechanical excitation system developed to excite and measure the resultant wing dynamics such that Frequency Response Functions (FRF) can be identified from recorded inertial data is detailed. The results of bench-top testing performed to characterize the purpose-built excitation pods is discussed. Specifically, ground-testing indicates that the excitation pod's instrumentation could accurately measure a perturbation force generated by a moving mass within 3% error of directly measured values. Further testing reveals that the excitation pod's instrumentation could measure the resultant wing acceleration response within 1 ms^{-2} error. A series of GVTs performed on various wing configurations demonstrate that the self-contained excitation pods generate a sufficiently powerful perturbation force to energize an elastic response. The structural modes experimentally identified through the excitation system and PFM estimation method closely match corresponding Finite Element Analysis (FEA) predictions. A flight-testing campaign was conducted to demonstrate the excitation system and PFM method in a free-flight environment. The results of this preliminary flight-test study indicate that the frequency characteristics associated with the

flexible wing's symmetric and anti-symmetric modes can be identified. Initial flight-testing results reveal the importance of properly defining the excitation signals such that the symmetric and anti-symmetric forcing functions are of comparable magnitude. After identifying the phase-crossover frequencies from FRFs measured through flight-testing, a flutter margin versus airspeed curve is developed for both the wing's symmetric and anti-symmetric modes. From these curves, an anti-symmetric flutter mechanism with an onset velocity of 43.2 ms^{-1} is identified. These results are in close agreement with predictions calculated from a corresponding FEA model. This study successfully demonstrates that the PFM method can be used in conjunction with a self-contained, flight-ready excitation system to safely and accurately measure an aircraft's flutter onset condition in the free flight environment.

This research is dedicated to my wife, Madelyn, who has offered her unending support and encouragement throughout this journey. Her patience knows no limit, as it has been thoroughly tested by my insistent tangents on aircraft design, Bode plots, and the ever elusive flutter onset speed.

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Chapter 1: Introduction

1.1 Motivation

Of the numerous setbacks that may occur during the certification process of an aircraft, the onset of unexpected aeroelastic flutter may be the most catastrophic. Aeroelastic flutter occurs when unsteady aerodynamic and structural forces couple in an unstable manner, resulting in unbounded oscillations of an aircraft's structure. In cases where corrective action is not taken in an appropriate amount of time, these unstable structural modes may lead to severe damage to the vehicle, loss of aircraft, or loss of life. While several different flight-testing techniques have been successfully utilized to identify an aircraft's flutter condition, existing methods may subject the vehicle to dangerous flight conditions close to the flutter boundary. Recent advances in the Parametric Flutter Margin (PFM) method have demonstrated the accuracy and improved safety of flutter identification in a series of wind tunnel experiments. By building upon these wind tunnel studies, this work aims at demonstrating the safety and accuracy of aeroelastic flight-testing conducted in the free-flight environment.

1.2 Traditional Aeroelastic Flight Testing Methods

For new or modified aircraft to pass certification requirements, flight-testing must be accomplished that demonstrates freedom from flutter, proper damping margins, and non-rapid reduction of damping over the proposed flight envelope. Past flight-testing programs have verified compliance with these requirements by subjecting the aircraft to higher dynamic pressures until flutter onset is perceived to be imminent [1]. Multiple data reduction techniques have been proposed and utilized over the past 90 years to predict an aircraft's flutter onset speed from collected flight-test data. Specifically, flight-test engineers have relied on methods such as damping extrapolation, Zimmerman-Weissenburger flutter margin, envelope functions, Nissim-Gilyard method, and Autoregressive Moving Average (ARMA) models to predict flutter prone flight conditions from subcritical test points. [2–6]. Unfortunately, all the listed methods may require an aircraft to operate at dynamic pressures dangerously close to flutter onset to collect data related to flutter margins. Consequently, any uncertainty in the predictive models, atmospheric turbulence, or deviation from the targeted flight conditions may result in inadvertently adding energy to the aircraft's structure and exciting an unstable aeroelastic response. As a result, these methods cannot be safely applied at the flutter onset velocity and are extrapolative in nature. With the growing desire to certify Unmanned Aerial Systems (UAS) and Urban Air Mobility (UAM) vehicles with unique configurations for commercial operations, safe and accurate flight-test methodologies which positively identify flutter margins without exposing the vehicle to flutter are needed.

1.3 Parametric Flutter Margin

The PFM method developed by Roizner and Karpel [7] allows data to be collected at flight conditions exceeding an aircraft’s nominal flutter boundary through the addition of a stabilizing element, such as a mass, located ahead of a lifting surface’s elastic axis. In this initial study, simulation techniques were utilized to compute the flutter onset condition of a simple three-degree-of-freedom airfoil and generic transport aircraft model. These numerical simulations showed that the flutter boundary calculated via the PFM method and classical flutter analysis techniques, such as the p-k method (NASTRAN) and g-method (ZAERO), agreed within 0.7%. This indicates that the PFM method can be used for frequency based analysis of an aeroelastic system at the flutter onset speed of the unmodified wing without the need for extrapolation to higher velocities.

The PFM method has been successfully applied to several experiments conducted within a controlled wind tunnel environment. Sodja *et al.* [8] performed the first physical demonstration by experimenting with a flexible 2D wing section. In this study, a small stabilizing mass was added to the wing tip’s leading edge and an impedance head was utilized to provide an excitation force to the structure. The computed Frequency Response Functions (FRFs) indicated that the nominal wing section’s flutter boundary identified via the PFM method and by traditional flutter testing techniques were within 6%. Furthermore, this experiment demonstrated that the addition of a stabilizing mass element positioned ahead of a wing’s elastic axis increases the dynamic pressure needed to excite aeroelastic instabilities.

Similarly, Roizner *et al.* [9] equipped a flexible half-wing model with a tip store containing a stabilizing mass element positioned well ahead of the wing’s leading

edge. In this experiment, a known excitation force was applied to the wing by abruptly cutting a pre-loaded tension chord attached to the tip store’s stabilizing mass element. By conducting wind tunnel testing, Roizner *et al.* demonstrated that the selected stabilizing parameter increased the wing’s flutter onset condition by approximately 14 ms^{-1} , which allowed for exceedance of the nominal flutter velocity without subjecting the test article to undue risk. The PFM method identified the flexible wing’s nominal flutter boundary within 7% error as compared to that determined by direct flutter testing. As with the previous study, Roizner *et al.* successfully applied the PFM method and identified the nominal wing’s flutter onset condition without relying on conventional aeroelastic models.

Aeroelastic testing was performed with a blended wing-body UAS at the University of Michigan’s Subsonic Low Turbulence wind tunnel to investigate the applicability of a self-contained electro-mechanical shaker pod for use in combination with the PFM method. Unlike previous studies, Karpel *et al.* [10] developed a custom, self-contained shaker mechanism responsible for generating a sinusoidal excitation force over a 3 Hz - 15 Hz frequency sweep. By placing the shaker mechanism near the wing’s leading edge, it not only provided the excitation force required to compute the aeroelastic system’s FRFs, but also acted as a stabilizing element. Wind tunnel testing conducted in accordance with the PFM method indicated a nominal flutter velocity agreeing within 6% of a tuned numerical model. This study also demonstrated the repeatability of the PFM method as wind tunnel testing was conducted in two parts over a 5 month period. Using the custom made shaker assembly, similar FRFs for zero airspeed conditions were computed for the blended wing-body on two different testing days.

Most recently, Heidersbach *et al.* [11] conducted the first flight-test evaluation of the PFM method for safe flutter testing. A detailed series of experiments were conducted to characterize the excitation pods, which provided the necessary input force for the PFM method. By characterizing the shaker assemblies used to excite the wing's elastic modes, it was shown that the excitation force can be resolved with minimal attenuation and phase lag. An initial GVT study of the flexible wing rigidly mounted in isolation was conducted to tune an FEA model. After adjusting the model's mass, stiffness, and damping characteristics, the computationally estimated natural frequencies closely aligned with those identified experimentally within 16%. Lastly, an initial demonstration of the PFM method and excitation pods were performed in the free-flight environment. Through this flight-testing, the symmetric bending and torsional natural frequencies identified via the PFM method agreed within 1.95 Hz and 0.4 Hz of the FEA predictions, respectively. Additionally, the experimental results indicated a predicted shift in the flexible wing's first symmetric bending and first anti-symmetric torsional aeroelastic frequencies with increased air-speed. Flight-test data also revealed challenges with recovering anti-symmetric modal characteristics due to the approximately time-aligned excitation functions. The results of the abbreviated flight-testing indicate initial promise in applying the PFM method to flutter flight-testing campaigns.

1.4 Thesis Outline

Chapter 2 discusses the theory associated with the dynamics of an aeroelastic system and provides details on the PFM method. This chapter begins with a derivation of an aeroelastic system's governing equations of motion. LaGrangian mechanics

are used to describe the induced motion of a flexible wing due to a generalized lifting force and pitching moment. The resulting equations of motion are coupled with Theodorsen's unsteady aerodynamic theory to provide a simplified aeroelastic model. The remainder of Chapter 2 is dedicated to describing the PFM method. The equations of motion associated with an aeroelastic system modified in accordance with the PFM method is provided to explain the physics utilized in this approach. The individual steps, equations, and definition required to identify a wing's flutter boundary from experimental data are then detailed.

While Chapter 2 focuses on the theory associated with aeroelasticity and the PFM method, Chapter 3 details the experimental equipment developed specifically to demonstrate the PFM method in the free-flight environment. Chapter 3 begins with an overview of the excitation pod system responsible for augmenting the aeroelastic system's stability, generating an excitation force to induce motion, and measuring the resulting structural dynamic response. Specifications and performance limitations of the fixed-wing vehicle used to demonstrate the PFM method for flight-test applications are then provided. Lastly, the test vehicle's flexible wing designed and built for this study is detailed. Due to undesirable frequency characteristics discovered during ground- and flight-testing, several different configurations of the flexible wing and excitation pod were utilized. The inertial and geometric modifications associated with each wing configuration are described.

Before conducting flight-tests, a series of ground-tests were performed to validate the excitation pod's performance and to determine the frequency characteristics of the flexible wing. Chapter 4 begins with a verification study aimed at ensuring that

the force generated by the excitation pod can be accurately measured using an embedded accelerometer in combination with Newton’s laws of motion. Further bench-top testing was conducted to verify that the wing’s dynamic response to external perturbations could be accurately estimated by the excitation pod’s inertial measurement units (IMU). After validating the excitation pod’s operation characteristics, a series of ground vibration tests (GVTs) were conducted for various wing configurations. From these GVTs, the natural frequencies associated with each tested wing was used to tune a corresponding FEA model later used to corroborate flight-testing results.

After building confidence in the excitation system and identifying the structural dynamic characteristics of the flexible wing through ground-testing, flight-testing was conducted to demonstrate the PFM method in the free-flight environment. Chapter 5 begins with providing the mathematical definitions of the symmetric and anti-symmetric FRFs used to characterize the structural response of the wing at a variety of different flight conditions. The testing methodology and flight profiles used to collect aeroelastic data at both low speed and high speed flight are then discussed. The FRFs measured in the free-flight environment are presented, and the identified structural modes and associated natural frequencies are noted. Finally, a flutter margin versus velocity curve is provided, which is utilized to extrapolate an estimated flutter onset velocity. The results identified through flight-testing are compared to a computation model of the test vehicle.

Primary conclusions of this study and potential areas of future work related to PFM flight-testing are discussed in Chapter 6. Details associated with generating an excitation force and measuring the resulting structural dynamic response using the excitation pods are reiterated. Flight-testing results are summarized to highlight the

ability to identify structural modes and the associated natural frequencies using the excitation pods in the free-flight environment. Best practices related to the excitation function's frequency sweep profile are discussed. The identification of the flutter mechanism and corresponding onset speed is again stressed. Finally, areas for future work to improve aeroelastic flight-testing performed in accordance with the PFM method are listed.

Chapter 2: Aeroelastic System Derivation and PFM Overview

This chapter discusses the theory used to characterize the aeroelastic behavior of the test vehicle utilized in this study. The governing aeroelastic equations of motion for a simplified 2D wing-section are derived in Section 2.1. Section 2.2 discusses the PFM method. In this section, the aeroelastic equations of motion presented in Section 2.1 are modified to account for the PFM method's salient features. A derivation of the PFM method that follows the work of Roizner and Karpel is presented. The equations required to identify a vehicle's flutter onset condition in accordance with the PFM method are discussed.

2.1 Aeroelastic EOM

To gain insight into the aeroelastic characteristics of the flight-test vehicle and flexible wing used in this study, a simplified analytical model was developed to estimate the anticipated flutter mechanism before flight-testing was conducted. By utilizing the K-Method algorithm and data collected through a GVT campaign, this analysis technique was able to calculate the aircraft's expected flutter velocity. Knowledge of this flight condition is crucial while designing the flight-testing matrix to ensure that test points are chosen such that the flutter onset condition can be explored without

subjecting the vehicle to undue risk. Additionally, this model was used in an unconventional manner to help design wing modifications such that the flutter velocity was within the test vehicle's achievable flight envelope. The following section discusses the derivation of the aeroelastic model used in this work.

A free body diagram (FBD) of a two-dimensional aeroelastic system is presented below in Figure 2.1. As shown in Figure 2.1, a representative airfoil cross-section is free to move along a pitching α and plunging h degree of freedom. Structural deflections caused by the lifting force L and pitching moment $M_{A.C.}$ are countered by a linear and torsional spring co-located along the wing's elastic axis. These restoring springs have spring constants K_h and K_α , respectively. In this FBD, b is the cross-section's semi-chord, ba represents the distance between the chord midpoint and elastic axis, and bx_a is associated with the distance between the chord midpoint and the lifting surface's center of gravity.

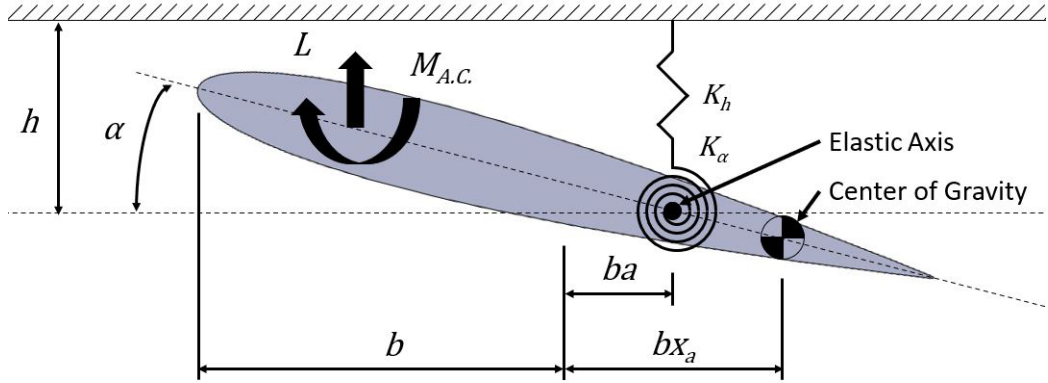


Figure 2.1: Free body diagram of a simplified 2D aeroelastic system.

A LaGrangian approach was used to derive the governing equations of motion associated with the aeroelastic system depicted above in Figure 2.1. Consequently, the cross-section's total kinetic and potential energies were required to be known as a function of the wing's pitching and plunging states. The proceeding equations are focused on relating the aeroelastic system's energy to it's degrees of freedom.

The aeroelastic system's total kinetic energy is defined by Equation 2.1 below. In Equation 2.1, the cross-section's kinetic energy T is computed in terms of the velocity $\dot{w}$ experienced by a differential mass element dm .

$$T = 0.5 \int \dot{w}^2 dm \quad (2.1)$$

By utilizing rigid body dynamics, Equation 2.1 was modified to compute the cross-section's kinetic energy with respect to the aeroelastic system's degrees of freedom. Since a given mass element's velocity is comprised of both a translational and rotational component, it can be calculated in terms of the cross-section's plunge velocity $\dot{h}$, pitch velocity $\dot{\alpha}$, and distance from the wing's elastic axis r .

$$T = 0.5 \int (\dot{h} + r\dot{\alpha})^2 dm \quad (2.2)$$

By evaluating the kinetic energy integral presented in Equation 2.2 and substituting appropriate identities (reference Appendix A), the algebraic expression presented in Equation 2.3 was derived. In Equation 2.3, m is the wing's mass per unit length, I_α corresponds to the mass moment of inertia about the wing's elastic axis, and S_A represents the static mass moment about the wing's elastic axis.

$$T = 0.5m\dot{h}^2 + 0.5I_\alpha\dot{\alpha}^2 + S_A\dot{h}\dot{\alpha} \quad (2.3)$$

As shown in Figure 2.1, the wing's restoring force and moment is generated by a linear and torsional spring with spring constants K_h and K_α , respectively. Consequently, Hook's Law was used in Equation 2.4 to compute the cross-section's total potential energy U resulting from a given plunge displacement h and torsional deflection α .

$$U = 0.5K_h h^2 + 0.5K_\alpha \alpha^2 \quad (2.4)$$

In order to proceed with deriving the aeroelastic system's governing equations of motion, the LaGrangian equation defined below in Equation 2.5 was used. In Equation 2.5, q_r refers to a given system state and $Q_r(t)$ symbolizes the associated generalized force, which is a function of time t .

$$\frac{d}{dt} \frac{\partial T}{\partial \dot{q}_r} - \frac{\partial T}{\partial q_r} + \frac{\partial U}{\partial q_r} = Q_r(t) \quad (2.5)$$

The second-order dynamic system described by Equations 2.6 and 2.7 were derived using the LaGrangian in conjunction with Equations 2.3 and 2.4, which describes the aeroelastic system's total kinetic and potential energy. In the following set of equations, the aerodynamic lifting force and pitching moment about the elastic axis are represented by L and $M_{E.A.}$, respectively.

$$m\ddot{h} + S_A\ddot{\alpha} + K_h\dot{h} = -L \quad (2.6)$$

$$S_\alpha\ddot{h} + I_\alpha\ddot{\alpha} + K_\alpha\dot{\alpha} = M_{E.A.} \quad (2.7)$$

To condense the aeroelastic system's governing equations of motion, Equations 2.6 and 2.7 were combined to develop the corresponding matrix representation shown in Equation 2.8.

$$\begin{bmatrix} m & S_\alpha \\ S_\alpha & I_\alpha \end{bmatrix} \begin{bmatrix} \ddot{h} \\ \ddot{\alpha} \end{bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix} \begin{bmatrix} \dot{h} \\ \dot{\alpha} \end{bmatrix} = \begin{bmatrix} -L \\ M_{E.A.} \end{bmatrix} \quad (2.8)$$

While the preceding equations define the cross-section's structural response to a generalized lifting force and pitching moment, a critical aspect missing from this derivation is a model detailing the aeroelastic system's unsteady aerodynamic behavior. Although several different unsteady aerodynamic models have been proposed, Theodorsen's unsteady aerodynamic theory was utilized in this analysis due to its high degree of accuracy for low values of reduced frequency and ease of implementation. Note that Theodorsen's aerodynamic model is dependant upon several governing assumptions, such that the flight envelope is restricted to the incompressible flow regime, the wing's structural response follows simple harmonic motion, and that thin airfoil theory is valid. Given the test vehicle's design characteristics and flight envelope as detailed in Chapter 3, the assumptions made in Theodorsen's unsteady aerodynamic model remain valid throughout the flight-testing study.

The unsteady lifting force and pitching moment about the wing's elastic axis is defined in Equations 2.9 and 2.10, respectively. In these equations originally derived by Theodorsen [12], ρ_∞ refers to the free-stream air density, u_∞ corresponds the free-stream airspeed, and $C(k)$ is Theodorsen's lift deficiency, which is a function of reduced frequency k .

$$L = \pi\rho_\infty b^2 \{-ba\ddot{\alpha} + \ddot{h} + u_\infty \dot{\alpha}\} + 2\pi\rho_\infty u_\infty bC(k)\{b(0.5 - a)\dot{\alpha} + \dot{h} + u_\infty \alpha\} \quad (2.9)$$

$$\begin{aligned} M_{E.A.} = \pi\rho_\infty b^2 \{-b^2(0.125 + a^2)\ddot{\alpha} + ba\ddot{h} - u_\infty b(0.5 - a)\dot{\alpha}\} \\ + 2\pi\rho_\infty u_\infty b^2(a + 0.5)C(k)\{b(0.5 - a)\dot{\alpha} + \dot{h} + u_\infty \alpha\} \end{aligned} \quad (2.10)$$

Since it was assumed that the wing's structure undergoes simple harmonic motion, the displacement, velocity, and acceleration associated with both wing pitching and plunging are expressed by Equations 2.11 - 2.16, respectively. In this set of equations, ω is a given state's frequency. Note that since the following equations were derived by

assuming a fully imaginary system pole, these identities are only valid at the flutter onset condition. All other flight conditions result in a system pole containing a real part due to non-zero damping.

$$\alpha(t) = \bar{\alpha}_0 e^{i\omega t} \quad (2.11)$$

$$\dot{\alpha}(t) = i\omega \bar{\alpha}_0 e^{i\omega t} = i\omega \alpha(t) \quad (2.12)$$

$$\ddot{\alpha}(t) = -\omega^2 \bar{\alpha}_0 e^{i\omega t} = -\omega^2 \alpha(t) \quad (2.13)$$

$$h(t) = \bar{h}_0 e^{i\omega t} \quad (2.14)$$

$$\dot{h}(t) = i\omega \bar{h}_0 e^{i\omega t} = i\omega h(t) \quad (2.15)$$

$$\ddot{h}(t) = -\omega^2 \bar{h}_0 e^{i\omega t} = -\omega^2 h(t) \quad (2.16)$$

Equations 2.11 - 2.16 were substituted into Equations 2.9 and 2.10 such that the unsteady lifting force and pitching moment are solely expressed in terms of the pitching and plunging displacement states. After substituting the unsteady lifting force and pitching moment into the governing equation of motion described in Equation 2.8, the dimensionless aeroelastic flutter equation was derived as presented below in Equation 2.17. In an attempt to describe the aeroelastic system's equation of motion in a simplified form, compact notation was used where large groupings of variables were defined as singular parameters. For example, in Equation 2.17 μ is the mass ratio, x_a represents the nondimensional static imbalance, and r_a symbolizes the nondimensional radius of gyration. Furthermore, ω_h and ω_α represent the wing's bending and torsional natural frequency at zero free-stream velocity as identified through GVT experiments. A complete list of variable definitions can be found in Appendix A. Lastly, structural damping, which was not originally accounted for in Figure 2.1 and

Equation 2.8 was added in an ad-hoc manner. The bending and torsional structural damping coefficient is represented by g_h and g_α , respectively.

$$\begin{bmatrix} \mu \left(1 - \frac{\omega_h^2}{\omega^2} (1 + i g_h) \right) - l'_h & \mu x_a - l'_\alpha \\ \mu x_a + m'_h & r_\alpha^2 \mu \left(1 - \frac{\omega_\alpha^2}{\omega^2} \right) (1 + i g_\alpha) + m'_\alpha \end{bmatrix} \begin{bmatrix} \left(\frac{h}{b} \right) \\ \alpha \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (2.17)$$

2.2 Parametric Flutter Margin (PFM) Method

The PFM method assumes that an aeroelastic system can be modified in a prescribed manner such that the nominal system's flutter onset condition is artificially increased by the addition of a stabilizing element and a hypothetical single-input single-output (SISO) feedback system. A known excitation force is applied coincidentally to the stabilizing mass's position and the resulting structural dynamics are measured to compute frequency response functions (FRFs) [8]. The phase-crossover frequency and corresponding flutter margin is recorded for increasing dynamic pressures to obtain a flutter margin versus velocity curve. Of particular interest is the velocity corresponding to a flutter margin of 0 dB, which indicates the nominal (unmodified) aeroelastic system's flutter boundary. Due to the addition of the stabilizing element, PFM testing can be conducted at airspeeds exceeding the unmodified vehicle's flutter speed, allowing for positive identification of the flutter boundary in a safe manner. While the PFM method is not a model-based approach, the following section details the simplified 2D equations used to describe this technique.

The aeroelastic system's governing homogeneous equation of motion denoted in Equation 2.17 can be expressed in the frequency domain as presented below in Equation 2.18. In Equation 2.18, $[A(i\omega)]$ is the system matrix containing the generalized

mass, stiffness, damping, and aerodynamic terms and $\{x(i\omega)\}$ is the modal displacement vector associated with the wing's pitch and plunge states.

$$[A(i\omega)]\{x(i\omega)\} = \{0\} \quad (2.18)$$

Since the PFM method requires FRFs to determine the flutter onset condition, a known force must be applied to the aeroelastic system to excite a structural response. The PFM method also mandates that the aeroelastic stability is positively augmented since the nominal system is neutrally stable at the flutter boundary. Due to these requirements, the nominal aeroelastic system is modified by the addition of a stabilizing element P_f and applied force u_f co-located near the wing's leading edge as shown below in Figure 2.2.

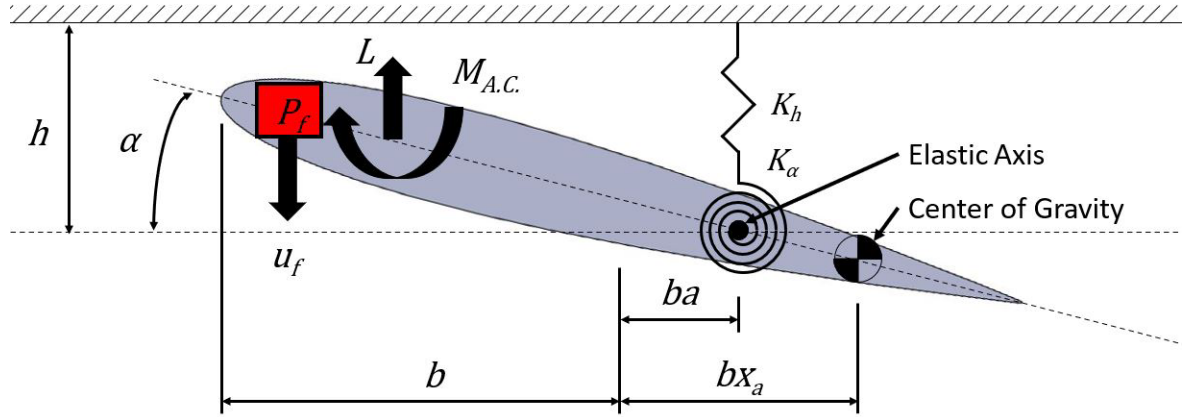


Figure 2.2: Free body diagram of simplified 2D aeroelastic system modified for PFM.

Equation 2.17 was modified to account for the PFM method's principle features. Most notably, the governing equation of motion is no longer homogeneous due to the excitation force required to generate a structural response. The right-hand side of

Equation 2.19 describes the effect of the applied force $u_f(i\omega)$ to each state as allocated through the input distribution vector $\{B_f\}$. A hypothetical single-input single-output (SISO) feedback system is added to remove the mass, stiffness, and damping effects associated with the stabilizing parameter P_f such that the aeroelastic characteristics of the nominal system can be determined. Lastly, the modified aeroelastic system's output scalar $y_f(i\omega)$ is related to the state vector $\{x(i\omega)\}$ as distributed through the plant output vector $\{C_f(i\omega)\}$.

$$\begin{aligned} [A(i\omega) + P_f\{B_f\}\{C_f(i\omega)\}]\{x(i\omega)\} &= \{B_f\}u_f(i\omega) \\ y_f(i\omega) &= \{C_f(i\omega)\}\{x(i\omega)\} \end{aligned} \tag{2.19}$$

For a known input force u_f , Equation 2.19 can be solved for the modal state vector $x(i\omega)$ and output scalar $y_f(i\omega)$ at all stable flight conditions. Note that due to the added stabilizing element, the flight condition associated with the nominal flutter boundary is now stable. A stabilization parameter must be selected to satisfy Equation 2.20, where ω_f denotes the flutter frequency.

$$u_f(i\omega_f) = P_f y_f(i\omega_f) \tag{2.20}$$

This condition can be proven by substituting Equation 2.20 into Equation 2.19. After performing this substitution, the external forcing function on the equation's right-hand side cancels the $P_f\{B_f\}\{C_f(i\omega)\}$ term required to account for the added stabilizing parameter. The resulting formula is the homogeneous flutter equation defined in Equation 2.17, which has a nontrivial solution associated with the vehicle's flutter boundary.

The stabilization parameter selected for this study is defined as the ratio between the shaker sub-assembly's static mass m_{st} and moving mass m_{mv} . Consequently, the aeroelastic system previously described in Equation 2.19 can be expressed in

terms of the moving mass's acceleration a_{mv} and static mass's acceleration a_{st} . This representation is presented below in Equation 2.21, where $\{\phi\}$ is defined as the modal displacement vector at the moving mass's point-of-action.

$$\begin{aligned} [A(i\omega) + \omega^2 m_{st} \{\phi^T\} [\phi]] \{x(i\omega)\} &= -m_{mv} \{\phi^T\} a_{mv} \\ a_{st}(i\omega) &= -\omega^2 [\phi] \{x(i\omega)\} \end{aligned} \quad (2.21)$$

To identify the nominal flutter boundary, data must be collected at various flight conditions within the vehicle's proposed flight envelope such that FRFs can be experimentally determined. These FRFs are displayed in a Bode plot format to depict trends in the aeroelastic system's gain and phase-angle as a function of the excitation force's input frequency. The system's gain and phase-angle is computed by Equations 2.22 and 2.23, respectively.

$$G(\omega) = 20 \log_{10} \left| \frac{P_f y_f(i\omega)}{u_f(i\omega)} \right| \quad (2.22)$$

$$\Phi(\omega) = \angle \frac{P_f y_f(i\omega)}{u_f(i\omega)} \quad (2.23)$$

By referencing an experimentally determined Bode plot, the phase-crossover frequency ω_{pco} , as defined in Equation 2.24, is identified. Using this definition, the phase-crossover frequency implies that the aeroelastic system's input and output signals are in-phase. Note that the aeroelastic system may have zero or multiple phase-crossover frequencies within the frequency range of interest.

$$\Phi(\omega_{pco}) = 0 \quad (2.24)$$

From the experimental Bode plot, the flutter margin, defined in Equation 2.25, is noted at each identified phase-crossover frequency. As stated by Roizner and Karpel, the flutter margin indicates the amount of stability parameter that must be removed

from the modified aeroelastic system to induce flutter at the associated flight condition. Consequently, a flutter margin of 0 dB corresponds to the unmodified system’s stability boundary. Flutter margins greater than 0 dB are analogous to a stable system while flutter margins less than 0 dB reveal that the original system is unstable at the given flight condition.

$$FM = -G(\omega_{pco}) \quad (2.25)$$

In order to determine an aircraft’s flutter boundary, FRFs are computed for multiple flight conditions spanning the proposed flight envelope. Using these FRFs, the phase-crossover frequency and corresponding flutter margin are recorded for each test point. A flutter margin versus airspeed curve is plotted to identify the condition resulting in a flutter margin of 0 dB, which is associated with the unmodified aeroelastic system’s flutter velocity. The PFM method allows for an aircraft’s flutter onset condition can be identified in a safe and accurate manner.

As noted above, a salient feature of the PFM method is the inclusion of a stabilizing element such that flight conditions exceeding the nominal flutter boundary can be safely explored. Additionally, the aeroelastic system’s dynamic response to an applied excitation force must be accurately measured such that FRFs can be identified. Like the system used by Karpel *et al.* [10], the additional stabilizing element, excitation mechanism, and measurement system utilized in this study was provided by a self contained, electro-mechanical excitation pod. The following section provides a detailed description of the excitation pods, flexible wing, and associated fixed-wing vehicle utilized throughout the ground- and flight-testing campaigns.

Chapter 3: Excitation Pod and Aircraft Design

As described in Chapter 2, the PFM method requires a known external force to be applied to an aeroelastic system to excite a structural response such that FRFs may be computed. In this chapter, the design of a self-contained electro-mechanical excitation pod responsible for the generation of this external force and measurement of the resultant response is discussed. Additionally, details of the fixed-wing vehicle developed specifically for PFM flight-testing are provided. This chapter describes the equipment utilized in the combined ground-testing and flight-testing campaigns aimed at demonstrating the PFM method in the free-flight environment.

3.1 Excitation Pod Design

The external force required for the PFM method is generated by an electro - mechanical excitation pod depicted in Figure 3.1. The excitation pod system consists of a shaker assembly and control board contained within a cylindrical fiberglass tube mounted to each wingtip of the flexible wing utilized in this study. The shaker assembly can be positioned at discrete locations inside the fiberglass tube ahead of a wing's elastic axis, which positively augments the aeroelastic system's stability. Consequently, the shaker assembly provides both the stabilizing element and mechanism for generating the excitation force required for the PFM method. An 82-gram moving

mass is driven by a linear voice coil and transverses a 60 mm magnetic guide shaft such that the resultant force acts perpendicular to the local chord line. A small MPU-9250 Inertial Measurement Unit (IMU) is attached to the moving mass such that the induced force applied to the wing's structure can be computed via Newton's laws of motion, $F = m_{mv}a_{mv}$. A closed-loop position controller implemented through the control board incorporates feedback from a Hall effect sensor and is utilized to track desired sinusoidal motion with input frequencies up to 20 Hz. The control board is equipped with an additional MPU-9250 such that the flexible wing's structural dynamic response may be measured. The control board is also equipped with a U-blox M8N GPS unit, which provides timing information such that a phase-angle offset can be incorporated between the left- and right-wing frequency sweep profiles. Experimental data associated with both the excitation pod and flexible wing's distributed accelerometers are stored to a local micro-SD card at a sampling rate of 200 Hz. As will be discussed in the following chapters, the wing's inertial properties required modification such that the flutter onset condition was within the test vehicle's achievable flight envelope. To this end, ballast could be secured to the pod's structure as shown in Figure 3.2.

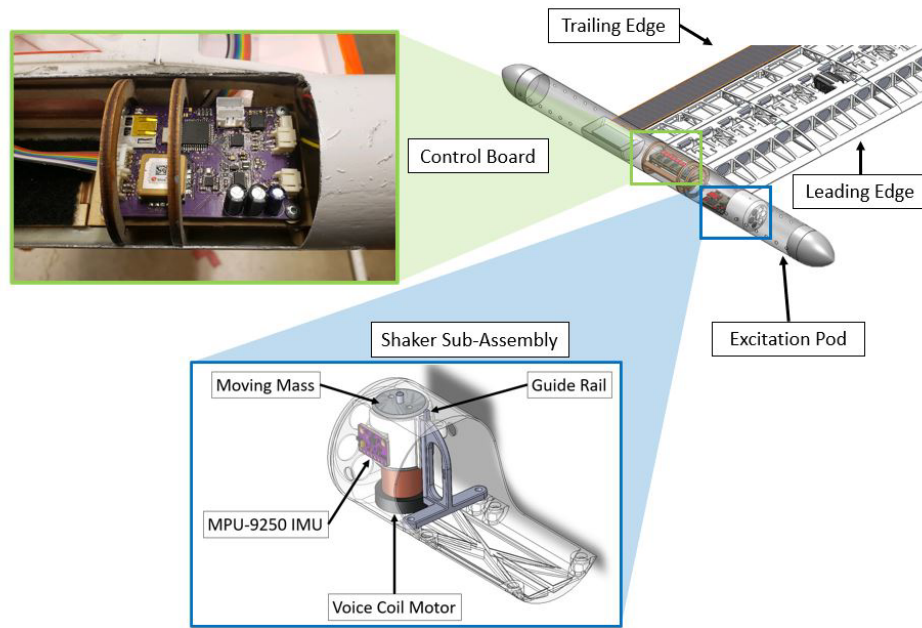


Figure 3.1: Rendering of the excitation pod, control board, and shaker assembly.

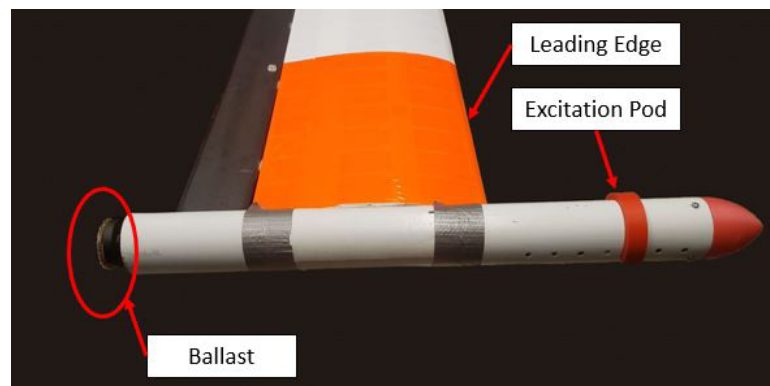


Figure 3.2: Excitation pod with rear ballast (wing configuration 2).

3.2 Flight-Test Vehicle Design

The fixed-wing flight-test vehicle shown in Figures 3.3 and 3.4 was developed to demonstrate the PFM method for small scale UAS operating in a free-flight environment. The vehicle is a traditional fixed-gear, tail-dragger monoplane featuring a high mounted wing and conventional empennage. The fuselage is constructed with birch plywood and is covered with an UltraKote polymer skin. Foam cores covered with fiberglass composite skin were utilized to build both the horizontal and vertical stabilizers. To mitigate engine vibrations, an electric propulsion system consisting of an E-flite Power 160 brushless motor, Castle Creation Edge High Voltage 80 Amp Electronic Speed Controller, two Zippy 5S 8000 mAh Lithium Polymer batteries, and a APC 19×12e propeller was selected to power the vehicle. A Castle Creation CC Battery Eliminating Circuit 2.0 is used to supply power from the propulsion system to the five HiTec HS-645MG servos responsible for deflecting the vehicle’s aerodynamic control surfaces. These servos are actuated by a Pixhawk 4 autopilot system, which serves as the guidance, navigation, and control flight computer for the vehicle. Additional geometric and performance specifications of the flight-test vehicle are listed in Table 3.1.

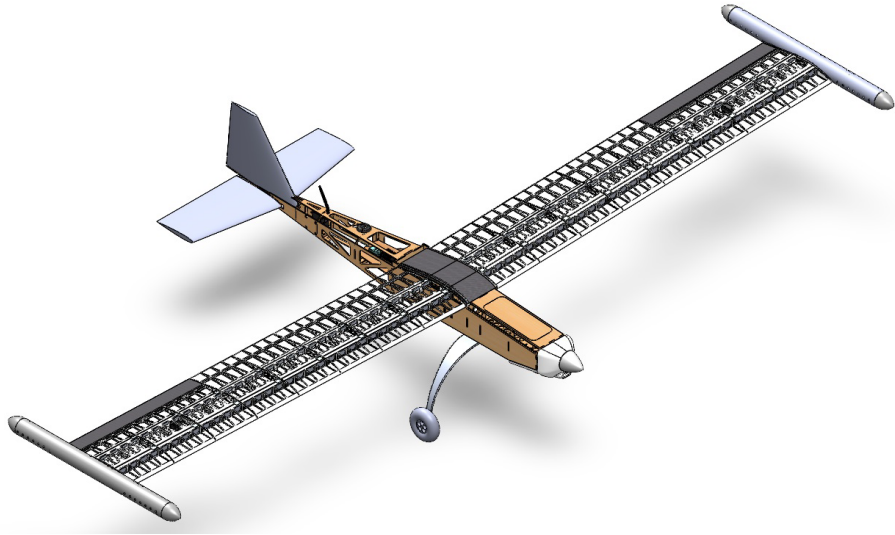


Figure 3.3: CAD rendering of the flight-test vehicle.



Figure 3.4: Flight-test vehicle in flight.

Characteristic	Metric	U.S. Customary
Length	1.7 <i>m</i>	65 <i>in</i>
Wingspan	3.4 <i>m</i>	134 <i>in</i>
Wing Area	1.15 <i>m</i> ²	1790 <i>in</i> ²
Horizontal Tail Area	0.17 <i>m</i> ²	260 <i>in</i> ²
Vertical Tail Area	0.06 <i>m</i> ²	100 <i>in</i> ²
Flying Weight	15.0 <i>kg</i>	33 <i>lbs</i>
Stall Speed	9 <i>ms</i> ⁻¹	29 <i>fts</i> ⁻¹
Cruise Speed	20 <i>ms</i> ⁻¹	66 <i>fts</i> ⁻¹
Maximum Speed	36 <i>ms</i> ⁻¹	118 <i>fts</i> ⁻¹

Table 3.1: Geometric and performance specification of the flight-test vehicle.

3.3 Flexible Wing Design

The vehicle’s flexible wing depicted in Figure 3.5 is 3.4 m long. This straight, untapered rectangular wing uses an SD-7043 airfoil and has an aspect ratio of 10. Additionally, a 2° dihedral angle was incorporated to improve lateral-directional handling qualities. The wing’s structure consists of 3D printed Nylon 12 airfoil sections joined together along a 6061-aluminum front and rear spar. Due to the allowable printing volume of the SLS machine, each wing-half is comprised of eight individual airfoil sections approximately 0.15 m in length. The front and rear spars span the entire length of the wing and have a 25.4 mm × 25.4 mm hollow square cross-section and 19.1 mm diameter hollow circular geometry, respectively. Furthermore, the left- and right-wing halves mount to the fuselage via a stainless-steel central wing joiner. The wing is equipped with 16 ADX333 3-axis accelerometers, distributed equally along the front and rear aluminum spars, as shown in Figure 3.5. Note that the spatial distribution of the accelerometers is offset by 0.2 m between the left- and right-wing

halves to reduce the possibility of locating a sensor at a structural node. As ground- and flight-testing proceeded, modifications to the flexible wing's design were incorporated to address identified issues as discussed in Chapters 4 and 5. Details of these configurations are presented below.

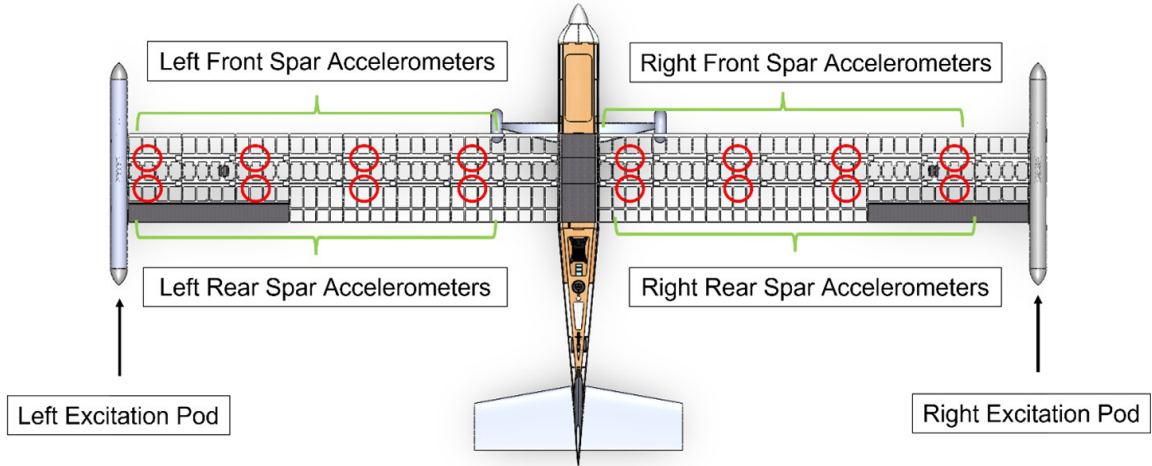


Figure 3.5: Top-down view of the flight-test vehicle and flexible wing.

3.3.1 Wing Configuration 1

This first wing configuration represents the original design intended to be utilized in the joint ground-testing and flight-testing study of the PFM method. As shown in Figure 3.6, horizontally oriented cross-braces between the front and rear spars were used in this design to increase the wing's torsional rigidity. Ground- and flight-testing of this configuration was performed with the excitation pod's shaker assembly located 135 mm (40% chord) ahead of the wing's leading edge as depicted in Figure 3.7. Aft-mounted ballast was not secured to the excitation pod's structure for testing conducting in this configuration.

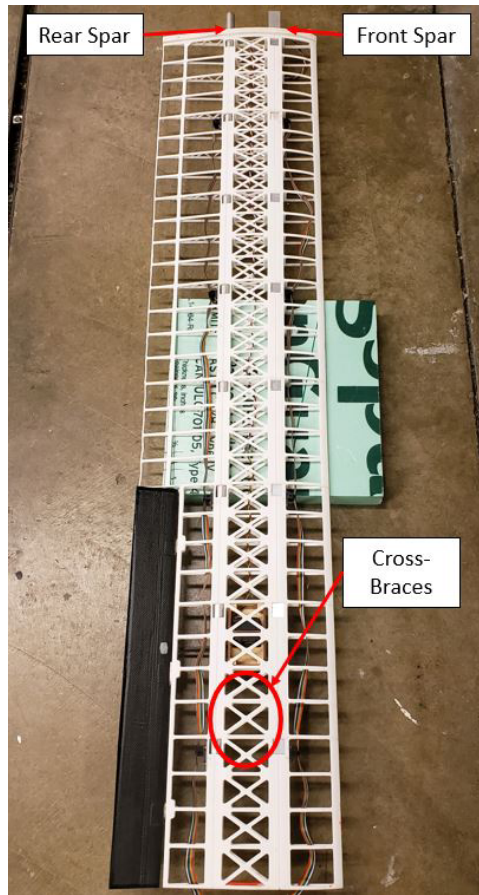


Figure 3.6: Wing configuration 1 with cross-braces.

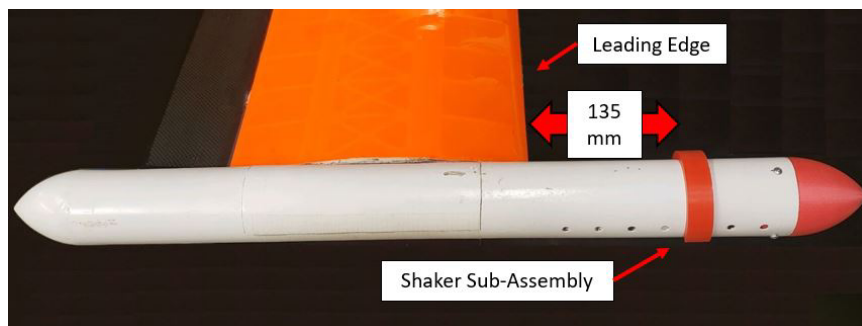


Figure 3.7: Shaker assembly location for wing configuration 1.

3.3.2 Wing Configuration 2

As will be discussed in Chapter 4, modifications to wing configuration 1 were required to alter the torsional mode's frequency characteristics to coincide with the excitation pod's performance limitations. In this new arrangement, referred to as wing configuration 2, the horizontal cross-braces connecting the front and rear spars were removed to lessen the structure's torsional rigidity as depicted in Figure 3.8. To lower the flight-test vehicle's flutter onset condition, 100 gram masses were secured to each excitation pod's structure 395 mm (116 % chord) behind the wing's leading edge. Like wing configuration 1, the excitation pod's shaker assembly was positioned 135 mm (40% chord) ahead of the wing's leading edge. The location of the added ballast and shaker assembly is depicted in Figure 3.9.

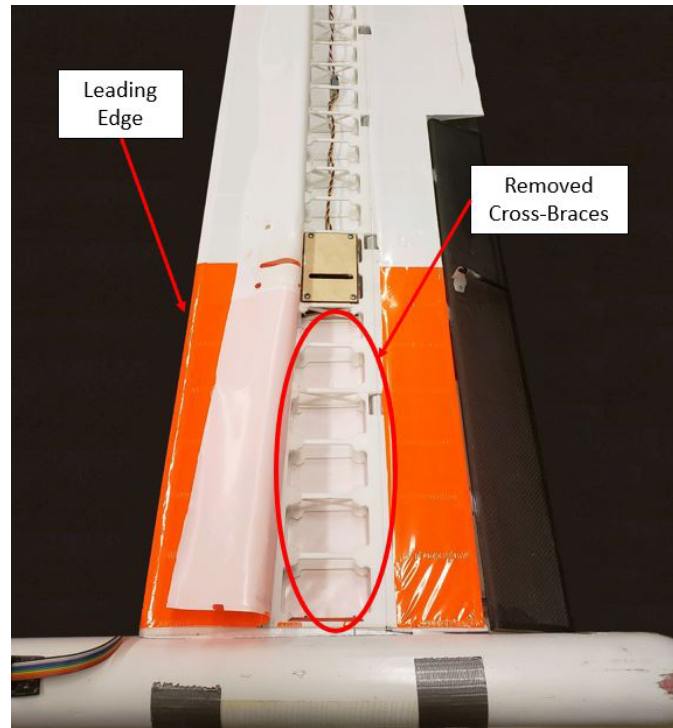


Figure 3.8: Wing configuration 2 with cross-braces removed.

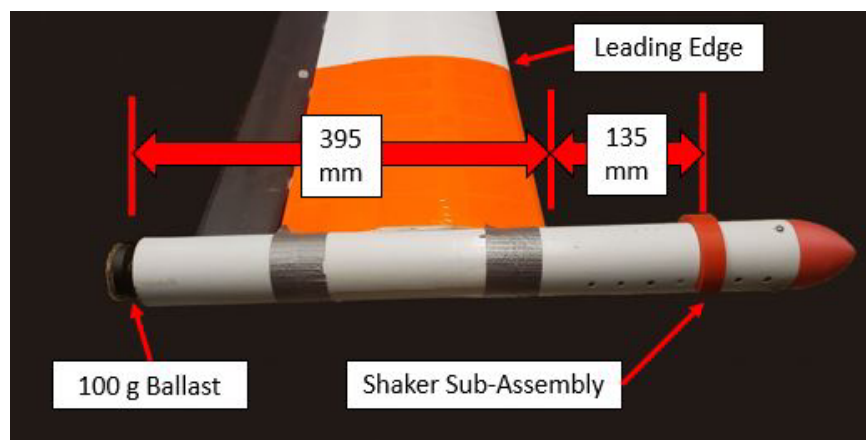


Figure 3.9: Ballast and shaker assembly location for wing configuration 2 (tail cone removed).

3.3.3 Wing Configuration 3

During flight-testing, it was discovered that the flutter velocity of wing configuration 2 well exceeded the test aircraft's maximum airspeed of 36 ms^{-1} . To lower the flutter onset speed to within the vehicle's achievable flight envelope, the wing's inertial properties were altered to reduce the torsional mode's natural frequency and to shift the wing's center of gravity further behind the elastic axis. As shown in Figure 3.10, wing configuration 3 featured an extended excitation pod measuring 0.76 m in length. Each excitation pod was equipped with a 220 g ballast located 630 mm behind the wing's leading edge. Similar to wing configuration 2, the shaker assembly was positioned 135 mm (40% chord) ahead of the wing's leading edge and the structure did not feature the cross-braces connecting the front and rear spars. While constructing this modification, it was noticed that the right-wing half used in wing configurations 1 and 2 had an unintended twist (wash-in) in the structure. To address this issue, wing configuration 3 features a rebuilt right-wing half with a similar twist profile of the matching left-wing half.

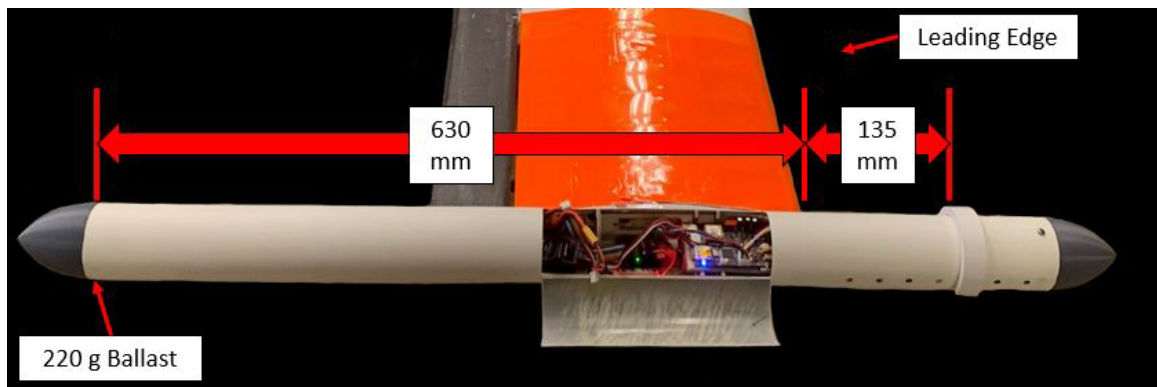


Figure 3.10: Ballast and shaker assembly location for wing configuration 3 (tail cone attached).

Chapter 4: Ground-Testing

Before advancing to the free-flight portion of this study, a thorough ground-testing campaign was performed to verify that the excitation pods operated as expected and to characterise the modal qualities of the flexible wing. This chapter begins with a calibration test of the excitation pod’s shaker assembly to ensure that the generated force can be accurately estimated by the embedded accelerometer. Additional testing was performed to verify that the control board’s IMU signals could be utilized in conjunction with rigid-body dynamics to measure the flexible wing’s response to a perturbation force. GVTs were conducted with each of the wing configurations described in Chapter 3 to identify the bending and torsional structural modes. Data collecting in the ground-testing phase of this study was used to build confidence in the experimental equipment and help guide the design of the flight-testing matrix.

4.1 Excitation Force Calibration Testing

As noted in Roizner *et al.* [9] and Karpel *et al.* [10], the PFM method requires precise knowledge of the external force applied to a wing’s structure to positively identify an aircraft’s flutter boundary. Consequently, the test-stand depicted below in Figure 4.1 was developed to verify that the force generated by the excitation pod’s shaker assembly could be applied to the wing’s structure with minimal attenuation

and phase lag. As shown in Figure 4.1, the test-stand consists of the excitation pod's shaker assembly described above in Chapter 3 and a Polylactic Acid (PLA) 3D printed clamp. In addition to securing the shaker assembly, the clamp is equipped with a base mounted 2-axis Futek MBA500 load cell rated for a maximum applied force and torque of 222 N and 5.65 Nm, respectively. The Futek load cell was connected to a custom signal conditioner and sampled at 4 kHz using a National Instrument's USB-6009 DAQ.



Figure 4.1: Excitation force calibration test-stand.

Equation 4.1 was used to calculate the external force, F_{ex} , generated by the shaker assembly in terms of the moving mass and the corresponding vertical acceleration component as measured by the embedded IMU. It should be noted that since the moving mass is constrained to translate in a direction perpendicular to the local chord line, it is assumed that both the acceleration and excitation force act along the same local z-axis.

$$F_{ex}(t) = -m_{mv}a_{mv,z}(t) \quad (4.1)$$

The transfer function described by Equation 4.2 was used to determine the gain and phase-angle characteristics of the excitation pod's external forcing function. In Equation 4.2, F_{ex} denotes the estimated force generated by the moving mass while $F_{ex,m}$ represents the applied force as directly measured by the Futek load cell (shown in Figure 4.1). The transfer function, $T_{Calibration}(i\omega)$, was calculated using MATLAB's *tfestimate* function.

$$T_{Calibration}(i\omega) = \frac{F_{ex}(i\omega)}{F_{ex,m}(i\omega)} \quad (4.2)$$

The resulting transfer function is depicted as the Bode plot in Figure 4.2. The experimental data used to identify this FRF was collected by performing a frequency sweep over the excitation pod's operational frequency range of 2 Hz – 20 Hz. As shown in Figure 4.2, the shaker assembly has a mean gain value of 1.03 ± 0.05 and mean phase-angle of $0.8^\circ \pm 2.1^\circ$ across the excitation pod's operational frequency range. The force as estimated by Equation 4.1 is approximately 3% greater than that measured from the base-mounted loadcell. The shaker's gain and phase-angle behavior appears independent of frequency. This finding verifies that the excitation force applied to the wing's structure throughout the GVT and flight-testing campaigns can be measured by utilizing the mounted accelerometer signals.

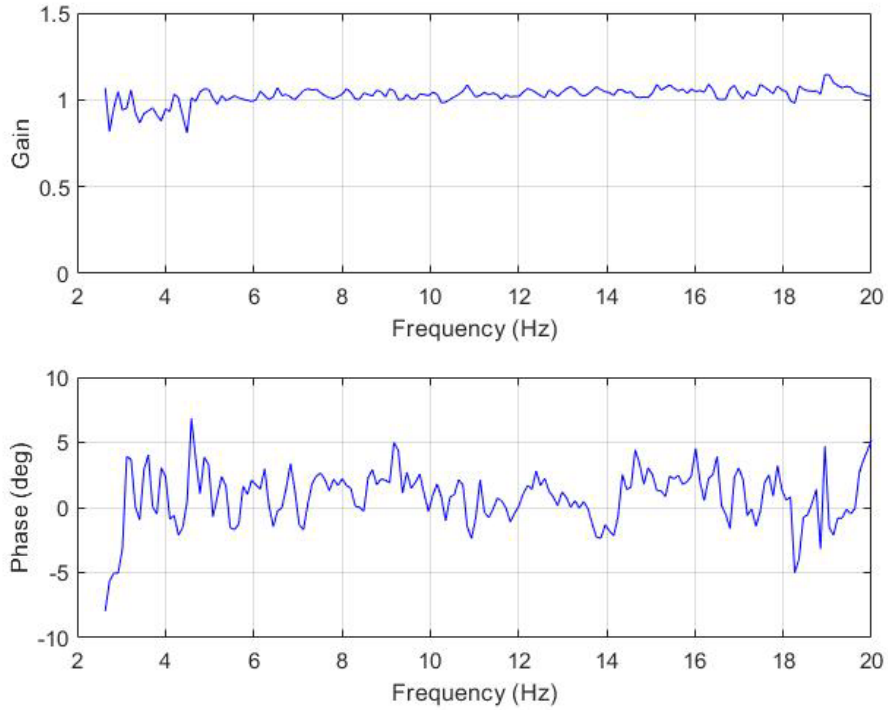


Figure 4.2: Base acceleration to resultant measured force Bode plot.

4.2 Static Mass Acceleration Verification

While the previous section demonstrated that the excitation force generated by the shaker assembly can be accurately measured by the IMU secured to the moving mass, the wing's response to this perturbation must also be known to compute the FRFs required for the PFM method. As explained in Chapter 2, the response signal is defined as the acceleration associated with the static mass added to stabilize the aeroelastic system. The following section details the testing and results associated with verifying that the aeroelastic system's response signal can be measured by the instruments contained within the excitation pod.

As previously shown in Figure 3.1, the components of the shaker assembly associated with the added static mass includes the 3D printed housing, linear voice coil motor, and magnetic guide rail. To be consistent with the PFM theory outlined in Chapter 2, it was assumed that the static mass's center of gravity is coincident with the location of the applied force generated by the moving mass. Due to geometric constraints, an accelerometer could not be placed at the voice coil motor's base to directly measure the static mass's acceleration. Instead, the acceleration signal was calculated using the control board's IMU in conjunction with rigid body dynamics as illustrated by the simplified 2D diagram presented in Figure 4.3.

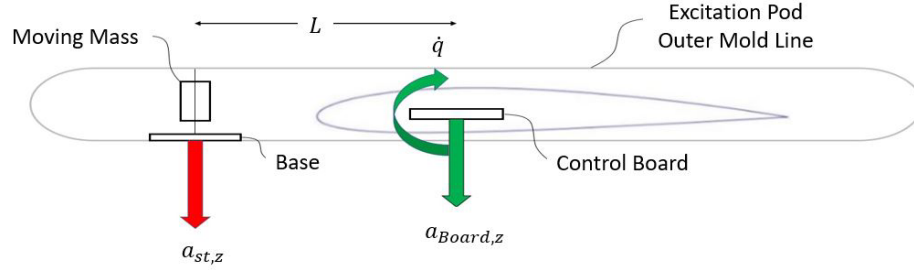


Figure 4.3: Simplified static mass acceleration illustration.

As described in Chapter 2, the PFM method requires precise knowledge of the aeroelastic system's dynamic response at the same location as the applied excitation force. Since the control board's IMU is positioned well behind the moving mass, a transformation was required to estimate the wing's response at the shaker assembly's centroid. By utilizing the excitation pod's control board accelerometer and gyroscope measurements in combination with rigid-body dynamics, the static mass's acceleration vector a_{st} was estimated by Equation 4.3 [13]. In Equation 4.3, a_{cb} corresponds

to the control board's rectilinear acceleration vector; X is the displacement vector between the control board's IMU and static mass's centroid; p_{cb} , q_{cb} , and r_{cb} represent the control board's body axis roll, pitch, and yaw angular velocities, respectively; and $\dot{p}_{cb}$, $\dot{q}_{cb}$, and $\dot{r}_{cb}$ are associated with the control board's body axis roll, pitch, and yaw angular accelerations, respectively.

$$\begin{bmatrix} a_{st,x} \\ a_{st,y} \\ a_{st,z} \end{bmatrix} = \begin{bmatrix} a_{cb,x} \\ a_{cb,y} \\ a_{cb,z} \end{bmatrix} + \begin{bmatrix} (q_{cb}^2 + r_{cb}^2) & -(p_{cb}q_{cb} - \dot{r}_{cb}) & -(p_{cb}r_{cb} - \dot{q}_{cb}) \\ -(p_{cb}q_{cb} + \dot{r}_{cb}) & (p_{cb}^2 + r_{cb}^2) & -(q_{cb}r_{cb} - \dot{p}_{cb}) \\ -(p_{cb}r_{cb} + \dot{q}_{cb}) & -(q_{cb}r_{cb} + \dot{p}_{cb}) & (p_{cb}^2 + q_{cb}^2) \end{bmatrix} \begin{bmatrix} X_x \\ X_y \\ X_z \end{bmatrix} \quad (4.3)$$

To verify that the static mass's acceleration could be computed using Equation 4.3, the experimental setup depicted in Figure 4.4 was constructed. As shown in Figure 4.4, the excitation pod was secured to the flexible wing previously described in Chapter 3. An ADX333 3-axis accelerometer was attached to the excitation pod's structure below the voice coil motor to directly measure the static mass's acceleration resulting from a frequency sweep. Vibrational movement was induced by performing a frequency sweep over the excitation pod's operation frequency range. Data collected during this experiment was recorded to the control board's micro SD card.

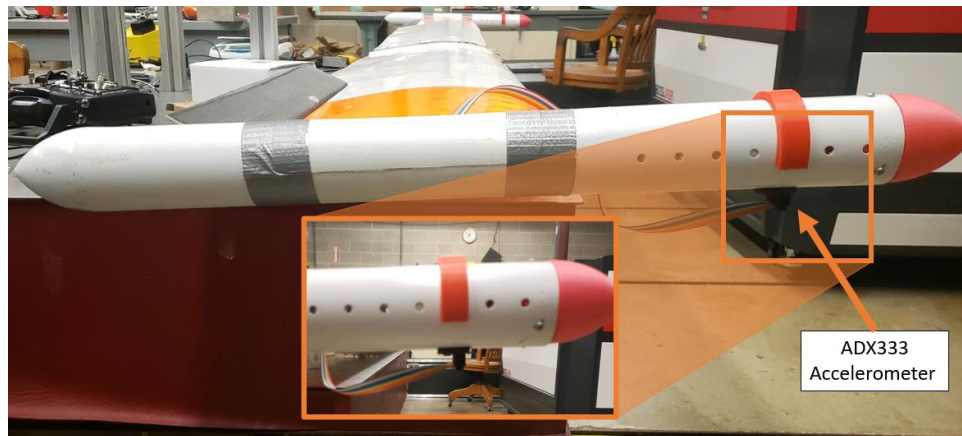


Figure 4.4: Setup used for validating the static mass acceleration calculation.

While verification of this measurement technique would normally warrant a frequency-domain analysis (i.e. Bode plot) to ensure accuracy over the excitation pod's operational envelope, the lack of motion generated at frequencies far from the wing's structural modes prevented this approach. After converting both the measured and calculated static mass's acceleration signals from the time-domain to Fourier-domain, the resultant gain coefficients associated with non-resonant frequencies were small in magnitude. Further division of the static mass's acceleration coefficients required to generate a Bode plot's gain information resulted in misleading values. Near-zero gains were computed for frequencies between the wing's bending and torsional modes. This is in contrast with the raw data, which indicates close agreement between the measured and calculated acceleration signals in the time-domain.

Instead, a time-domain approach was utilized to verify that Equation 4.3 accurately computes the static mass's acceleration. After debiasing and passing both acceleration signals through a lowpass filter with a cutoff frequency of 20 Hz, the filtered values were plotted against time as shown in Figures 4.5 and 4.6. Both plots indicate that the computed static mass's acceleration signal qualitatively matches that as measured directly by the external accelerometer. The standard deviation of the difference between the two signals is 1.0 ms^{-2} , which is approximately equal to the IMU's stated calibration error tolerance of 0.8 ms^{-2} . It can be conclusively stated that Equation 4.3 accurately estimates the static mass's acceleration.

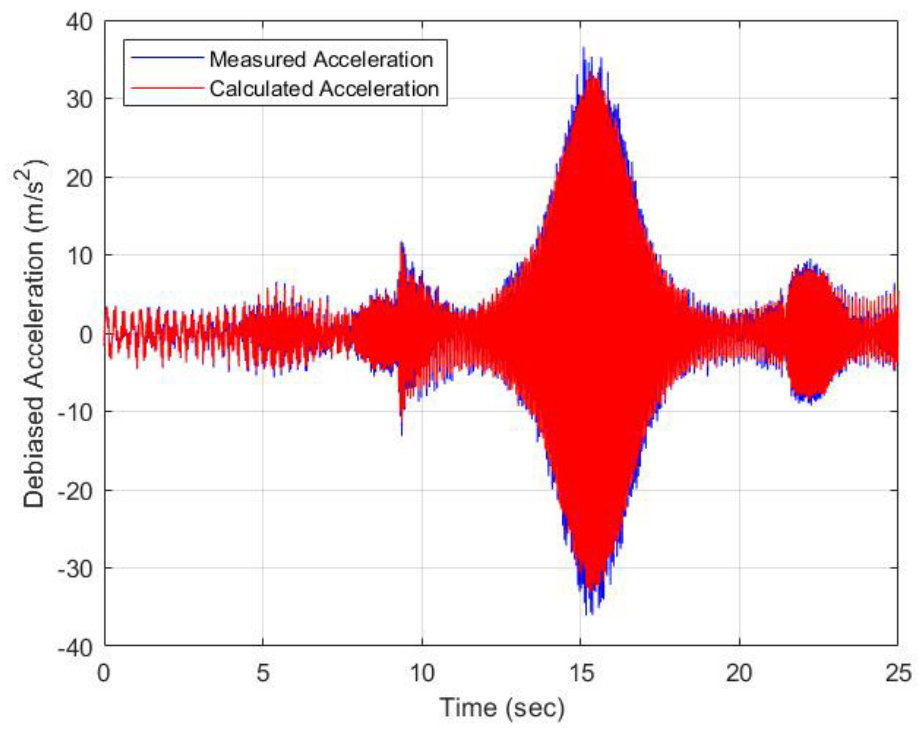


Figure 4.5: Static mass's acceleration signal in the time-domain.

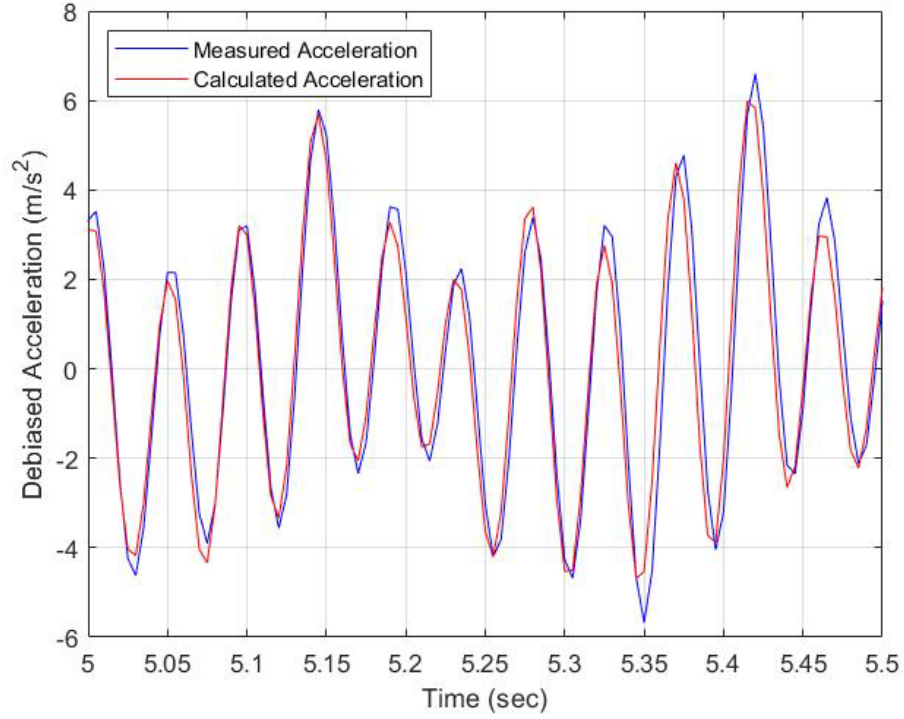


Figure 4.6: Close-up of the static mass's acceleration signal in the time-domain.

4.3 Ground Vibration Testing

Although the PFM method is not model based, the development of flight-testing procedures was aided by knowledge of the anticipated flutter mechanism as estimated from the K-method analysis method and an approximate FEA model described in Heidersbach *et al* [11]. To obtain the frequency characteristics of the flexible wing, ground vibration tests were conducted before preceding with flight-tests. GVTs were performed such that FRFs could be identified through measured acceleration data and compared to the computational model. Significant variations between the experimental and computational FRFs were addressed by tuning the FEA model's mass,

stiffness, and damping properties. By validating the model’s structural dynamic response at zero airspeed, better estimates of the projected flutter onset speed could be determined. This, in turn, was used to establish the flight conditions targeted during flight-testing and validate the results of the PFM method.

Throughout the GVT campaign, the flexible wing was rigidly mounted to an optical table. By securing the flexible wing to a significantly more massive base, the measured acceleration response data could be directly attributed to the wing’s structural dynamics. In comparison, if the GVTs were conducted with the entire flight-test vehicle supported by its own landing gear, the acceleration data would include artifacts associated with the fuselage’s and landing gear’s structural modes. It should be noted that left- and right-wing GVT trials were performed separately to mitigate coupled dynamics and better isolate each wing half’s frequency response characteristics.

The transfer function described in Equation 4.4 was used to define the FRFs associated with the GVTs. In Equation 4.4, the static mass’s vertical acceleration component, $a_{st,z}$, was computed by Equation 4.3 and the moving mass’s vertical acceleration, $a_{mv,z}$, was directly measured by the shaker assembly’s MPU-9250 IMU. It should be noted that a negative sign is included to account for the reaction force generated from the moving mass onto the wing’s structure.

$$T_{GVT}(i\omega) = \frac{-a_{st,z}(i\omega)}{a_{mv,z}(i\omega)} \quad (4.4)$$

4.3.1 Wing Configuration 1

The GVT results associated with wing configuration 1 are presented below in Figure 4.7. By referencing the gain subplot, acute peaks corresponding to the first bending mode are observed at 3.9 Hz and 4.1 Hz for the left- and right-wing, respectively. The phase-angle subplot also indicates a phase-angle change of 180° for input frequencies around each wing's identified bending mode. Additional maximum gain values occur at input frequencies of 18.5 Hz and 19.9 Hz for the left- and right-wing's first torsional modes, respectively. The identified structural modes for wing configuration 1 are summarized in Table 4.1 below.

Since the shaker assembly can only generate sinusoidal excitation forces between 2 Hz and 20 Hz, frequency characteristics cannot be identified outside of this operational range. As shown in Figure 4.7, a trending decrease in the right-wing's gain plot is not observable beyond 19.9 Hz, and it is possible the torsional mode's natural frequency is faster than the capabilities of the excitation pod. Due to this finding, the structural modifications associated with wing configuration 2 were incorporated to lower the torsional mode's frequency characteristics to within the excitation pod's performance envelope.

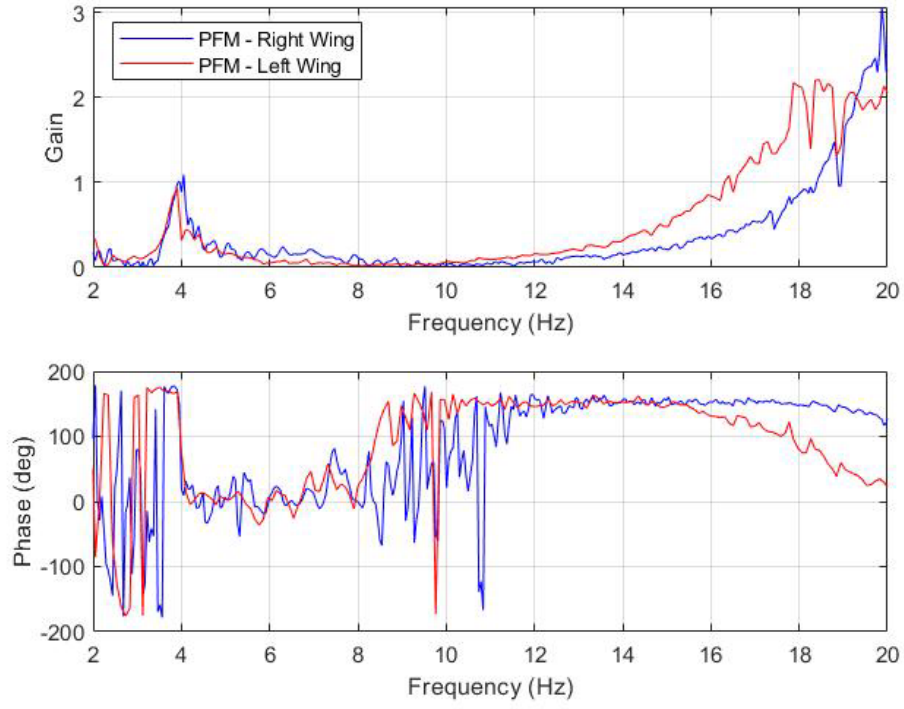


Figure 4.7: Wing configuration 1 GVT.

Wing-Half	1 st Bending Mode Natural Frequency	1 st Torsional Mode Natural Frequency
Left-Wing	3.9 Hz	18.5 Hz
Right-Wing	4.1 Hz	19.9 Hz

Table 4.1: GVT results of wing configuration 1.

4.3.2 Wing Configuration 2

Figure 4.8 below shows the FRFs of the isolated flexible wing associated with experimental GVT testing for wing configuration 2. Distinct gain peaks are observed at input frequencies of 3.7 Hz and 3.8 Hz and correspond to the natural frequencies of the left- and right wing's first bending modes. The associated phase-angles also indicate a change of 180° in the vicinity of these gain peaks. These experimental results closely match the estimated bending frequency predicted by the FEA model described in Heidersbach *et al.* [11]. Additional gain peaks at higher input frequencies of 17.6 Hz and 17.0 Hz are also present for the left- and right-wings and correspond to the first torsional mode. The modifications incorporated into wing configuration 2 successfully lowered the torsional modes' natural frequencies by approximately 1 Hz and 3 Hz for each wing. Lastly, a phase angle-change of 180° can be observed at an input frequency of approximately 8 Hz for both wings without a corresponding gain peak. Like Configuration 1, spectral data from the x-axis acceleration indicates a slight peak at 8.5 Hz corresponding to a weak fore-aft mode. It should be noted that the corresponding FEA model did not predict this in-plane mode. Table 4.2 below lists the identified structural modes for wing configuration 2.

When comparing the two experimental FRFs presented in Figure 4.8, the identified bending and torsional modes have slightly different natural frequencies and corresponding gain values between the left- and right-wings. Although both wings were built utilizing the same construction techniques, it is possible that the stiffness, damping, and mass attributes of each wing differs due to slight variations in material properties associated with the spar structure and 3D printed airfoil sections.

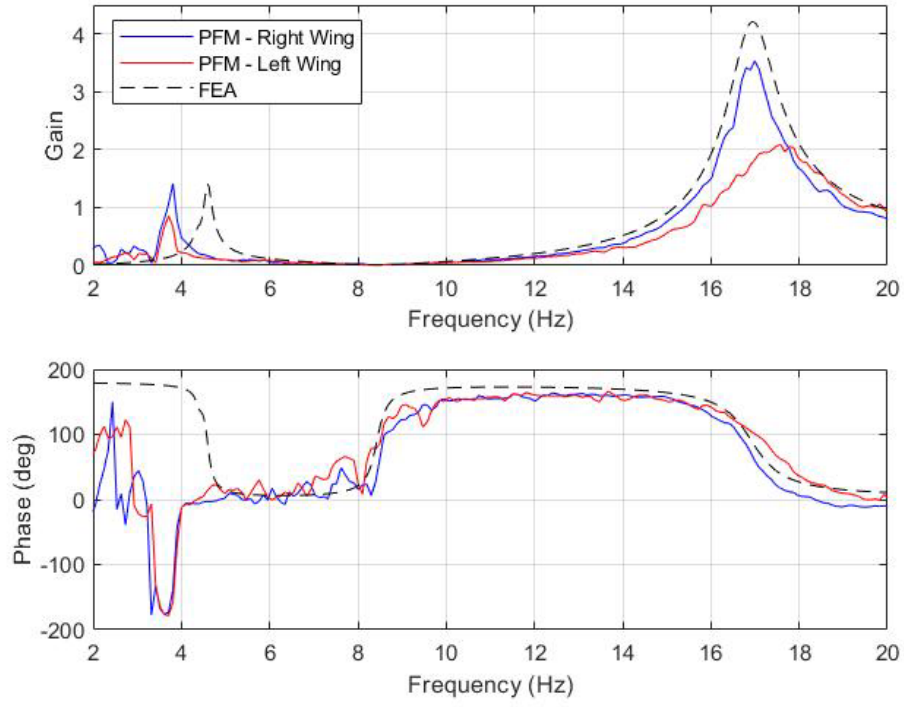


Figure 4.8: Wing configuration 2 GVT.

Wing-Half	1 st Bending Mode Natural Frequency	1 st Torsional Mode Natural Frequency
Left-Wing	3.7 Hz	17.6 Hz
Right-Wing	3.8 Hz	17.0 Hz

Table 4.2: GVT results of wing configuration 2.

After tuning the FEA model associated with wing configuration 2 to better align with the experimental GVT results present in Figure 4.8, a computational analysis indicated a corresponding flutter velocity in excess of 60 ms^{-1} . Since this estimated flutter onset velocity exceeded the flight-test vehicle's maximum achievable flight speed of 36 ms^{-1} , significant modifications to the wing's structure were required to lower the flutter boundary. This finding prompted the inertial property changes associated with wing configuration 3 as discussed in Section 3.3.3 to further reduce the structure's torsional stiffness and lower the wing's flutter velocity.

4.3.3 Wing Configuration 3

The frequency characteristics identified through ground vibration testing conducted with wing configuration 3 is presented below in Figure 4.9. As shown in the gain subplot, distinct gain peaks occur at an input frequency of 3.4 Hz for both the left- and right-wing halves and correspond to each wing's bending mode. Additional gain peaks associated with each wing's torsional mode manifest at an input frequency of 9.9 Hz. As depicted in the phase-angle subplot, a 180° phase change occurs in the vicinity of the noted structural modes. Like wing configurations 1 and 2, an additional 180° phase-angle change not related to either the bending or torsional mode occurs at an intermediate frequency of approximately 6 Hz. By performing a Fourier transformation of the measured x-axis acceleration signal, it was concluded that this phase-angle increase is linked to a weak fore-aft mode. Table 4.3.3 below lists the identified structural modes for wing configuration 3.

The torsional modes' natural frequencies associated with wing configuration 3 are significantly less than that identified for either wing configurations 1 or 2. Due to the

addition of the 220 g ballast associated with wing configuration 3, the torsional mode's natural frequency was slowed by approximately 9 Hz and 7 Hz when compared to wing configurations 1 and 2, respectively. Since the torsional modes' natural frequencies were reduced, this indicates that the modification undertaken to lower the flutter onset condition were successful.

As shown in the Bode plot associated with wing configuration 3, it can be noted that the left- and right-wing halves have much better symmetry than that identified in previous wing configurations. Specifically, the bending and torsional modes' natural frequencies exactly match between the left- and right-wing halves. This is in contrast to the Bode plots associated with wing configurations 1 and 2, which featured up to a 0.2 Hz and 1.4 Hz difference between the left- and right-wings' bending and torsional natural frequencies. This improvement in symmetry can be explained by noting that wing configuration 3 featured a rebuilt right-wing half to better match the twist profile of the left-wing counterpart. It is possible that the unintended wing twist strained the structure and augmented the torsional rigidity; this would further cascade into augmenting the wing's torsional natural frequency.

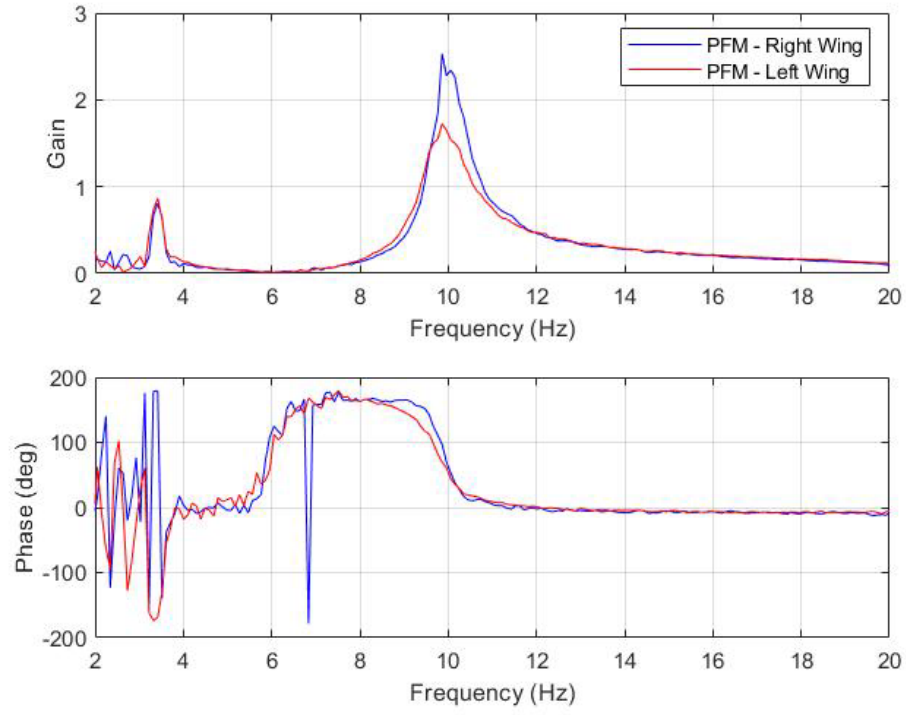


Figure 4.9: Wing configuration 3 GVT.

Wing-Half	1 st Bending Mode Natural Frequency	1 st Torsional Mode Natural Frequency
Left-Wing	3.4 <i>Hz</i>	9.9 <i>Hz</i>
Right-Wing	3.4 <i>Hz</i>	9.9 <i>Hz</i>

Table 4.3: GVT results of wing configuration 3.

Chapter 5: Flight-Testing

Following the completion of the ground-testing campaign, flight-testing was conducted using the vehicle outlined in Chapter 3 to demonstrate the PFM method in the free-flight environment. The series of flight-tests discussed below were performed to estimate FRFs for wing configurations 2 and 3 at different airspeeds and identify trends relating the flexible wing’s structural modes to increasing dynamic pressures. The resulting Bode plots associated with wing configuration 3 were then utilized to determine the aeroelastic system’s gain margin at a variety of airspeeds near the estimated flutter boundary. The nominal system’s flutter onset velocity was then identified by noting the airspeed resulting in a flutter margin of 0 dB. These experimental results are in close agreement with those calculated by a tuned FEA model developed by Heidersbach *et al.* [11].

5.1 Frequency Response Function Definitions

Unlike the GVTs, which conveyed the frequency characteristics of each wing half in isolation, the results associated with flight-testing rely on decomposing the flexible wing’s structural response into separate symmetric and anti-symmetric FRFs. The symmetric, u_{sym} , and anti-symmetric, u_{asym} , forcing functions used to excite a response from the aeroelastic system were computed by Equations 5.1 and 5.2,

respectively. These equations are expressed in terms of the moving mass' vertical acceleration signals as directly measured by the IMUs attached to the left- and right-shaker assemblies.

$$u_{sym}(t) = \frac{a_{mv,z}^R(t) + a_{mv,z}^L(t)}{2} \quad (5.1)$$

$$u_{asym}(t) = \frac{a_{mv,z}^R(t) - a_{mv,z}^L(t)}{2} \quad (5.2)$$

Similarly to Equations 5.1 and 5.2, the flexible wing's symmetric response signal, y_{sym} , and anti-symmetric response signal, y_{asym} , were calculated via Equations 5.3 and 5.4, respectively. In Equations 5.3 and 5.4, $a_{st,z}^L$ and $a_{st,z}^R$ denote the static mass's vertical acceleration components as calculated by Equation 4.3 for both the left- and right-wings.

$$y_{sym}(t) = \frac{a_{st,z}^R(t) + a_{st,z}^L(t)}{2} \quad (5.3)$$

$$y_{asym}(t) = \frac{a_{st,z}^R(t) - a_{st,z}^L(t)}{2} \quad (5.4)$$

The transfer functions presented below in Equations 5.5 and 5.6 were numerically evaluated via MATLAB's *tfestimate* function to calculate the flexible wing's symmetric and anti-symmetric FRFs.

$$T_{sym}(i\omega) = \frac{-y_{sym}(i\omega)}{u_{sym}(i\omega)} \quad (5.5)$$

$$T_{asym}(i\omega) = \frac{-y_{asym}(i\omega)}{u_{asym}(i\omega)} \quad (5.6)$$

5.2 Low Speed Flight-Testing Profile

After completing an intensive ground study of the excitation pods and wing structure as described in Chapter 4, flight-testing began with a low speed investigation of the aeroelastic system. As the majority of the test conditions associated with

low speed flight-testing were far from the anticipated flutter boundary, the primary objective of this test series was to understand the performance limitations of the excitation pods and to ensure that the instrumentation could identify the wing's modal characteristics in the free-flight environment. This section details the flight profile and testing techniques utilized during low speed flight-testing.

A standardized flight profile similar to the path depicted below in Figure 5.1 was repeatably flown throughout the majority of flight-testing to collect aeroelastic data in the free-flight environment. As shown in Figure 5.1, a typical flight in the low speed testing regime consisted of 3 rectangular patterns flown around the airport supporting flight operations. Each rectangular pattern had an associated targeted true airspeed (TAS) setpoint such that FRFs could be identified at different flight conditions. For the low speed flight-testing portion of this study, TAS setpoints between 26 ms^{-1} and 36 ms^{-1} were targeted in 2 ms^{-1} increments. Due to the voltage drop that accompanies battery usage, the fastest TAS conditions associated with a given flight were targeted first as the most amount of power was required to achieve these test points. This is in contrast with typical flight-testing philosophy, which advocates targeting aggressive test points only after clearing less risky flight conditions. Except during take-off and landing phases, flight operations for slower speed flight conditions were conducted at a constant altitude of 400 ft above ground level (AGL) to abide federal regulations outlined in 14 CFR Part 107.

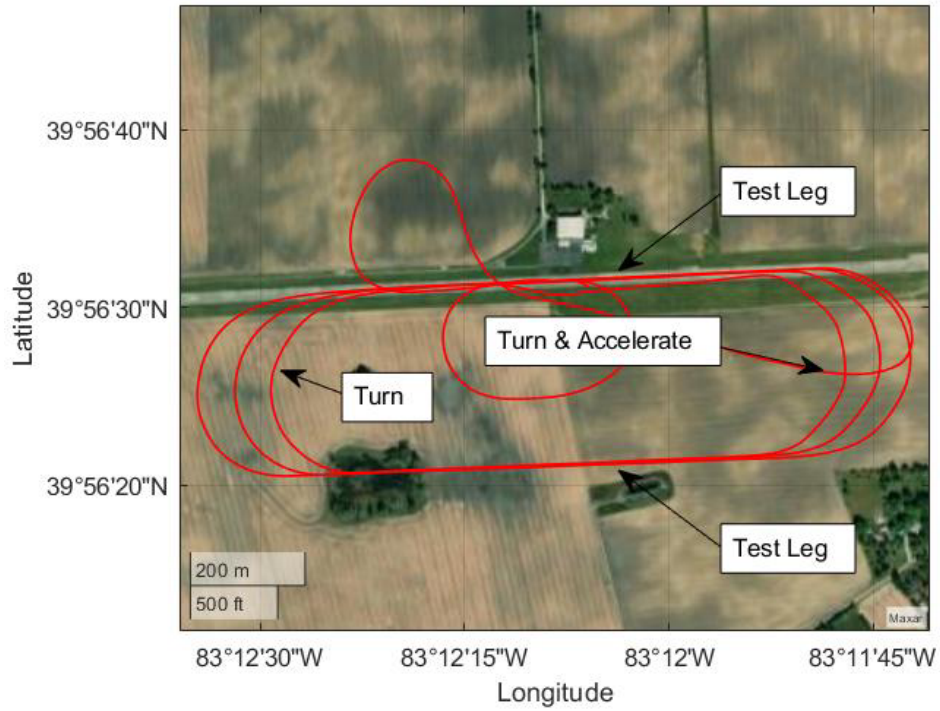


Figure 5.1: Typical flight path for low speed aeroelastic flight-testing.

The excitation pod's frequency sweep was manually activated while the vehicle tracked the straight east- and west-bound legs depicted in Figure 5.1. Since these flight operations were conducted under 14 CFR Part 107, visual contact with the vehicle had to be maintained throughout the flight. This limited the amount of time on condition associated with each test point. For example, at the maximum achievable airspeed of 36 ms^{-1} , the vehicle would travel approximately 1 km over a 30 second period. Assuming that the UAS operator was positioned near the center of the rectangular flight path, this would result in a line-of-sight distance in excess of 0.5 km. As this proved to be a limiting factor, the excitation pods' frequency sweeps were required to be completed within a 30 second window. The east- and west-bound

legs of the rectangular flight pattern were sized to support a 30 second long frequency sweep for each TAS condition previously noted.

During the frequency sweeps, it was desirable for the flight-test vehicle to maintain a steady, level, unaccelerated flight condition to mitigate the effects of unsteady loading on the airframe. This was achieved by utilizing the Pixhawk autopilot system described in Chapter 3. Except during take-offs and landings, the flight-test vehicle was controlled with the Pixhawk's "mission" mode, which tracked a desired flight path, airspeed, and altitude as defined by the operator. An example trace of the vehicle's attitude and TAS is presented in Figure 5.2. As shown in this figure, the vehicle maintained a relatively constant attitude throughout the sample test point. Large roll angles are noted at the initiation and termination of the example test point. These excursions correspond to the 180° turns depicted in Figure 5.1. Figure 5.2 also indicates that the Pixhawk autopilot system was able to track a desired TAS setpoint. A comprehensive survey of a particular flight revealed that the Pixhawk was able to track desired flight conditions with an average roll, pitch, and TAS standard deviation of 2.7°, 0.6°, and 0.4 ms⁻¹, respectively. By utilizing the Pixhawk autopilot system, desired flight conditions were able to be tracked in a smooth, repeatable manner.

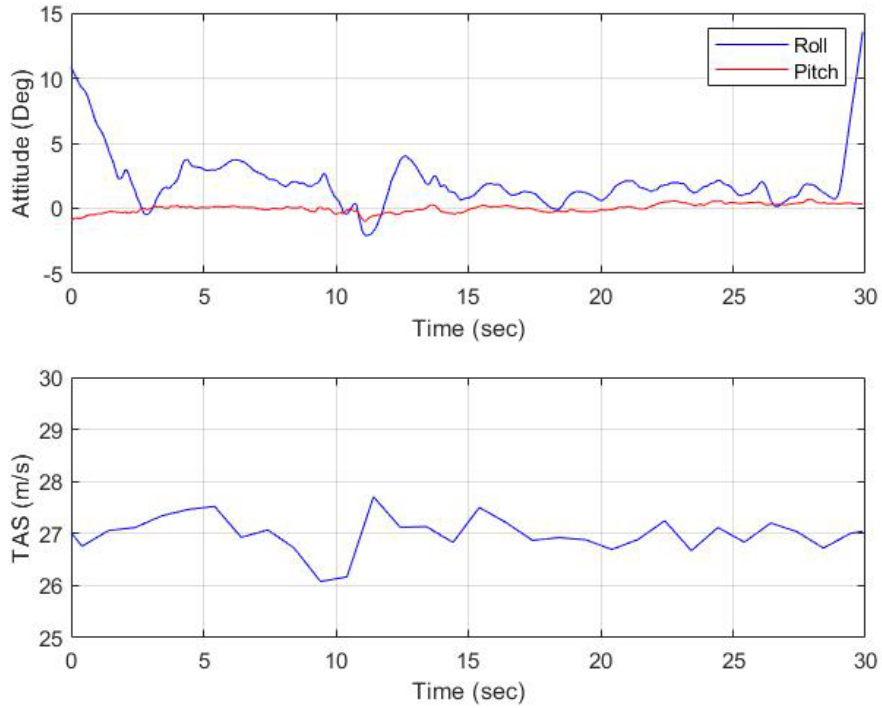


Figure 5.2: Example attitude and speed trace during low speed flight-testing.

5.3 High Speed Flight-Test Profile

While the previous section detailed the standard flight profile utilized during low speed aeroelastic flight-testing, a different approach was required to achieve higher airspeeds closer to the vehicle's flutter boundary. After verifying that the excitation pods' instrumentation could identify frequency characteristics in the free-flight environment, a high speed flight profile was used to measure FRFs at airspeed near the anticipated flutter boundary. From these FRFs, the phase-crossover frequency and corresponding flutter margin were noted such that the aeroelastic system's flutter speed could be experimentally identified in accordance with the FPM method.

Since the test vehicle's maximum level-flight speed was below the expected flutter boundary as estimated from a corresponding FEA model, a different flight-testing approach was required to achieve faster test points. Specifically, the vehicle's maximum level-flight speed of 36 ms^{-1} was slower than the estimated flutter onset condition of 40 ms^{-1} . To address this performance shortcoming, a new testing approach traded potential energy stored in terms of altitude for increased kinetic energy and correspondingly faster airspeeds. Diving profiles were utilized to collect aeroelastic flight-test data for wing configuration 3 at TAS setpoints of 38 ms^{-1} , 40 ms^{-1} , and 42 ms^{-1} .

To conduct the high speed portion of the flight-testing campaign, the test vehicle flew a profile similar to the flight path depicted below in Figure 5.3. As shown in this figure, a typical high speed flight-test consisted of two rectangular patterns flown around the airport of origin. After take-off, the Pixhawk autopilot's "mission" mode was activated and commanded the aircraft to track the southern most leg of the pre-loaded flight path. During this phase of flight, the aircraft climbed to an initial altitude of 400 ft AGL and maintain a targeted TAS of 36 ms^{-1} . Upon reaching the northwestern corner's "perch" position, the Pixhawk's "mission" mode would be deactivated in favor of a "stabilized" control law, which provided control inputs to track a desired pitch and roll attitude setpoint. After applying full throttle, a push-over maneuver was performed to re-orient the aircraft to a nose-down pitch attitude proportional to the test point's desired TAS. The excitation pods' frequency sweep was activated once the aircraft reached a steady-state airspeed. To prevent controlled flight into terrain, the dive profile and frequency sweep was terminated at an altitude of approximately 100 ft AGL and a slight climb was initiated. The

Pixhawk's "mission" mode would be activated to reposition aircraft for another dive attempt.

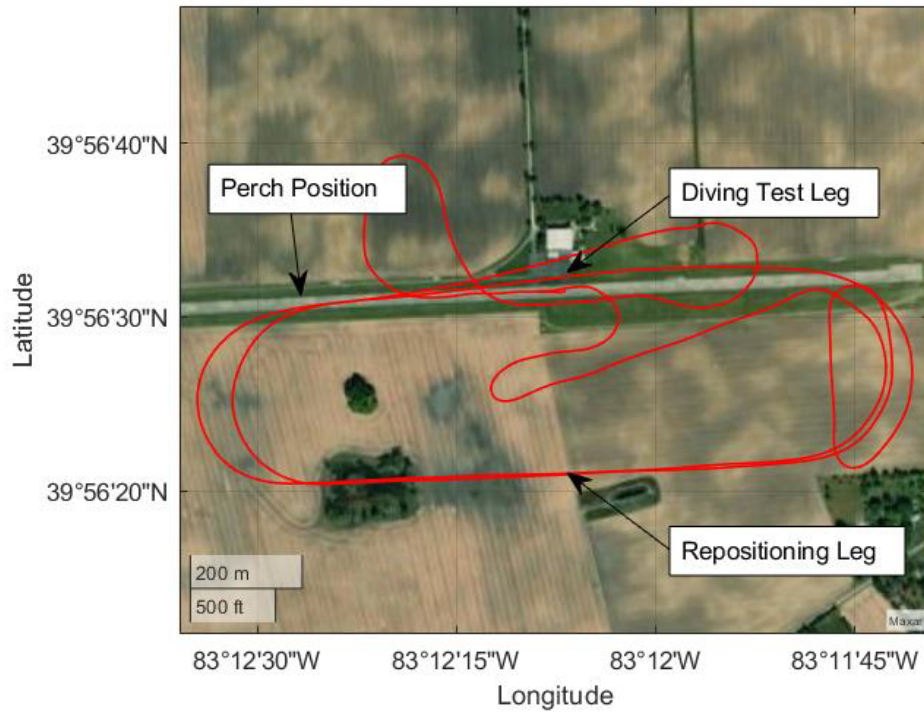


Figure 5.3: Typical flight path for high speed aeroelastic flight-testing.

Early attempts to perform high speed dives were met with noisy wing acceleration signals, which cascaded to equally noisy FRFs and inconclusive flutter margins. During these early attempts, the aircraft's operator actively manipulated the controls to track a desired TAS. If the TAS exceeded the targeted test condition, the operator commanded a shallower dive angle. Conversely, if the TAS was slower than the desired condition, the operator commanded a more aggressive nose-down attitude. While this "closed loop" approach was successful in tracking desired TAS setpoint,

the constant changes in pitch attitude caused the aircraft to experience a time-varying load factor. During more extreme pitch attitude changes, the resulting load factors would overpower the shaker assemblies' voice coil motors and effectively pin the moving masses to either the top or bottom of the shaker guide rails. This phenomena corrupted the corresponding FRFs as the aeroelastic system was not excited with a continuous sweep of input frequencies.

While the "closed-loop" approach required the aircraft operator to consistently make corrective control inputs, a smoother "open-loop" alternative was investigated where the dive maneuver could be performed without operator input. In this approach, the aircraft's pitch trim was intentionally set to a nose-down value. While operating in the "stabilized" mode, the Pixhawk autopilot system interpreted the command signal as a nose-down setpoint. The autopilot system made the required elevator control inputs to track a negative pitch angle associated with the high speed dive profiles. This resulted in a much smoother load factor and did not cause the moving masses to bind during the dive profiles.

Figure 5.4 below compares the load factor experienced during a sample "closed-loop" and "open-loop" diving maneuver. The load factor associated with the "closed-loop" approach had a standard deviation of 0.45 G and had peak acceleration measurements as much as 1 G different than the mean value. This is in contrast to the 0.08 G standard deviation experienced during the "open-loop" diving maneuver. By comparing the traces plotted in Figure 5.4, it is clear that the "open-loop" testing approach provided a smoother and more consistent load factor.

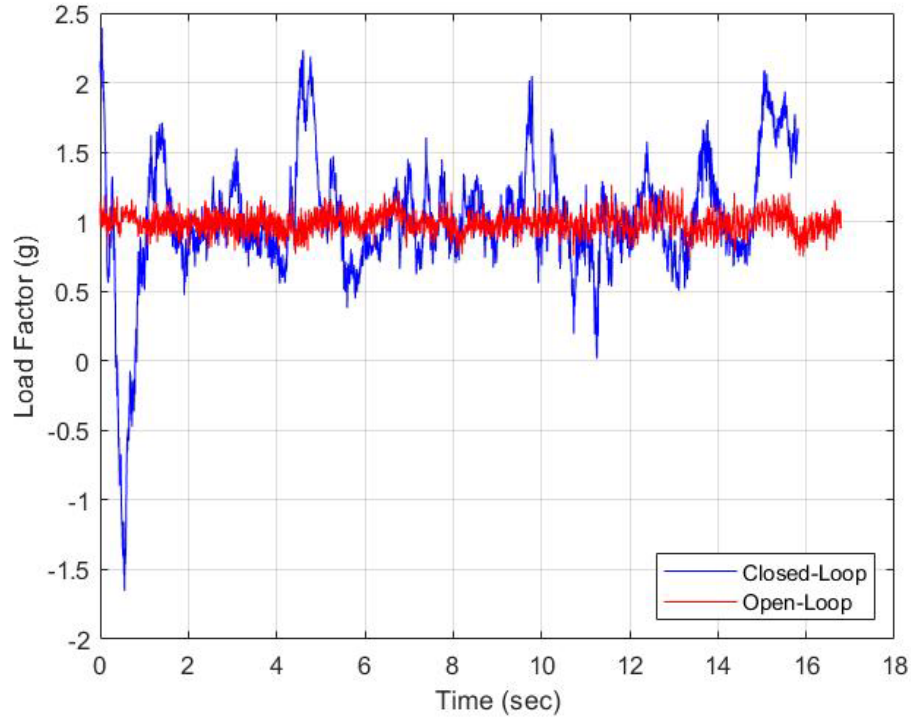


Figure 5.4: Load factor for open-loop and closed-loop dive profiles.

An example trace of the test vehicle's attitude and TAS as measured during an "open-loop" diving profile is depicted in Figure 5.5 below. As shown in Figure 5.5, a steady-state pitch attitude of approximately -5° was required to achieve a TAS of 40 ms^{-1} . For the sample dive, which is representative of a typical high speed test profile, the standard deviation of the roll, pitch, and TAS states was 2.7° , 0.4° , and 0.5 ms^{-1} , respectively. These values are nearly identical to the standard deviations experienced during the low speed flight-testing, which indicates that the rigid-body flight dynamics experienced during both stages of flight-testing are comparable. Note that the sample trace's relatively large roll motion is a result of pilot inputs required for course corrections needed to combat crosswinds at altitude.

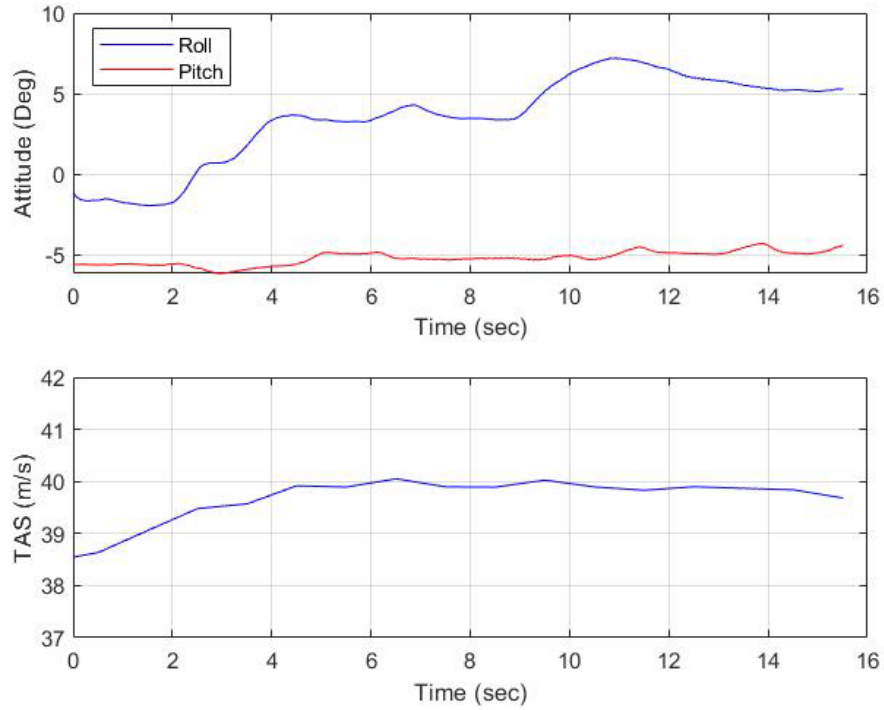


Figure 5.5: Example attitude and speed trace during high speed flight-testing.

5.4 Flight-Testing Results: Wing Configuration 2

Although the estimated flutter boundary associated with wing configuration 2 well exceeded the test vehicle's performance specifications, flight-testing proceeded to ensure that modal characteristics could be identified in the free-flight environment using the excitation pods described in Chapter 3. Specifically, the objective of this test series was to ensure that the excitation pods' moving masses provided a sufficiently strong perturbation force to excite a structural response from the wing. A second objective was to determine if the excitation pods' instrumentation was adequately sensitive such that changes in the wing's modal characteristics could be identified

at different flight conditions. The following section details the flight-testing results associated with wing configuration 2.

The forcing functions applied to the wing's structure during flight-testing of wing configuration 2 were generated simultaneously by both the left- and right-wings' excitation pods. Both excitation pods performed a frequency sweep between 2 Hz – 20 Hz over a 30 second test window. As shown in Figures 5.6 and 5.7, the command signals were approximately time-aligned throughout the duration of the frequency sweeps. While both command signals appear to be in-phase, it should be stressed that there was no active feedback mechanism to track a desired phase-angle shift between each pod's excitation functions. Instead, both the left- and right-shaker assemblies followed a pre-programmed command signal that varied the input frequency as a function of elapsed time. Any phase offset between the shaker assemblies' frequency sweeps was a consequence of a different triggering time caused by the update rate of each pod's radio receiver.

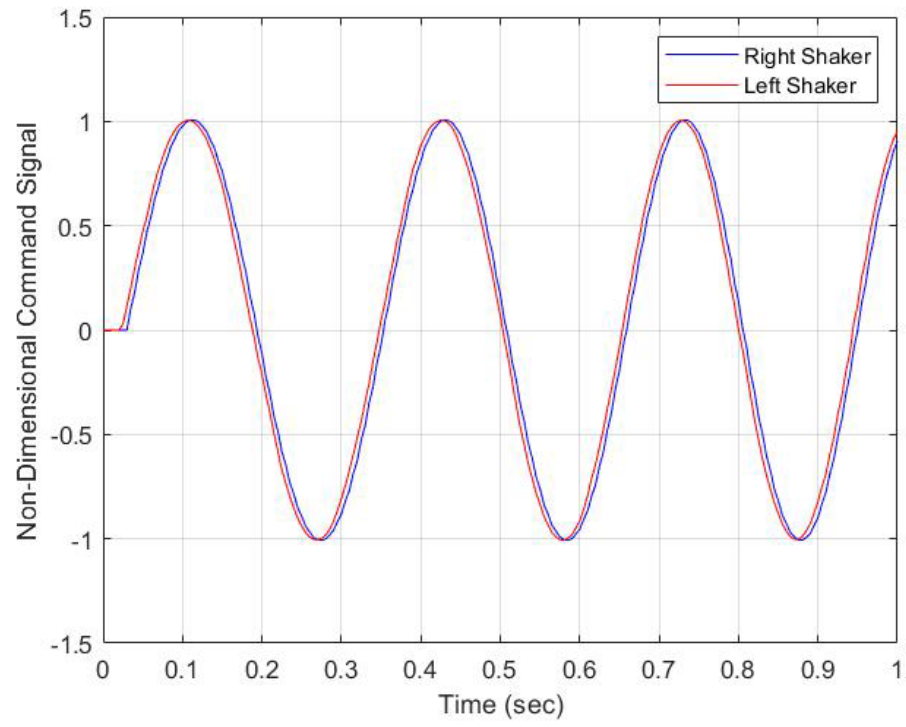


Figure 5.6: Left- and right-shaker command signals at lower input frequencies ($\omega = 3.2$ Hz).

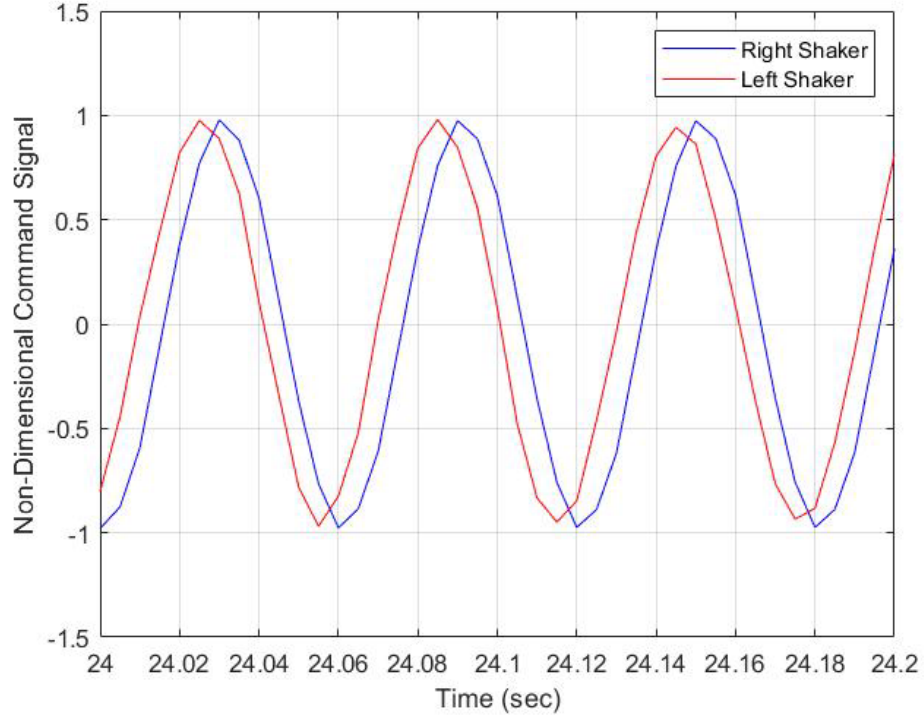


Figure 5.7: Left- and right-shaker command signals at higher input frequencies ($\omega = 16.6$ Hz).

The symmetric FRFs for wing configuration 2 are shown in Figure 5.8 for TAS conditions of 26.7 ms^{-1} , 31.8 ms^{-1} , and 35.2 ms^{-1} . As indicated by the gain subplot, the low frequency range of the FRFs are quite noisy, but peak amplitudes are observed at input frequencies of 3.7 Hz, 4.5 Hz, and 4.6 Hz for each of the tested airspeeds. Comparing the phase behavior with the GVT results presented in Chapter 4 indicates these natural frequencies correspond to the wing's first symmetric bending mode. Additional peak amplitudes associated with the first symmetric torsional mode are observed at higher frequencies of 16.2 Hz for each explored flight condition.

Although the first symmetric bending mode’s natural frequency increased with air-speed, the natural frequency of the first symmetric torsional mode remained constant. A phase-angle shift is observed at input frequencies around 7 Hz for each of the test conditions explored. Similar to the GVT FRFs, the gain subplot does not indicate a corresponding gain peak. Given the similarities between the flight-test and GVT FRFs, this phase-angle shift likely represents an in-plane structural mode that does not have a significant impact on the flutter mechanism. Figure 5.9 shows the anti-symmetric FRFs from the same flight-tests. The gain peaks occur at input frequencies of 16.1 Hz, 15.6 Hz, and 15.9 Hz, corresponding to the first anti-symmetric torsional mode’s aeroelastic frequencies. Results for the symmetric and anti-symmetric natural frequencies are listed below in Table 5.4 for direct comparison to predicted FEA results calculated by Heidersbach *et al.* [11].

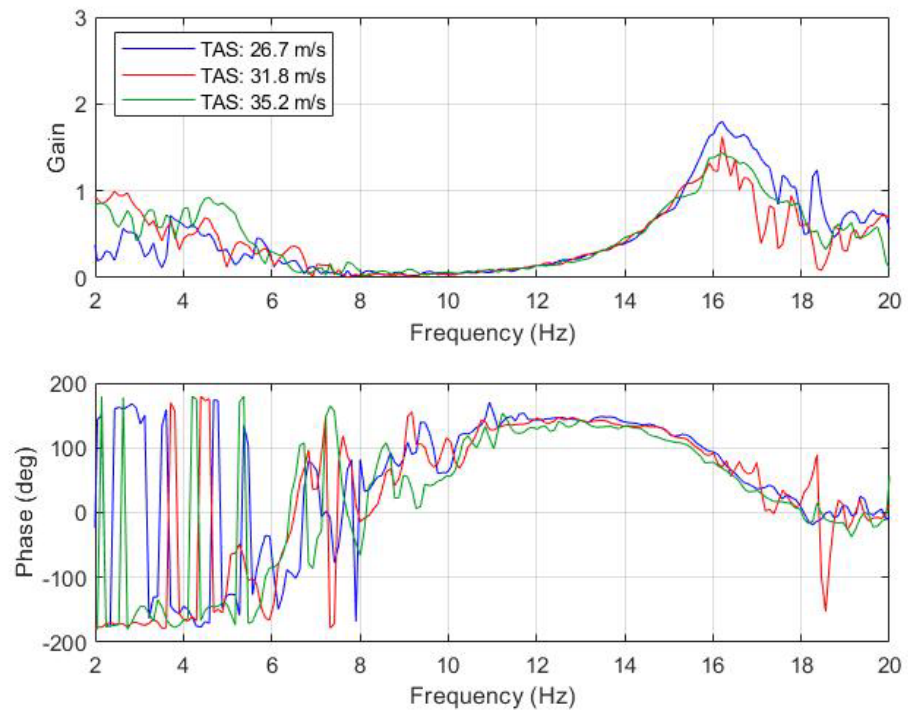


Figure 5.8: Flight-test symmetric FRFs for wing configuration 2.

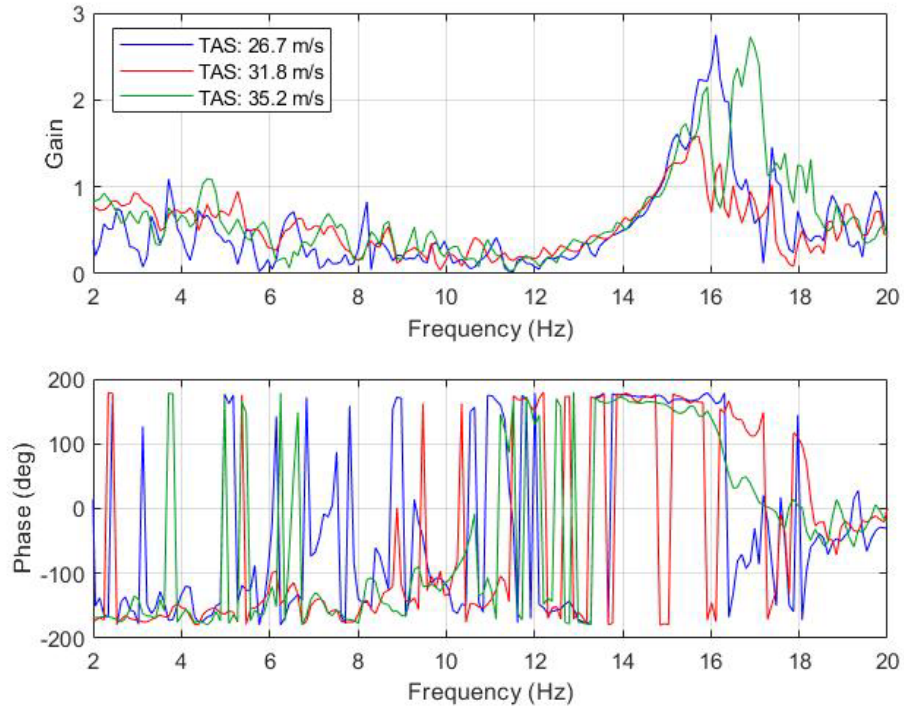


Figure 5.9: Flight-test anti-symmetric FRFs for wing configuration 2.

True Airspeed (ms^{-1})	1 st Symmetric Bending Mode (Hz)		1 st Anti-Symmetric Torsional Mode (Hz)		1 st Symmetric Torsional Mode (Hz)	
	Flight-Test		FEA		Flight-Test	
	FEA	FEA	FEA	FEA	FEA	FEA
26.7	3.7	5.65	16.1	16.2	16.2	18.5
32.8	4.5	5.70	15.7	15.9	16.2	18.0
35.2	4.6	5.75	15.9	15.5	16.2	17.5

Table 5.1: Identified aeroelastic frequencies of wing configuration 2.

Compared to the symmetric modes, the anti-symmetric gain and phase subplots shown in Figure 5.9 have substantially higher noise. This behavior can be explained by referencing Equation 5.2, which computes the anti-symmetric input function, u_{asym} , in terms of the difference between the left- and right-moving mass' measured acceleration signals. As previously mentioned, both excitation pods were approximately time aligned throughout the execution of the flight-test. Consequently, the left- and right-moving mass' measured acceleration signals were nearly identical in the time domain, resulting in a significantly attenuated anti-symmetric forcing function. Since the resultant anti-symmetric response signal was comprised of a diluted component associated with the excitation pods' applied force, the FRFs were dominated by external anti-symmetric accelerations from atmospheric turbulence or aircraft rigid-body dynamics. As a result, the uncertainty associated with the anti-symmetric FRFs as estimated by flight-test data are significantly higher than their symmetric counterparts. Even so, the measured gains are small when the phase angles are close to zero, implying being far from nominal system's flutter speed.

5.5 Flight-Testing Results: Wing Configuration 3

Although the flight-tests conducted with wing configuration 2 verified that modal characteristics could be identified in the free-flight environment using the excitation pods' instrumentation, the anticipated flutter boundary well exceeded the vehicle's maximum level-flight speed. In order to experimentally pinpoint the vehicle's flutter onset velocity, modifications associated with wing configuration 3 were made to lower the flutter speed to a value contained within the vehicle's achievable flight envelope. This section begins with detailing alterations made to the excitation pods' frequency

sweep profile to address the shortcomings noted in Section 5.4. Sample FRFs identified at different flight- conditions are then presented. From these FRFs, the natural frequencies associated with different structural modes are tabulated. Finally, a flutter margin versus TAS curve is presented, which is used to extrapolate an estimated flutter onset velocity.

As discussed in Section 5.4, the approximately time-aligned frequency sweeps performed by the left- and right-wings' excitation pods resulted in an attenuated anti-symmetric forcing function as compared to the symmetric counterpart. Since the resulting anti-symmetric FRFs had higher uncertainty, it was desired to modify the frequency sweep profiles such that the symmetric and anti-symmetric forcing functions were of comparable magnitude. This was achieved by introducing an intentional 90° phase-angle offset between the left- and right-wing's forcing functions. This phase-angle offset was incorporated by linking the frequency sweep command signal to the excitation pods' GPS clock. By referencing the GPS clock, the left- and right-wing's excitation pods equated the elapsed time to a pre-programmed command signal state. It should be stressed the the left- and right-wing's excitation pods had slightly different functions relating elapsed time to the command signal state to introduce the 90° phase-angle offset. Like the previous frequency sweep profile discussed in Section 5.4, input frequencies between 2 Hz and 20 Hz were targeted over a 30 second window. An example frequency sweep's command signal is plotted in the time domain for both low and high input frequencies in Figures 5.10 and 5.11, respectively.

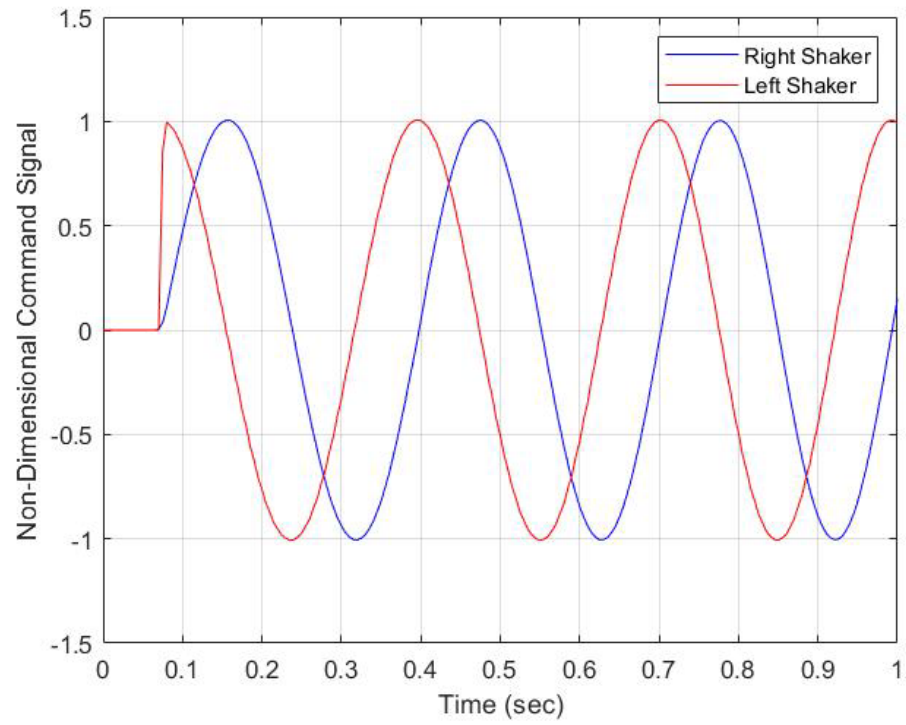


Figure 5.10: Left- and right-shaker command signals at lower input frequencies ($\omega = 3.1$ Hz) w/ phase offset.

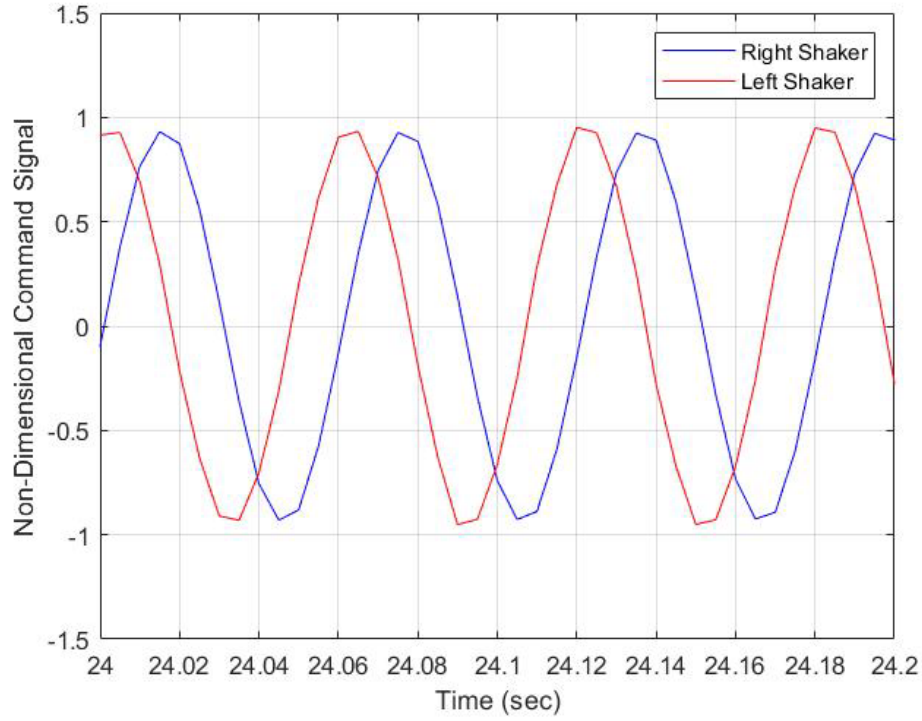


Figure 5.11: Left- and right-shaker command signals at higher input frequencies ($\omega = 16.7$ Hz) w/ phase offset.

A Bode plot relating the left- and right-excitation pods' sinusoidal command signals is presented below in Figure 5.12. As shown in this figure, a relatively constant mean gain value of 1.0 characterizes the relationship between the left- and right-wing's command signals for input frequencies between 3 Hz and 19 Hz. Additionally, the desired phase-angle offset of 90° is introduced for input frequencies between 3 Hz and 16 Hz. For input frequencies above 16 Hz, the resulting phase-angle begins to decrease and reaches a minimum value of approximately 80° at 20 Hz. This decrease in phase-angle at higher input frequencies is likely a result of the control board's 200 Hz update rate. At faster input frequencies, the 5 millisecond delay between command

signal updates equates to a increasingly larger percentage of a sinusoidal period. The time delay associated with the control board's update rate introduces a quantization effect, which may be inadvertently changing the phase-angle offset between the left- and right-wing's command signals.

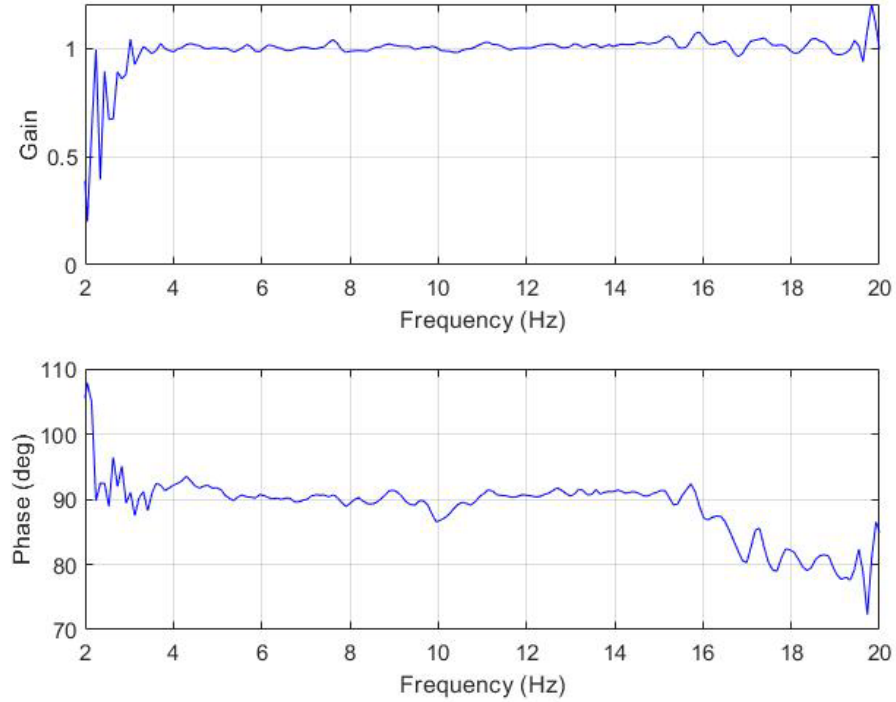


Figure 5.12: Bode plot between the left- and right-wing's command signal

The symmetric FRFs associated with wing configuration 3 at TAS conditions of 30.1 ms^{-1} , 34.4 ms^{-1} , 35.7 ms^{-1} , and 38.8 ms^{-1} are presented below in Figure 5.13. As shown by the soft peak in the gain subplot, the wing's 1st symmetric bending mode has a natural frequency of approximately 2 Hz - 4 Hz for the flight conditions explored. While it is difficult to identify an exact peak due to the gradual change in gain and

noisiness of the FRFs, the overall trend does indicate the presence of the 1st symmetric bending mode at low frequency. More distinctive gain peaks corresponding to the 1st symmetric torsional mode are noted at moderate frequencies. Specifically, peak gain values are observed at 10.2 Hz, 9.3 Hz, 9.0 Hz, and 10.1 Hz for TAS conditions of 30.1 ms⁻¹, 34.4 ms⁻¹, 35.7 ms⁻¹, and 38.8 ms⁻¹, respectively. Additionally, the 180° phase-angle change that occurs between 8 Hz - 12 Hz provides further evidence of the 1st symmetric torsional mode. While the torsional mode's natural frequency generally decreases with TAS, the abrupt increase that accompanies the 38.8 ms⁻¹ flight condition does not follow the expected trend. Further testing at this high speed flight condition is required to determine the repeatability of this observation.

Wing configuration 3's anti-symmetric FRFs at discrete TAS conditions between 30.1 ms⁻¹ and 38.8 ms⁻¹ are presented below in Figure 5.14. By referencing the anti-symmetric gain subplot, distinct peaks related to the 1st anti-symmetric torsional mode are noted at 7.9 Hz, 8.0 Hz, 8.1 Hz, and 9.6 Hz. Furthermore, a 180° phase-angle change occurs between 8 Hz - 10 Hz that accompanies the 1st anti-symmetric torsional mode's natural frequency. It should be stressed that the observed increase in the anti-symmetric torsional mode's natural frequency with TAS is in contrast with the expected trend calculated by the corresponding FEA model. More gradual gain peaks occur at higher frequencies of 17.9 Hz, 18.0 Hz, 17.2 Hz, and 17.7 Hz for each of the analyzed airspeeds. These peaks and corresponding phase-angle changes are associated with the 2nd anti-symmetric bending mode.

Table 5.2 below tabulates the identified natural frequencies associated with each observed symmetric and anti-symmetric mode at a variety of different airspeed conditions.

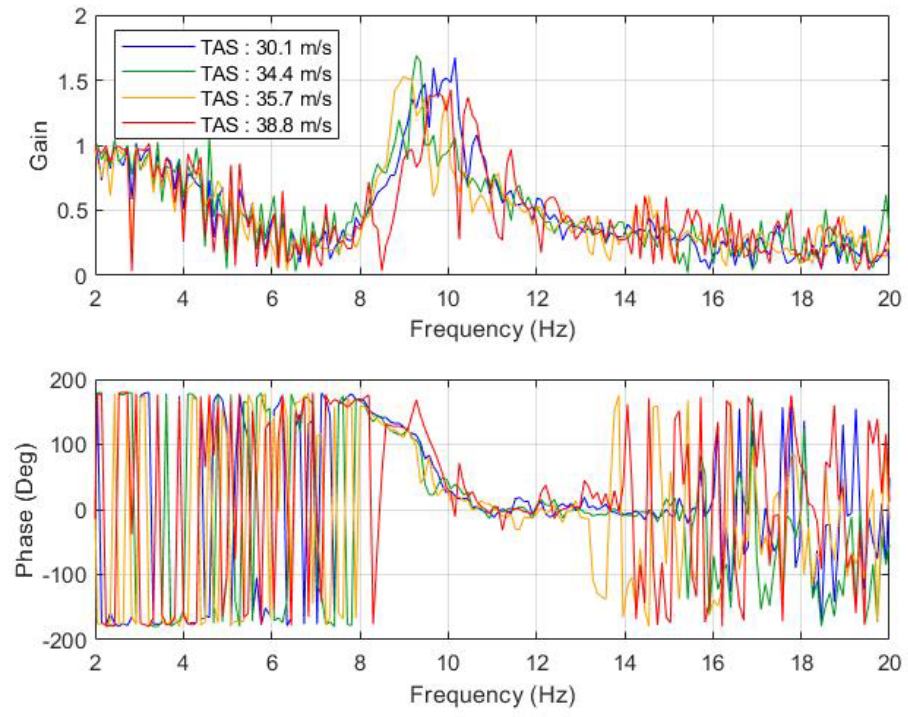


Figure 5.13: Flight-test symmetric FRFs for wing configuration 3.

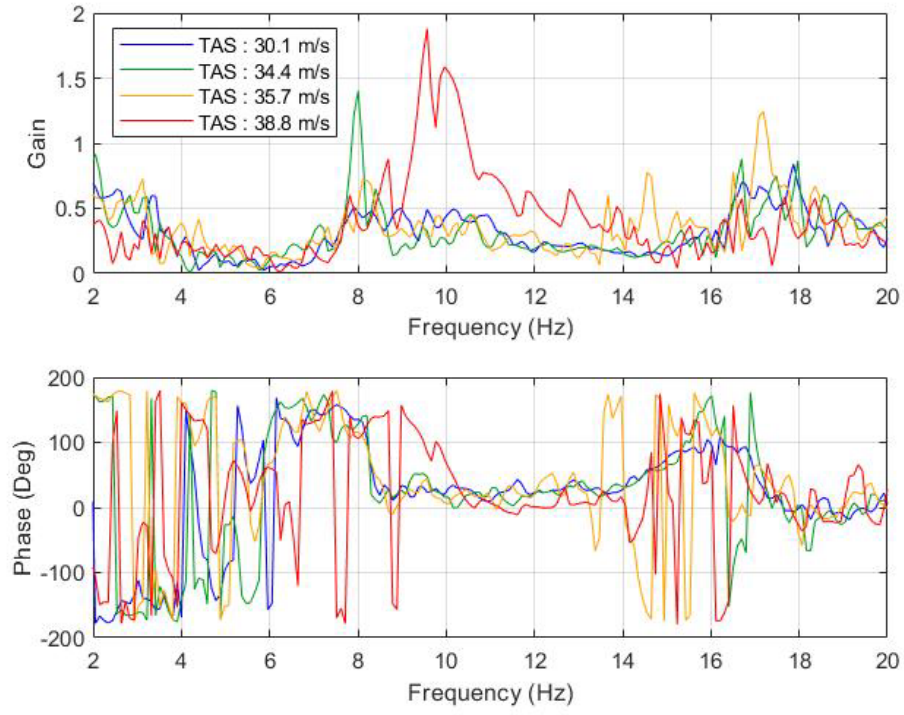


Figure 5.14: Flight-test anti-symmetric FRFs for wing configuration 3.

True Airspeed (ms^{-1})	1 st Symmetric Torsional Mode (Hz)	1 st Anti-Symmetric Torsional Mode (Hz)	2 nd Anti-Symmetric Bending Mode (Hz)
30.1	10.2	7.9	17.9
34.4	9.3	8.0	18.0
35.7	9.0	8.1	17.2
38.8	10.1	9.6	17.7

Table 5.2: Identified aeroelastic frequencies of wing configuration 3.

The FRFs measured at different airspeeds were analyzed to identify the phase-crossover frequency and flutter margin associated with the given flight condition. Figures 5.15 and 5.16 are sample symmetric and anti-symmetric FRFs associated with a TAS of 42.5 ms^{-1} . By referencing the phase-angle subplots, a phase-crossover frequency of 11.4 Hz and 8.4 Hz were identified with a corresponding gain value of 0.41 and 0.95 for the symmetric and anti-symmetric modes, respectively. Since $F.M. = 1 - G(\omega_{pco})$, the symmetric and anti-symmetric flutter margins for the sample flight condition are 0.59 and 0.05. All FRFs used to identify the phase-crossover frequency and flutter margin are provided in Appendix D.

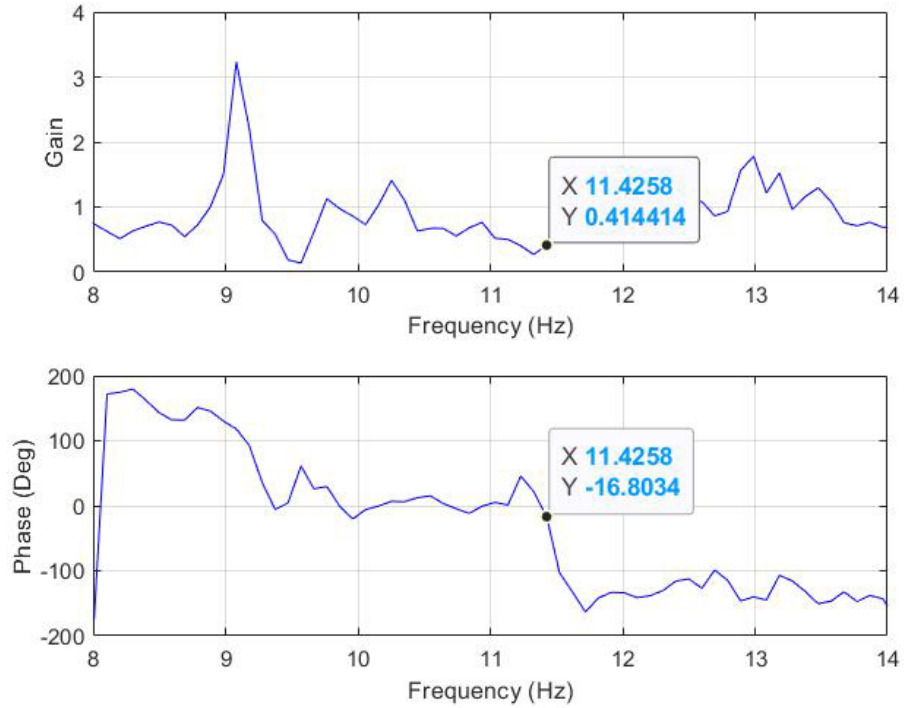


Figure 5.15: Symmetric mode's ω_{pco} and gain value for TAS = 42.5 ms^{-1} .

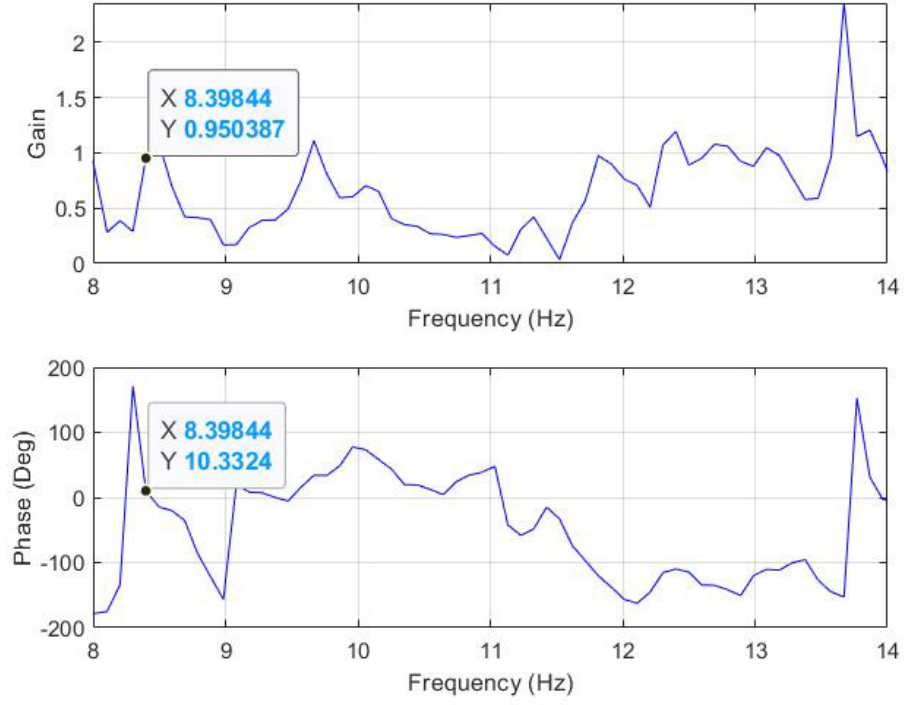


Figure 5.16: Anti-symmetric mode's ω_{pco} and gain value for TAS = 42.5 ms⁻¹.

After identifying the phase-crossover frequencies, the corresponding flutter margins were plotted against TAS for both the symmetric and anti-symmetric structural modes in Figures 5.17 and 5.18, respectively. The 12 different FRFs collected over 7 independent flight were analyzed to determine the symmetric and anti-symmetric flutter margins for airspeeds between 34 ms⁻¹ and 43 ms⁻¹. In the figures below, a linear best-fit curve is co-plotted with the experimental data to highlight trends between flutter margin and TAS. While flutter margin is not expected to vary linearly with airspeed, the resulting best-fit line is used in-lieu of higher speed experimental data to extrapolate a possible flutter onset velocity, if applicable.

As shown in Figure 5.17, the symmetric mode's flutter margin dose not decrease with TAS between 34 ms^{-1} and 43 ms^{-1} . While the flutter margin data points have a large amount of spread, the values generally increase with airspeed. This trend indicates that the aeroelastic system's symmetric mode become more stable with airspeed for TAS between 34 ms^{-1} and 43 ms^{-1} . The large spread of the symmetric mode's identified flutter margins is likely related to the noisiness of the corresponding FRFs. If smoother FRFs were able to be measured, the repeatability of the identified flutter margins may be improved. Since the gain values do not reveal a strong tendency to approach zero with increasing TAS, it can be concluded that flutter onset due to a symmetric mechanism is not expected to occur for neighboring airspeeds.

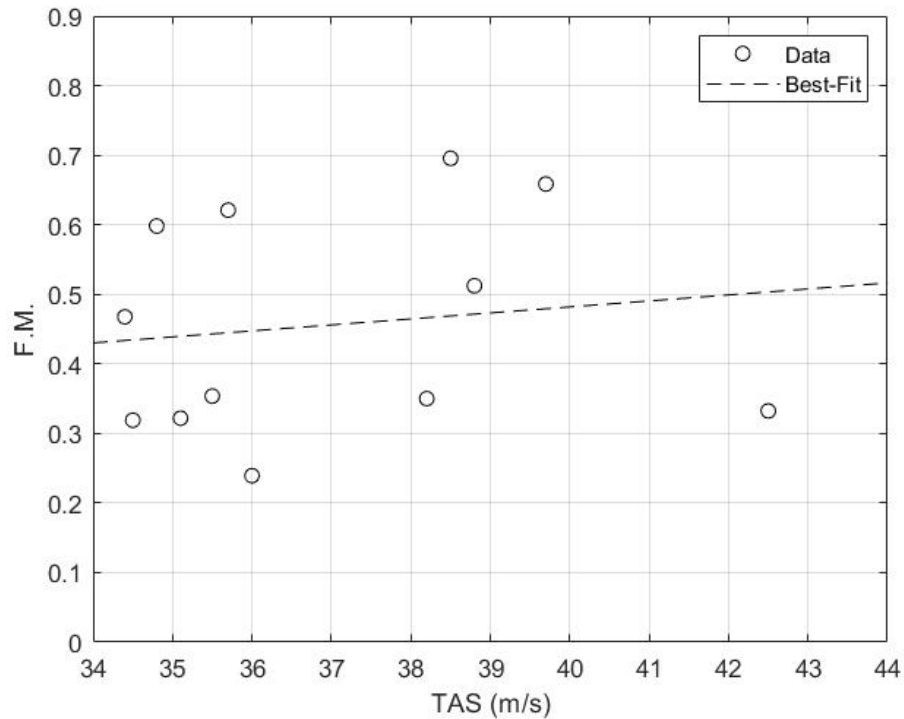


Figure 5.17: Symmetric flutter margin versus TAS.

Figure 5.18 below plots the anti-symmetric mode's flutter margin values identified through flight-testing for TAS between 34 ms^{-1} and 43 ms^{-1} . As shown in this graph, the anti-symmetric gain values strongly decrease with airspeed for the explored flight conditions. Since the flutter margins trend towards zero with increasing TAS, it may be concluded that wing's anti-symmetric response is directly related to the anticipated flutter mechanism. By referencing the best-fit line, the wing's nominal flutter velocity of 43.2 ms^{-1} may be extrapolated. These finding are in close agreement with the FEA results, which calculated an anti-symmetric flutter mode at a TAS of 40 ms^{-1} . The experimental flutter onset velocity identified through flight-testing conducted in accordance with the PFM method is within 8% error of the computational results.

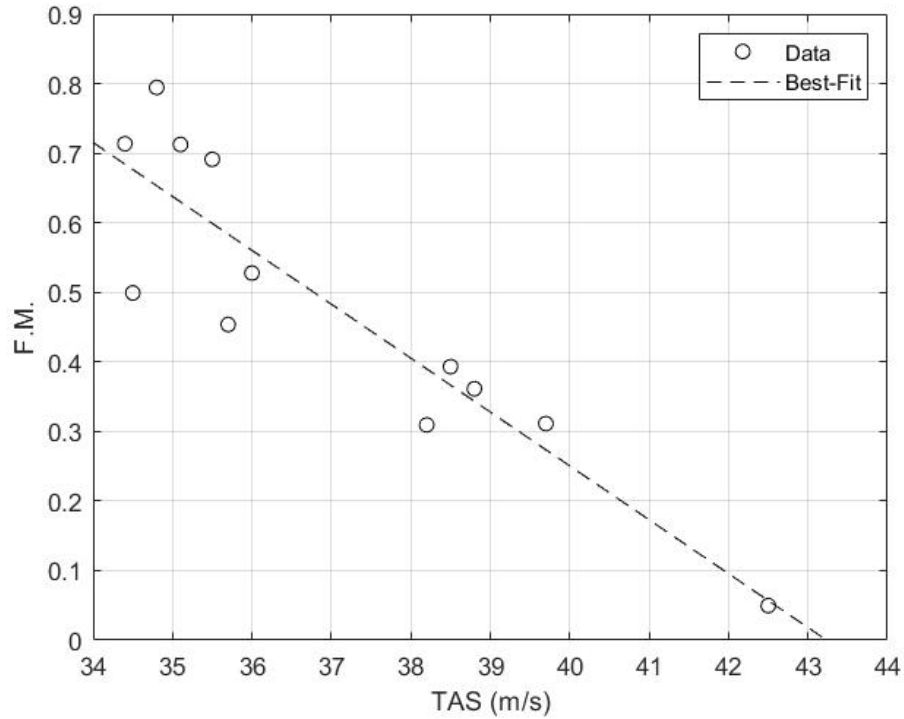


Figure 5.18: Anti-symmetric flutter margin versus TAS.

Chapter 6: Conclusion and Future Work

This work provides a summary of the first free-flight demonstration of aeroelastic flight-testing conducted in accordance with the PFM method. Throughout the course of this study, a self-contained electromechanical excitation pod was designed, built, and characterized that positively augmented the system's stability and excited an aeroelastic response from the wing under investigation. A series of GVTs were conducted to identify the structural dynamic modes and corresponding natural frequencies of a purpose-built flexible wing. This data was used in a relate project to develop a FEA model of the test vehicle such that flight-testing results could be compared to computational findings. Flight-test data indicated that the excitation pods' shaker assembly was sufficiently strong to excite a measurable aeroelastic response in the free-flight environment. Most importantly, the test vehicle's flutter boundary was able to be extrapolated from the FRFs measured through flight-testing. While there are several area that may be improved upon to enhance the accuracy of PFM flight-testing, this study indicates that the PFM method can be successfully applied in the free-flight environment.

6.1 Primary Conclusions

As discussed in Chapter 4, a detailed series of bench top experiments were conducted to characterize the excitation pods purposely built for PFM flight-testing. By performing a calibration of the shaker assemblies used to excite the wing's elastic modes, it was shown that the excitation force can be accurately measured with the embedded IMU secured to the moving mass. The excitation force's magnitude as estimated by the accelerometer and Newton's laws of motion are comparable to the measurements directly taken with an external load cell. Most importantly, it was demonstrated that the excitation force can be resolved with minimal attenuation and phase lag over an operational range of 2 Hz - 20 Hz. Furthermore, ground-testing indicated that the wing's dynamic response to the excitation force can be accurately measured using the control board's IMU signals in conjunction with rigid body dynamic principles. Bench top testing proved that this approach calculates acceleration values comparable to those taken directly by an external accelerometer. The error between these measurement techniques were small with a standard deviation of 1.0 ms^{-2} . Since the PFM method requires precise knowledge of the forcing function and resulting dynamic response, it is crucial to ensure that the excitation pods can accurately measure these signals.

A series of GVTs were performed to identify the modal characteristics of each wing configuration tested in this study. The GVT results associated with wing configuration 1 revealed the presence of a bending mode and torsional mode at approximately 4.0 Hz and 19.2 Hz, respectively. Since the torsional mode coincided with the upper limit of the excitation pod's frequency sweep, modifications to the wing's

structure were required to lower the natural frequency. GVT testing of wing configuration 2 showed that these modifications were successful as the bending and torsional modes' natural frequencies lowered to 3.8 Hz and 17.3 Hz, respectively. After adjusting the corresponding FEA model's mass, stiffness, and damping characteristics, the computationally estimated natural frequencies closely aligned with those identified experimentally within 16%. Since the estimated flutter onset velocity of wing configuration 2 well exceeded the achievable flight envelope of the test vehicle, further wing modifications were performed to lower the flutter boundary by adding 220-gram ballast to rear-most station of each wing's excitation pods. As shown by the GVT results associated with wing configuration 3, the torsional mode's natural frequency was significantly lowered to 9.9 Hz. The wing's bending mode's frequency also experienced a drop to 3.4 Hz. While identifying each mode's natural frequency through the GVTs is required to characterize the wing's frequency response, it is also important to note that the excitation pods used in this campaign were able to excite a measurable dynamic response using hardware contained within an independent, flight-ready package.

During an initial series of flight-tests, it was demonstrated that the excitation pods were capable of exciting and measuring a structural response of the flexible wings in the free-flight environment. Through flight-testing conducted with wing configuration 2, the symmetric bending and torsional natural frequencies identified via the PFM method agreed within 1.95 Hz and 0.4 Hz of the FEA predictions, respectively. The experimentally identified anti-symmetric torsional mode's natural frequency aligned with the computational results within 0.4 Hz. Additionally, flight-test data indicated a predicted shift in the flexible wing's first symmetric bending and

first anti-symmetric torsional aeroelastic frequencies with increased airspeed. Similar conclusions were drawn from the results associated with wing configuration 3. The excitation pods were able to identify the presence of a symmetric bending mode, symmetric torsional mode, and two anti-symmetric torsional modes throughout the explored flight envelope. While not always in agreement with expected trends, flight-testing data associated with wing configuration 3 did indicate a shift in the modes' natural frequencies with airspeed. By utilizing the excitation pods, frequency characteristics of the wing configurations tested in this study were able to be identified through data collected in the free-flight environment.

Challenges encountered during early flight-testing attempts revealed the necessity to properly define the forcing functions used to excite both a symmetric and anti-symmetric response from the lifting-surface under investigation. As shown in data collected with wing configuration 2, the approximately time-aligned excitation signals associated with the left- and right-wing's shaker assemblies resulted in a highly attenuated anti-symmetric forcing function. As a result, the wing's anti-symmetric modes were not adequately energized, and the resulting FRFs contained a high amount of uncertainty. To address this issue, a phase-angle offset was introduced between the left- and right-wing's excitation signals. For this study, a 90° phase-angle offset was selected such that the symmetric and anti-symmetric forcing functions were of equal magnitude. For aeroelastic flight-testing applications, it is highly recommended that the forcing functions used to excite a response from a lifting-surface are properly defined such that both symmetric and anti-symmetric structural modes are sufficiently energized.

Most importantly, this study demonstrated that aeroelastic flight-testing conducted with the PFM method can be used to identify an aircraft's flutter onset condition. Using the purpose-built excitation pods, the wing's symmetric and anti-symmetric FRFs were able to be measured at a variety of flight conditions. From these FRFs, the phase-crossover frequency and corresponding flutter margin were noted for each test condition. As previously discussed, flight-testing data indicated that the anti-symmetric mode's flutter margin sharply decreased with airspeed between 34 ms^{-1} and 42 ms^{-1} . Since the anti-symmetric mode's gain margin trended towards zero, it was concluded that the anticipated flutter mode involved an anti-symmetric mechanism. By using a best-fit line, a flutter onset velocity of 43.2 ms^{-1} was extrapolated from the flight-test results. The identified flutter boundary agrees within 8% of that estimated from the associated FEA model. This study successfully demonstrated that the PFM method can be utilized to identify an aircraft's flutter onset speed in the free-flight environment.

6.2 Future Work

Continuation of this study is currently in progress and is exploring best practices to improve the safety and accuracy of aeroelastic flight-testing performed with the PFM method. While this study demonstrated that the purpose-built excitation pods were capable of measuring an aeroelastic system's FRFs in the free-flight environment, the resulting Bode plots were negatively impacted by the associated noise. Improvements in flight-testing techniques should be investigated to generate smoother, more accurate FRFs. Additionally, experimental verification of the estimated flutter onset condition is in-progress to assess the accuracy of the flutter boundary identified

through PFM flight-testing. While this study demonstrated that the PFM method can be successfully used in free-flight, further fine-tuning of the flight-test methodology may improve the accuracy of the identified aeroelastic characteristics.

By reducing the uncertainty of the FRFs measured through flight-testing, more accurate phase-crossover frequencies, corresponding flutter margins, and resulting flutter boundary estimates may be determined from data collected in the free-flight environment. One potential method to reduce the noise associated with the flight-testing FRFs is to collect more aeroelastic response data to be processed by system identification techniques. This may be achieved by increasing the time required to perform a frequency sweep between two fixed boundaries. To address line-of-sight limitations associated with 14 CFR Part 107 operations, longer time-on-conditions may be achieved by performing a circular flight path with a small bank angle to approximate steady, level flight conditions. For example, at a TAS of 40 ms^{-1} and bank-angle of 15° , a turn radius of 610 m and load-factor of 1.04 G would result. Such a flight profile may approximate steady, level flight and allow the operator to maintain visual contact with the vehicle. Additionally, for testing conducted with limited time-on-condition, more targeted frequency sweeps focused around the estimated phase-crossover frequency could result in smoother FRFs. For this study, an alternative frequency sweep between 8 Hz - 15 Hz performed over 30 second would allow more aeroelastic data to be collected for frequencies of interest. This may improve the accuracy of FRFs identified through flight-test data. Lastly, the noise levels afflicting FRFs may be reduced by utilizing a stronger, more powerful excitation signal to energize an aeroelastic response. The response associated with a stronger excitation signal may surpass the effects associated with atmospheric turbulence and rigid body

flight dynamics. While a stronger excitation function may require a more powerful and heavier actuator, this additional weight may be incorporated into the stabilizing parameter, further improving the safety associated with aeroelastic flight-testing conducted with the PFM method.

Further flight-testing is in-progress to verify the flutter onset condition identified through the PFM method. Specifically, flight-testing at conditions in excess of 43.2 ms^{-1} is required to determine if the resulting flutter margins are less than zero. This would allow the flutter boundary to be positively identified through interpolation of flight-test data as compared to the extrapolation method used in this study. Additionally, flight-testing of the nominal aeroelastic system without the shaker assembly should be performed. This would allow the flutter onset condition to be calculated through accepted methods, such as ARMA. Lastly, it is desirable to conduct flight-testing of the nominal aeroelastic system at the anticipated flutter boundary to ensure that an unstable aeroelastic response manifests. Through the use of direct testing, the flutter onset condition calculated through the PFM method can be compared to the true value. Such testing would prove that the PFM method could accurately and safely identify an aircraft's flutter velocity in the free-flight environment.

Appendix A: Nomenclature, Abbreviations, and Acronyms

Symbol	Definition (and units when applicable)
A	Aeroelastic system matrix
a	Acceleration, m/sec^2
B_f	Input modal distribution vector
b	Semi-chord distance, m
b_a	Chord midpoint to elastic axis distance, m
bx_a	Chord midpoint to center of gravity distance, m
C	Theodorsen's lift deficiency
C_f	Aeroelastic system plant output vector
F	Force, N
G	Gain
g	Structural damping coefficient
h	Aeroelastic system's plunge state, m
I_α	Mass moment of inertia about elastic axis, kgm^2
i	Imaginary unit vector
K	Spring constant, N/m
k	Reduced frequency
L	Lift, N
M	Moment, Nm
m	Mass, kg
	Mass per length, kg/m
P_s	Stabilization parameter
p	Body-axis roll rate, rad/sec
Q_r	LaGrangian generalized force
q	Body-axis pitch rate, rad/sec
q_r	LaGrangian generalized system state
r	Distance from elastic axis, m
	Body-axis yaw rate, rad/sec
r_a	Non-dimensional radius of gyration
S_a	Static moment about elastic axis, Nm
T	Transfer function

t	Kinetic energy, J
U	Time, sec
u	Potential energy, J
u_{inf}	Input signal
X	Free-stream velocity, m/sec
x	Distance between CB's IMU and shaker centroid, m
x_a	Modal displacement vector
y_f	Non-dimensional static imbalance
α	Output scalar
μ	Aeroelastic system's pitch state, rads
ρ_{inf}	Non-dimension mass ratio
ϕ	Free-stream fluid density, kg/m^3
ω	Phase-angle, rads
	Frequency, rad/sec

Subscripts:

X_{AC}	Aerodynamic center
X_{asym}	Anti-symmetric
X_{cb}	Control board
X_{EA}	Elastic axis
X_{ex}	External
X_{FT}	Force transmissibility
X_{GVT}	Ground vibration test
X_h	Aeroelastic system's plunge state
X_m	Measured
X_{mv}	Moving mass component
X_{pco}	Phase-crossover
X_{st}	Static mass component
X_{sym}	Symmetric
X_x	Body frame x-axis, forward
X_y	Body frame y-axis, spanwise
X_z	Body frame z-axis, up
X_{alpha}	Aeroelastic system's pitch state

Superscripts:

X^L	Left
X^R	Right

Mathematical Operators:

$\dot{X}$	First derivative with respect to time
$\ddot{X}$	Second derivative with respect to time
dX	Calculus differential term

Abbreviations and Acronyms:

ARMA	Autoregressive moving average
CAD	Computer aided design
FBD	Free body diagram
FEA	Finite element analysis
F.M.	Flutter margin
FRF	Frequency response function
GPS	Global positioning system
GVT	Ground vibration test
IMU	Inertial measurement unit
PFM	Parametric flutter margin
SISO	Single input single output
TAS	True airspeed
UAM	Urban air mobility
UAS	Unmanned aerial system

Appendix B: K-Method Code

```
1 %Author: Ross W. Heidersbach
2 %Title: K-Method Flutter Analysis
3 %Date: 11/28/2021
4
5 %Compute a Typical Section's Damping Vs. Velocity Graph and Flutter
6 %via "k" Method Approach (Utilized Theodorsen's Unsteady ...
   Aerodynamics and
7 %Accounts for Structural Damping)
8
9 clc
10 clear
11 close all
12
13 %%
14 %Define Provided Structural and Aerodynamic Data
15
16 %Define Wing-Shaker Configuration's Geometric Properties
17 b = 0.17; %Wing's Semi-Chord Length (m)
18 L = 3.4036; %Wing Span (m)
19 X_Wing_CG = 0.13; %Location of Wing's CG Relative to Wing's ...
   Leading Edge (m)
20 X_EA = 0.136; %Location of Wing's Elastic Axis Relative to Wing's ...
   Leading Edge (m)
21 %X_Point_Mass = 0.395; %Location of Point Mass Relative to Wing's ...
   Leading Edge (m)
22 %X_Point_Mass = 0.555; %Location of Point Mass Relative to Wing's ...
   Leading Edge (m)
23 X_Point_Mass = 0.702; %Location of Point Mass Relative to Wing's ...
   Leading Edge (m)
24
25 %Define Wing-Shaker Configuration's Mass Properties
26 Mass_Wing = 6.7; %Total Wing Mass as Experimentally Measured (kg)
27 Mass_Point = 0.4; %Total Point Mass as Experimentally Measured (kg)
28
29 %Define Nominal Wing-Shaker Configuration's Mass Moment of ...
   Interia Properties
```

```

30 I_Wing = 0.0835; %Wing's Mass Moment of Interia @ Elastic Axis ...
    Determined via Solidworks (kg/m^2)
31
32 %Define Wing-Shaker Configuration's Frequency Properties
33 W_h = 21.7; %Wing-Bending Natural Frequency (rad/sec)
34 W_a = 49.7; %Wing-Torsion Natural Frequency (rad/sec)
35
36 %Define Wing-Shaker Configuration's Damping Properties
37 g_h = 0.025; %Assumed Wing-Plunge Structural Damping
38 g_a = 0.025; %Assummed Wing-Pitch Structural Damping
39
40 %Define Wing-Shaker Configuration's Aerodynamic Properties
41 C_La = 2*pi; %Lift-Curve Slope (CL/rad) in Accordance with Thin ...
    Airfoil Theory
42
43 %Define Environmental Properties
44 rho = 1.225; %Air Density @ Sea Level (kg/m^3)
45
46 %%
47 %Compute Structural and Aerodynamic Variables
48
49 %Compute Wing-Shaker Configuration's Geometric Properties
50 X_CG = ((Mass_Wing*X_Wing_CG) + ...
    (Mass_Point*X_Point_Mass))/(Mass_Wing + X_Point_Mass); ...
    %Compute the Wing-Shaker Configuration's Center of Gravity ...
    Location (m)
51 a = (X_EA-b)/b; %Compute the Wing-Shaker Configuration's Elastic ...
    Axis - Midchord Offset Percentage
52 x_a = (X_CG-X_EA)/b; %Compute the Wing-Shaker Configuration's ...
    Center of Gravity - Elastic Axis Offset Percentage
53
54 %Compute Wing-Shaker Configuration's Mass Properties
55 Mass = Mass_Wing + Mass_Point; %Compute Wing-Shaker ...
    Configuration's Total Mass (kg)
56
57 %Compute Wing-Shaker Configuration's Inertial Properties
58 I_xx = I_Wing + ((Mass_Point)*((X_Point_Mass-X_Wing_CG)^2)); ...
    %Compute Wing-Shaker Configuration's Moment of Inertia @ ...
    Elastic Axis (kg/m^2)
59
60 %Compute Wing-Shaker Configuration's Non-Dimensonal Properties
61 m = Mass/L; %Compute Wing-Shaker Configuration's Specific Mass(kg/m)
62 mu = (m)/(pi*rho*(b^2)); %Compute Wing-Shaker Configuration's ...
    Dimensionless Mass Ratio (Dimensionless)
63 r_a = sqrt((I_xx)/(m*(b^2))); %Compute Wing-Shaker ...
    Configuration's Dimensionless Radius of Gyration (Dimensionless)
64
65 %%
66 %Algorithm for Computing Damping as a Function of Airspeed via K ...
    Method
67

```

```

68 %Define Reduced Frequency Matrix (Range is Determined by Trial ...
    and Error)
69 K_Array = [0.01:0.001:0.4];
70
71 %Initialize Symbolic Variable Omega
72 syms Omega;
73
74 %For-Loop Executable per Each Pre-Defined Reduced Frequency
75 for index = 1:length(K_Array)
76
77     %Select Current Reduced Frequency
78     k = K_Array(index);
79
80     %Compute Current Theodorsen's Lift Deficiency
81     C = Theodorsen_Lift_Deficiency(k);
82
83     %Compute Current L_H and l_h (Ignoring Prime to Avoid ...
        Confusion with Transpose)
84     L_H = -1 + ((2*i*C)/(k));
85     l_h = L_H;
86
87     %Compute Current L_A and l_a (Ignoring Prime to Avoid ...
        Confusion with Transpose)
88     L_A = -0.5 + ((i*(1+2*C))/(k)) + ((2*C)/(k^2));
89     l_a = L_A - ((0.5+a)*L_H);
90
91     %Compute Current M_H and m_h (Ignoring Prime to Avoid ...
        Confusion with Transpose)
92     M_H = 0.5;
93     m_h = M_H + ((0.5+a)*L_H);
94
95     %Compute Current M_A and m_a (Ignoring Prime to Avoid ...
        Confusion with Transpose)
96     M_A = 0.375 - (i/k);
97     m_a = M_A - ((0.5+a)*(-L_A+M_H)) - (((0.5+a)^2)*L_H);
98
99     %Construct Current System Matrix A (Defined in Flutter E.O.M. ...
        w/ T.T. Notes)
100    A(1,1) = ((mu)*(1 - ((W_h^2)/(W_a^2)))*(1+i*g_h)*Omega) - l_h;
101    A(1,2) = (mu*x_a) - l_a;
102    A(2,1) = (mu*x_a) + m_h;
103    A(2,2) = (r_a^2)*(mu)*(1-(1+i*g_a)*Omega) + m_a;
104
105    %Compute Current Characteristic Equation of System Matrix
106    Characteristic_Equation = det(A);
107
108    %Isolate Coefficients of Current Charactersitic Equation
109    Coefficient_Array = flip(coeffs(Characteristic_Equation, Omega));
110
111    %Compute Current Omega
112    Roots = double(roots(Coefficient_Array));

```

```

113
114     %Isolate Real (X) and Imaginary (Y) Parts of Current Omega
115     X_Array(:,index) = real(Roots);
116     Y_Array(:,index) = imag(Roots);
117
118     %Compute Current Natural Frequency
119     W_Array(:,index) = W_a./sqrt(X_Array(:,index));
120
121     %Compute Current Damping Ratio
122     G_Array(:,index) = Y_Array(:,index)./X_Array(:,index);
123
124     %Compute Current Airspeed
125     U_Array(:,index) = (b*W_a)./(k*sqrt(X_Array(:,index)));
126
127     %Display Percentage Complete
128     %clc
129     %fprintf('Percent Complete: %.1f %% \n', ...
130             (index/length(K_Array))*100)
131 end
132
133 %Clear Command Window
134 clc
135
136 %%
137 %Identify Flutter Velocity
138
139 %Determine Array Indices Corresponding to Change of Damping Polarity
140 G_Polarity_Change_Indices_Mode_1 = find(diff(G_Array(1,:)<0));
141 G_Polarity_Change_Indices_Mode_2 = find(diff(G_Array(2,:)<0));
142
143 %Identify Airspeed Corresponding to Change of Damping Polarity
144 U_Flutter_Mode_1 = U_Array(1,G_Polarity_Change_Indices_Mode_1);
145 U_Flutter_Mode_2 = U_Array(2,G_Polarity_Change_Indices_Mode_2);
146
147 %Identify Damping Corresponding to Change of Damping Polarity
148 G_Flutter_Mode_1 = G_Array(1,G_Polarity_Change_Indices_Mode_1);
149 G_Flutter_Mode_2 = G_Array(2,G_Polarity_Change_Indices_Mode_2);
150
151 %Identify Frequency Corresponding to Change of Damping Polarity
152 W_Flutter_Mode_1 = W_Array(1,G_Polarity_Change_Indices_Mode_1);
153 W_Flutter_Mode_2 = W_Array(2,G_Polarity_Change_Indices_Mode_2);
154
155 %Combine Velocity and Damping Arrays for Flutter Points of Mode 1 ...
156     & Mode 2
157 U_Flutter_Array = [U_Flutter_Mode_1 U_Flutter_Mode_2];
158 G_Flutter_Array = [G_Flutter_Mode_1 G_Flutter_Mode_2];
159 W_Flutter_Array = [W_Flutter_Mode_1 W_Flutter_Mode_2];
160
161 %Extract Airspeed Corresponding to Slowest Flutter Velocity
162 [Flutter_Velocity, Temporary_Index] = min(U_Flutter_Array);

```

```

162 Flutter.Damping = G.Flutter.Array(Temporary_Index);
163 Flutter.Frequency = W.Flutter.Array(Temporary_Index);
164
165 %%
166 %Construct Plots of Results
167
168 %Plot Damping Versus Airspeed
169 figure(1)
170 plot(U_Array(1,:),G_Array(1,:))
171 hold on
172 plot(U_Array(2,:),G_Array(2,:))
173 plot(Flutter.Velocity,Flutter.Damping,'k*')
174 hold off
175 grid on
176 xlabel('Airspeed (m/s)')
177 ylabel('Damping (g)')
178 title('Damping Versus Airspeed (200 g w/ 12" Pod Extension)')
179 legend('Torsional Mode','Bending Mode','Flutter Condition', ...
        'location','Northwest')
180 axis([0 120 -0.7 0.7])
181
182 %Plot Frequency Versus Airspeed
183 figure(2)
184 plot(U_Array(1,:),W_Array(1,:)/(2*pi))
185 hold on
186 plot(U_Array(2,:),W_Array(2,:)/(2*pi))
187 plot(Flutter.Velocity,Flutter.Frequency/(2*pi),'k*')
188 hold off
189 grid on
190 xlabel('Airspeed (m/s)')
191 ylabel('Frequency (Hz)')
192 title('Frequency Versus Airspeed (200 g w/ 12" Pod Extension)')
193 legend('Torsional Mode','Bending Mode','Flutter Condition')
194 axis([0 120 0 30])
195
196 %Display Flutter Velocity in Command Window
197 fprintf('Flutter Velocity: %.1f m/s \n',Flutter.Velocity)

```

Appendix C: K-Method Results

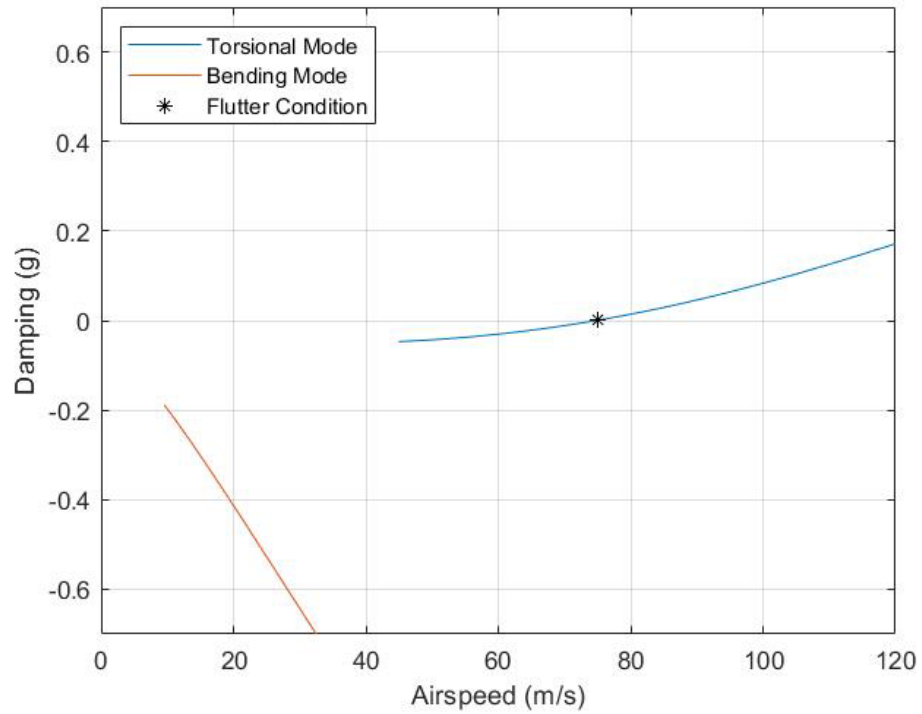


Figure C.1: Damping - TAS curve for wing configuration 2

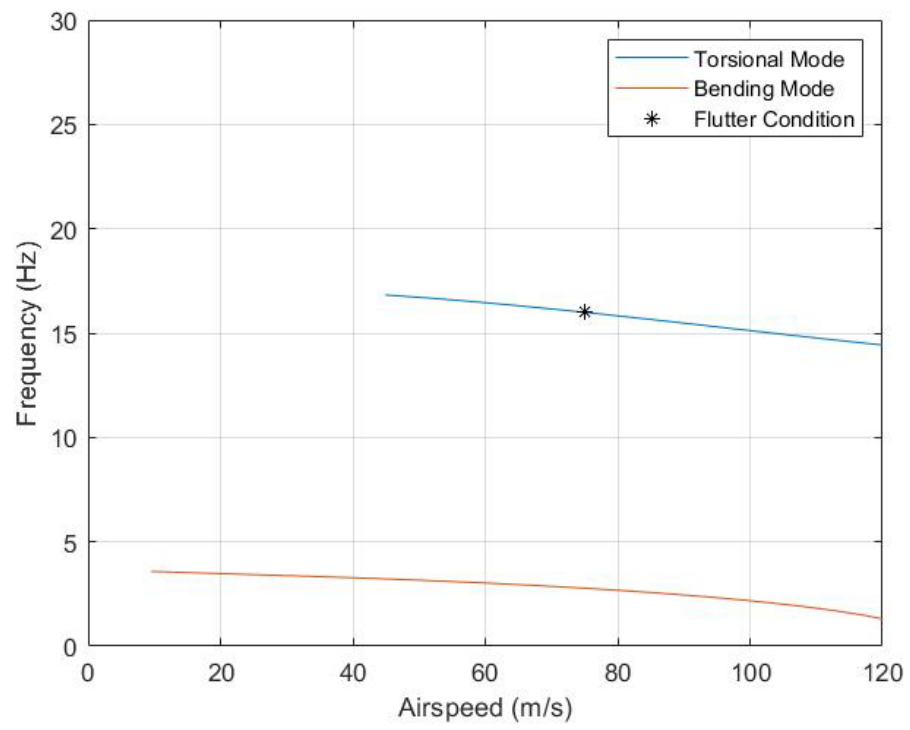


Figure C.2: ω - TAS curve for wing configuration 2

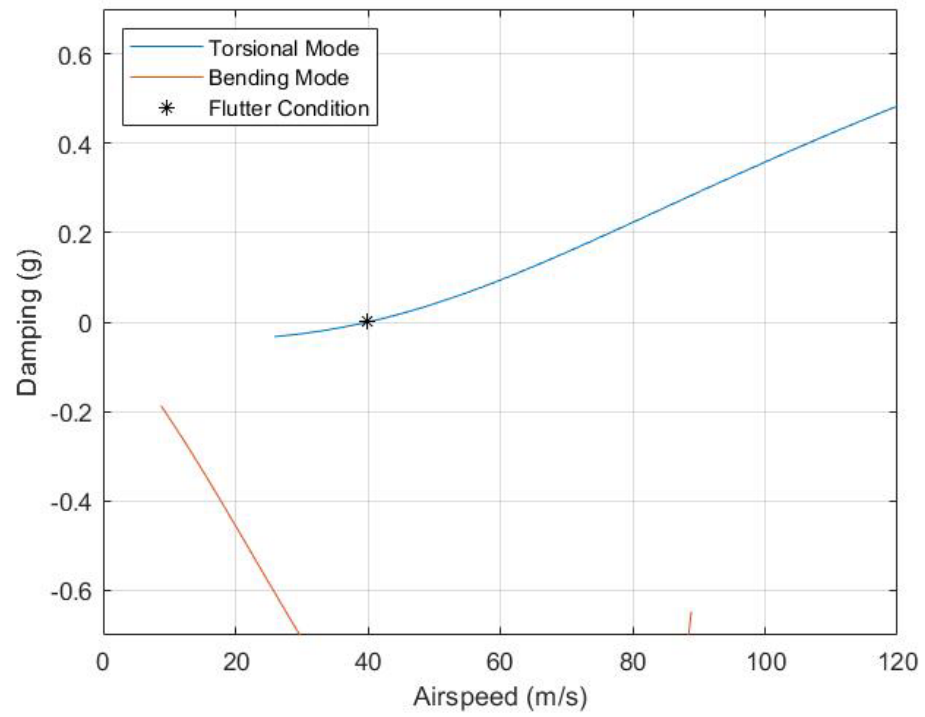


Figure C.3: Damping - TAS curve for wing configuration 3

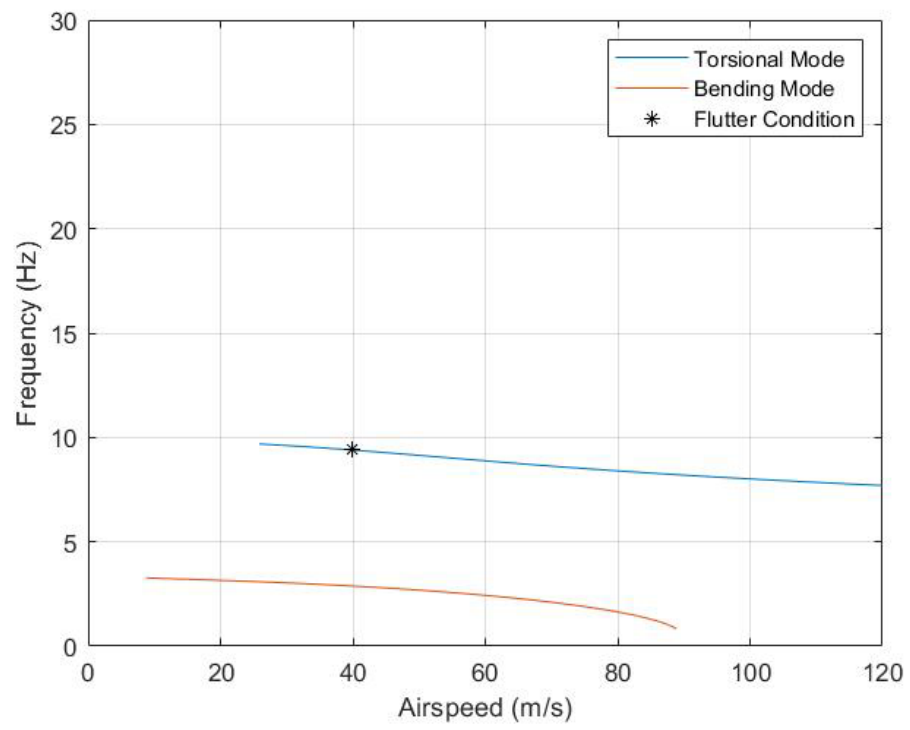


Figure C.4: ω - TAS curve for wing configuration 3

Appendix D: Flight-Testing FRFs of Wing Configuration 3

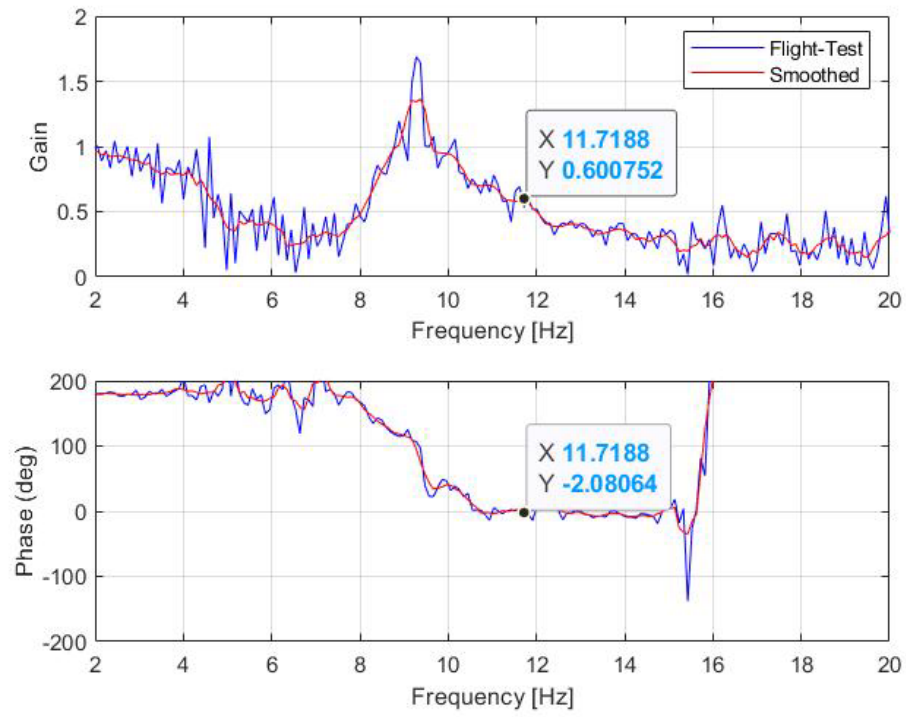


Figure D.1: Symmetric FRF for $TAS = 34.4 \text{ ms}^{-1}$

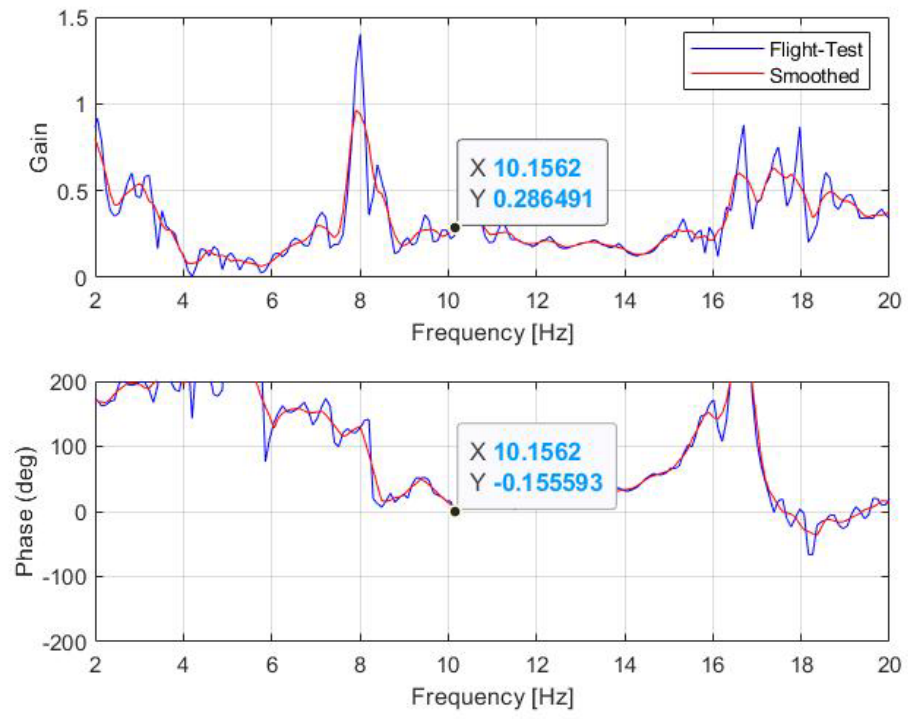


Figure D.2: Anti-symmetric FRF for $TAS = 34.4 \text{ ms}^{-1}$

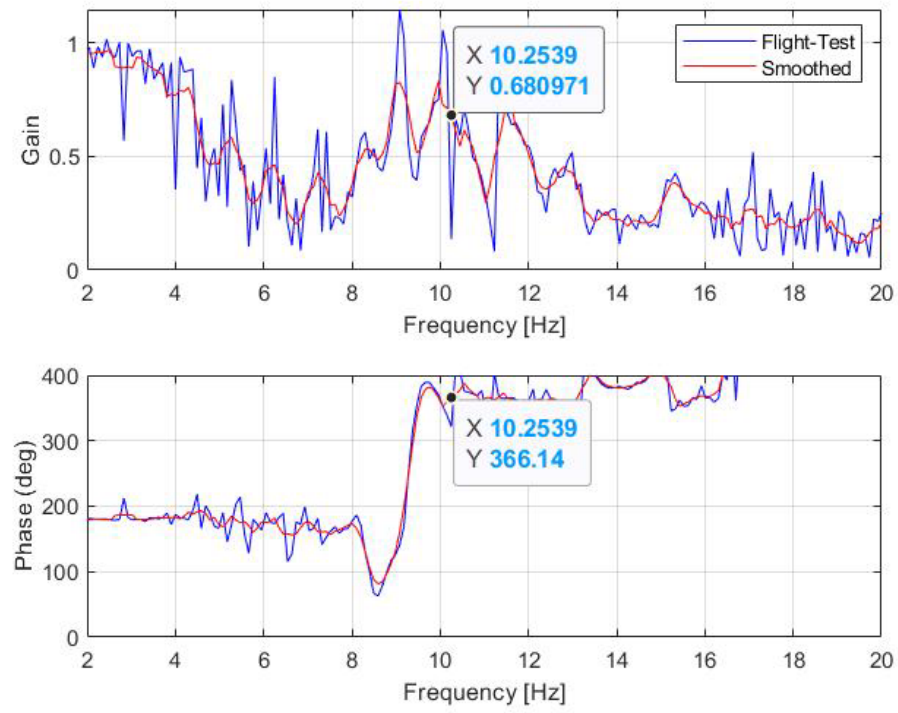


Figure D.3: Symmetric FRF for $TAS = 34.5 \text{ ms}^{-1}$

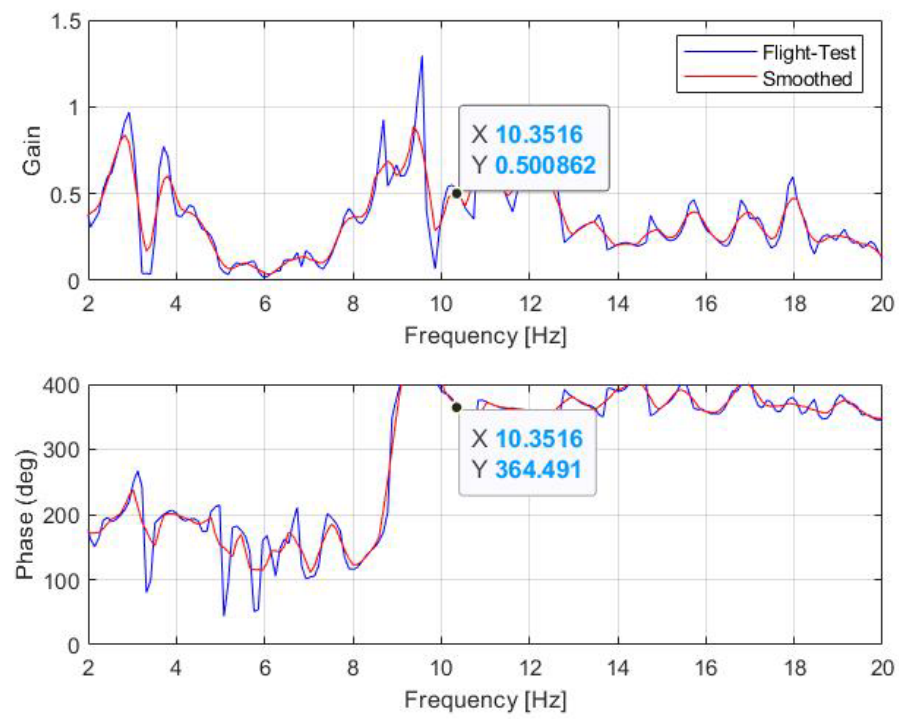


Figure D.4: Anti-symmetric FRF for $TAS = 34.5 \text{ ms}^{-1}$

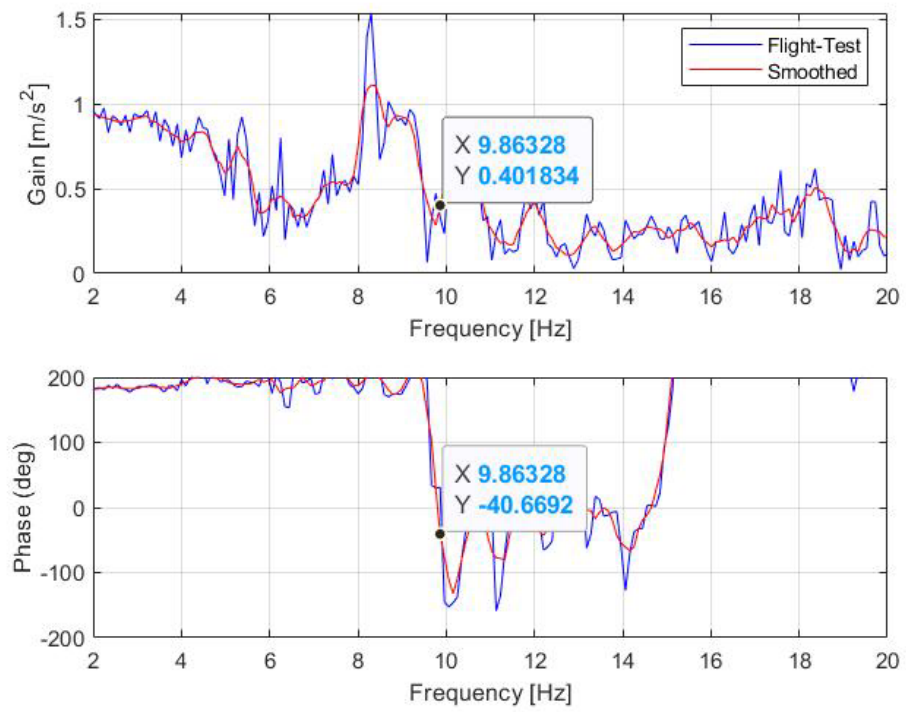


Figure D.5: Symmetric FRF for $TAS = 34.8 \text{ ms}^{-1}$

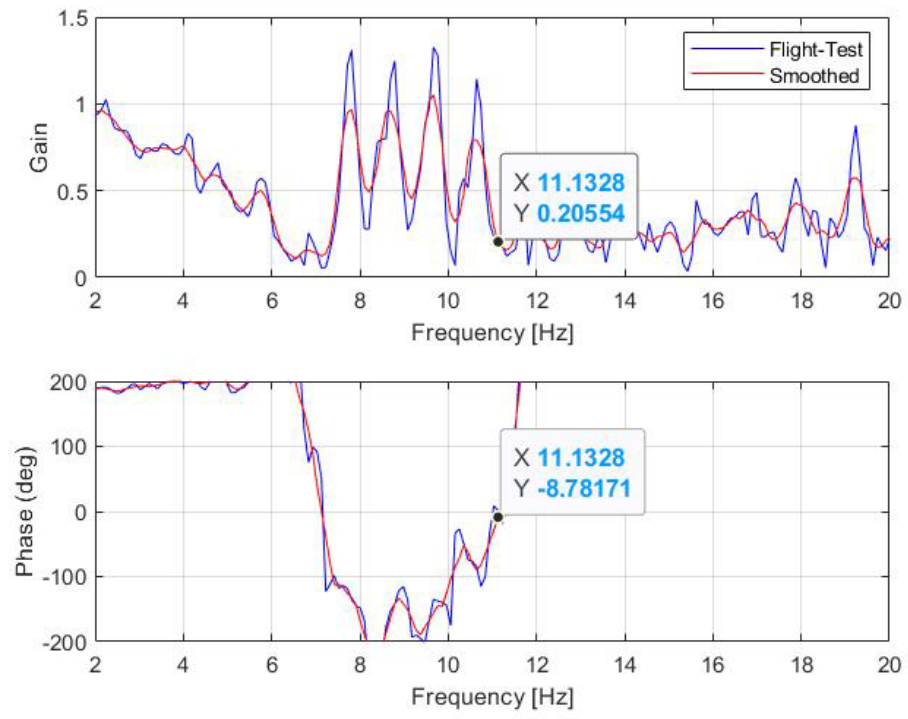


Figure D.6: Anti-symmetric FRF for $TAS = 34.8 \text{ ms}^{-1}$

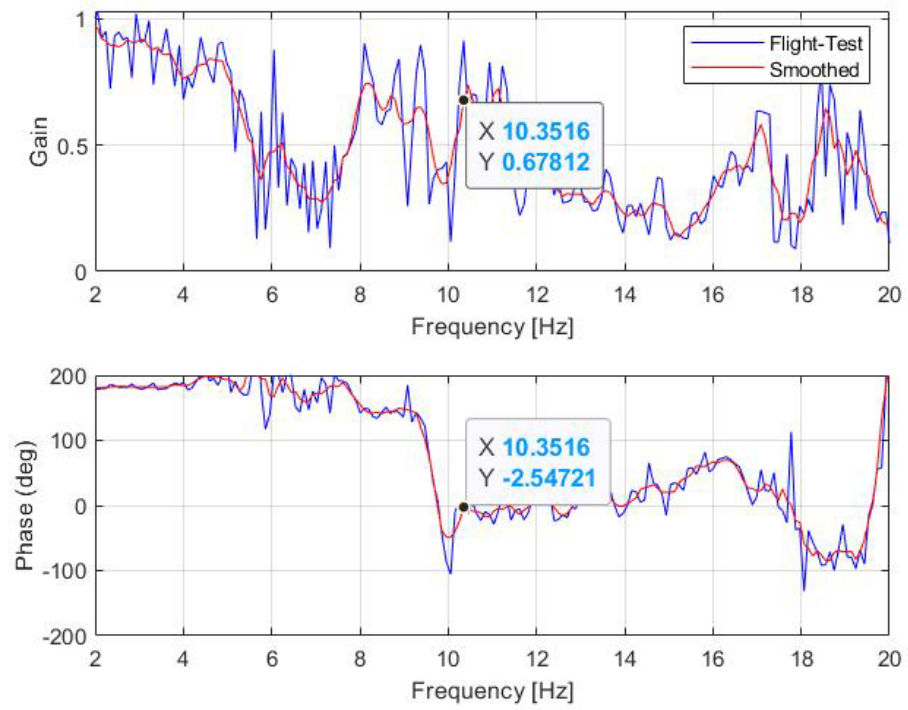


Figure D.7: Symmetric FRF for $TAS = 35.1 \text{ ms}^{-1}$

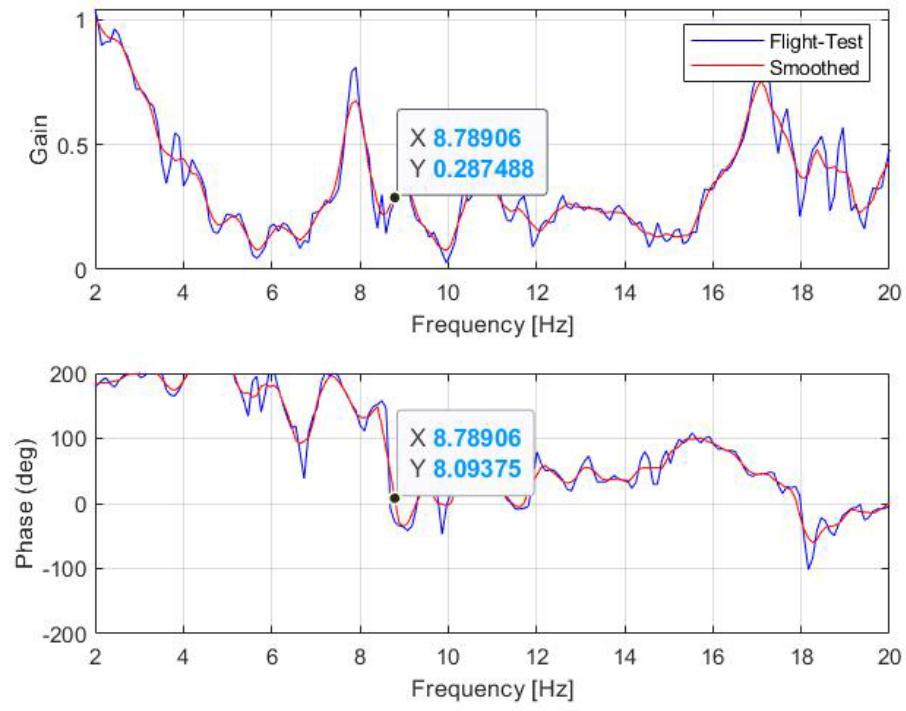


Figure D.8: Anti-symmetric FRF for $TAS = 35.1 \text{ ms}^{-1}$

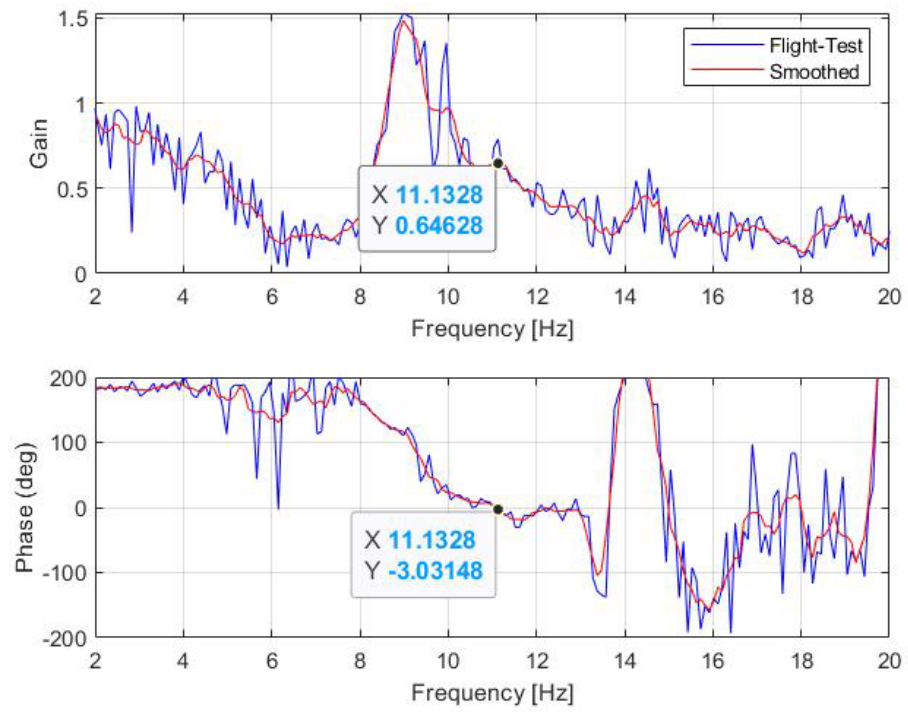


Figure D.9: Symmetric FRF for $TAS = 35.5 \text{ ms}^{-1}$

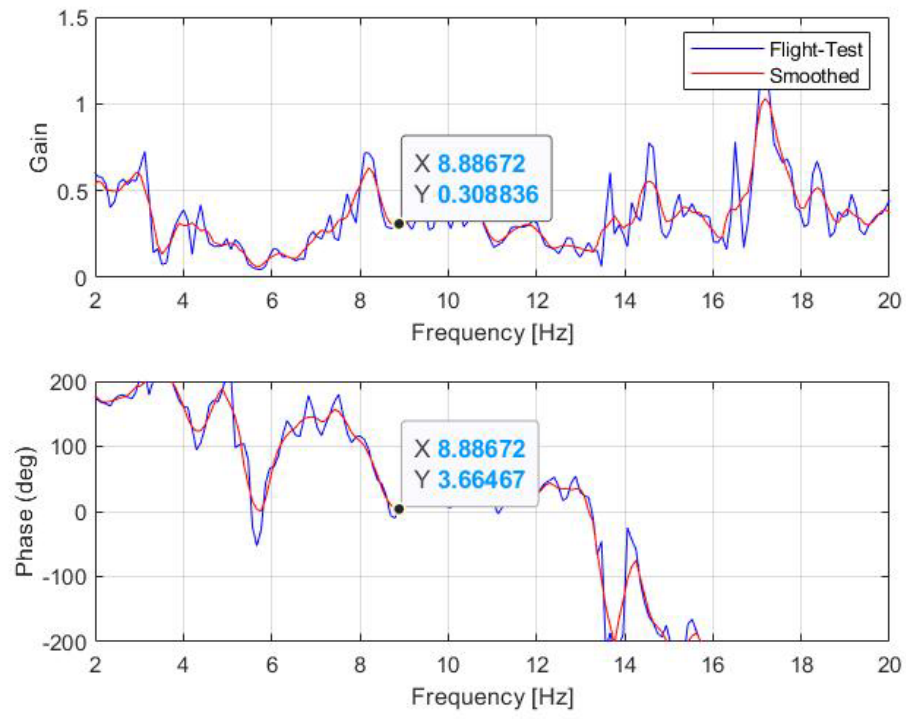


Figure D.10: Anti-symmetric FRF for $TAS = 35.5 \text{ ms}^{-1}$

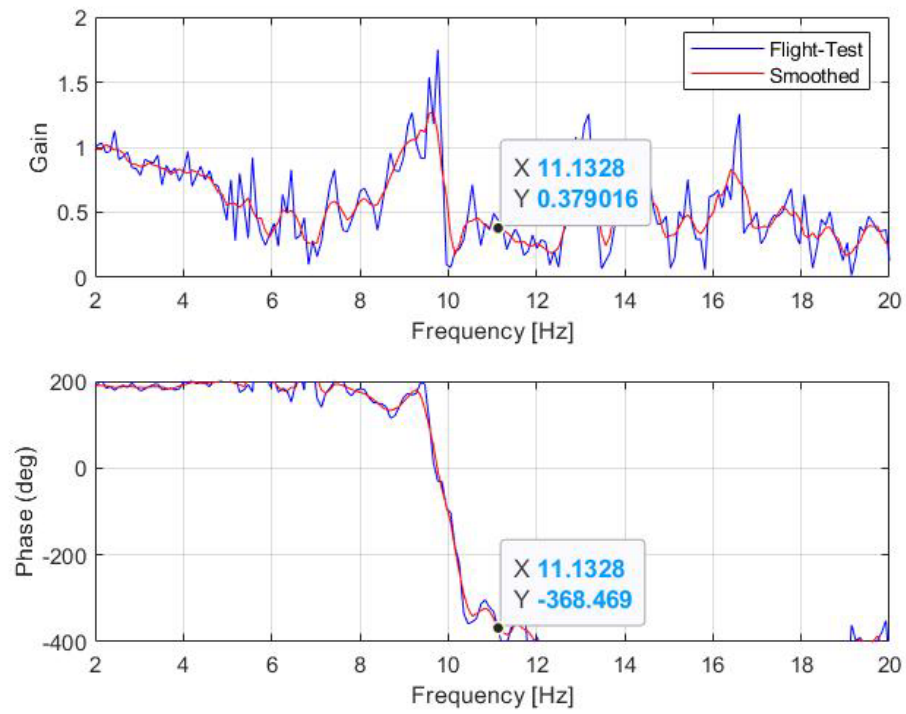


Figure D.11: Symmetric FRF for $TAS = 35.7 \text{ ms}^{-1}$

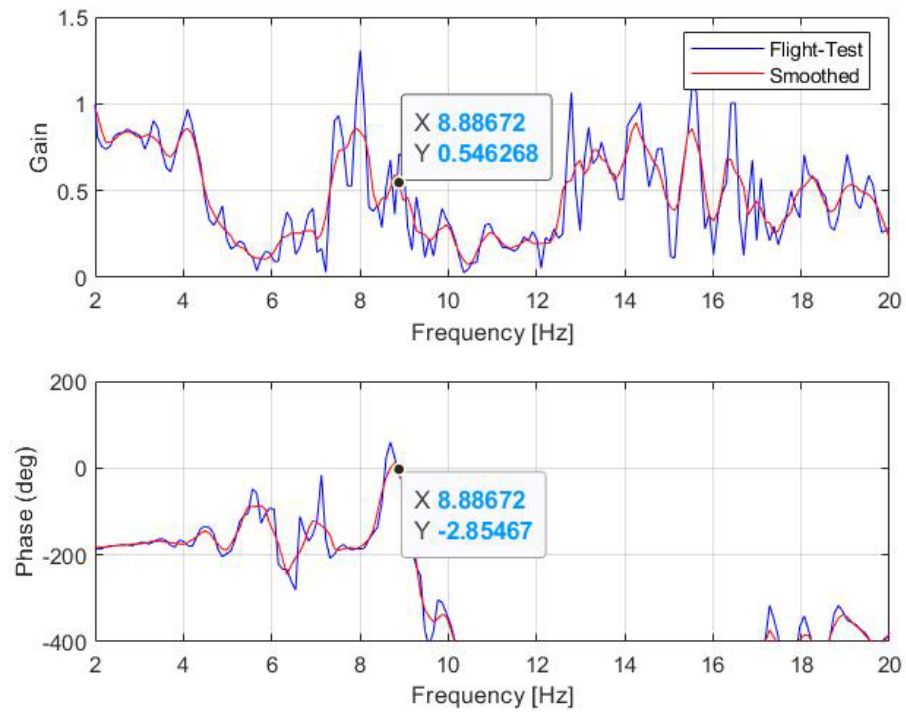


Figure D.12: Anti-symmetric FRF for $TAS = 35.7 \text{ ms}^{-1}$

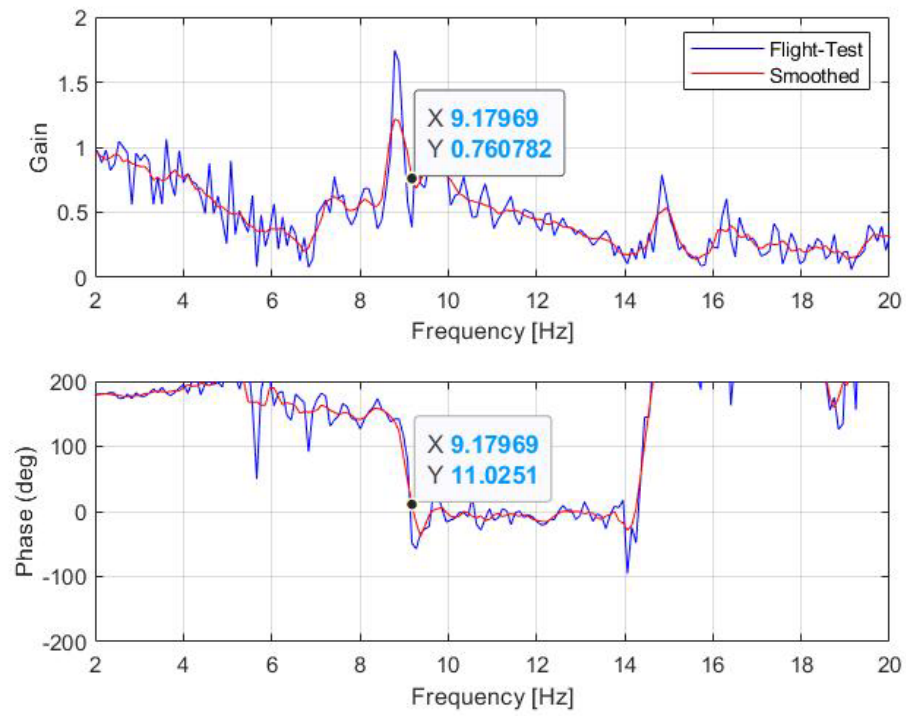


Figure D.13: Symmetric FRF for $TAS = 36.0 \text{ ms}^{-1}$

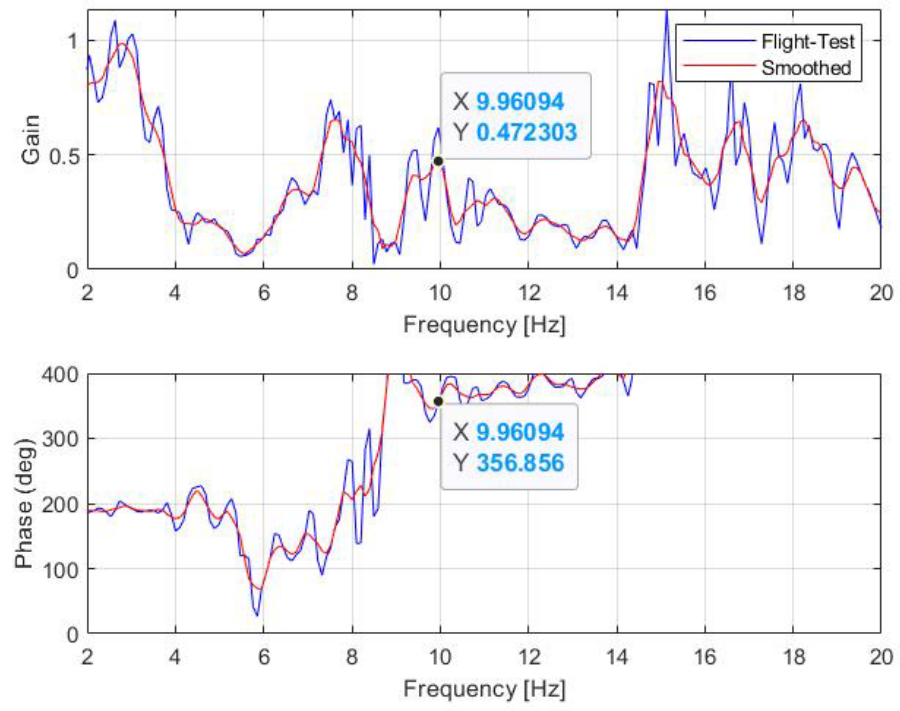


Figure D.14: Anti-symmetric FRF for $TAS = 36.0 \text{ ms}^{-1}$

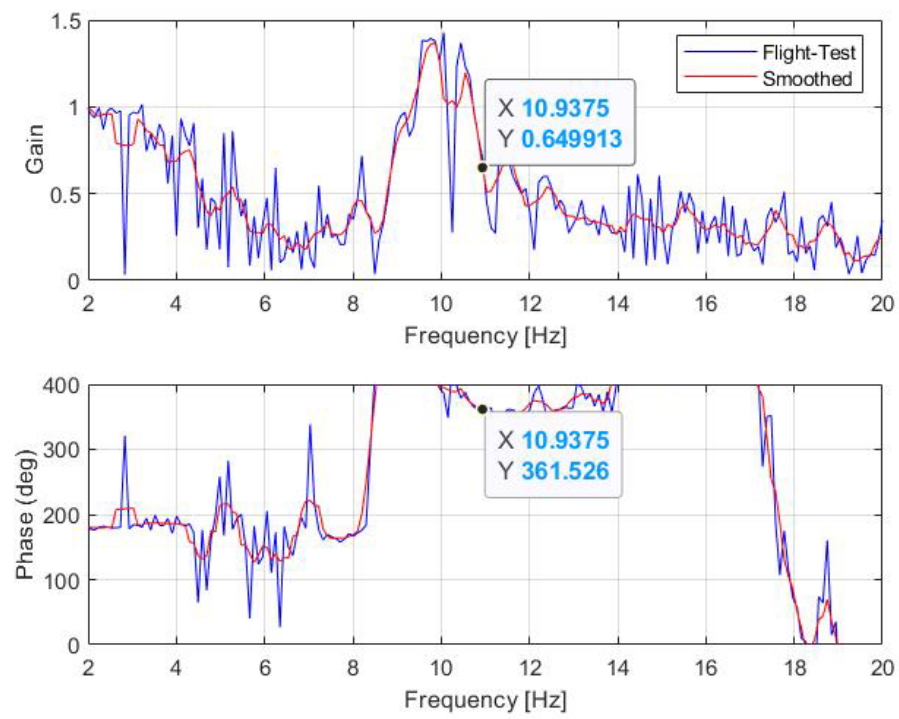


Figure D.15: Symmetric FRF for $TAS = 38.2 \text{ ms}^{-1}$

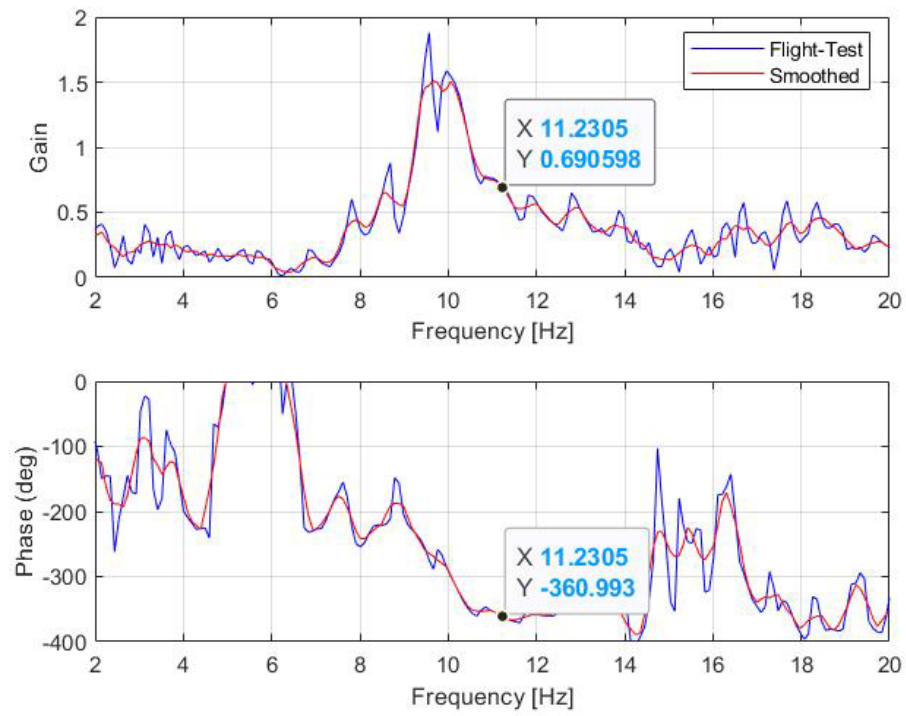


Figure D.16: Anti-symmetric FRF for $TAS = 38.2 \text{ ms}^{-1}$

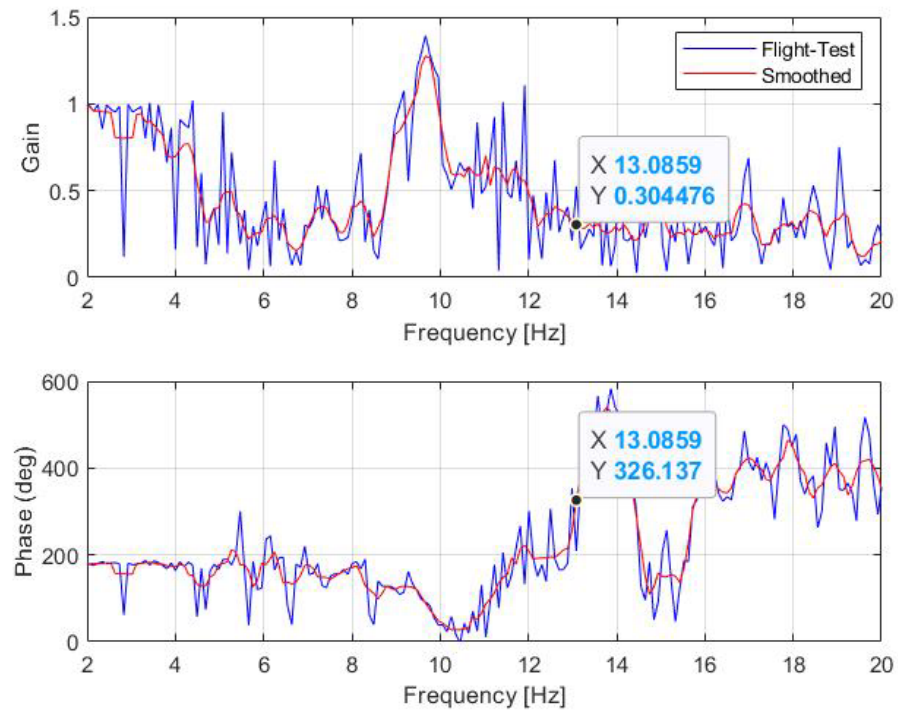


Figure D.17: Symmetric FRF for $TAS = 38.5 \text{ ms}^{-1}$

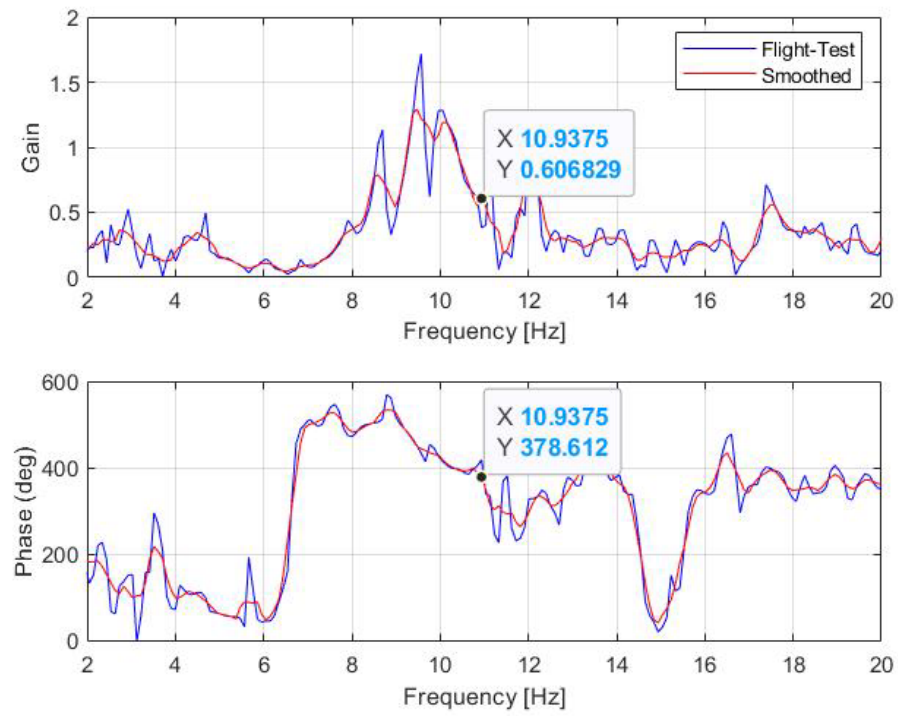


Figure D.18: Anti-symmetric FRF for $TAS = 38.5 \text{ ms}^{-1}$

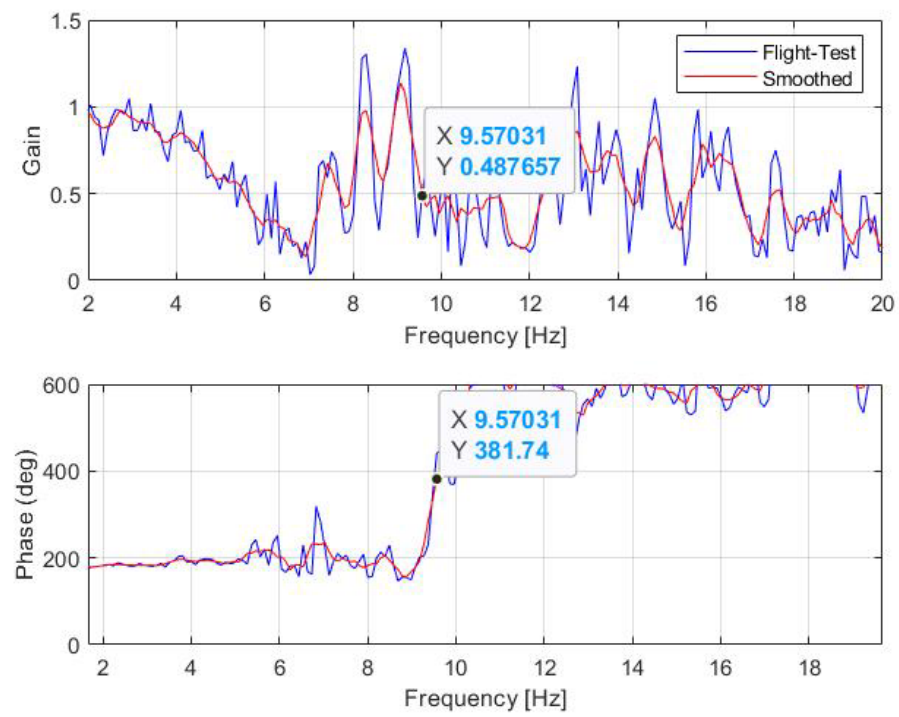


Figure D.19: Symmetric FRF for $TAS = 38.8 \text{ ms}^{-1}$

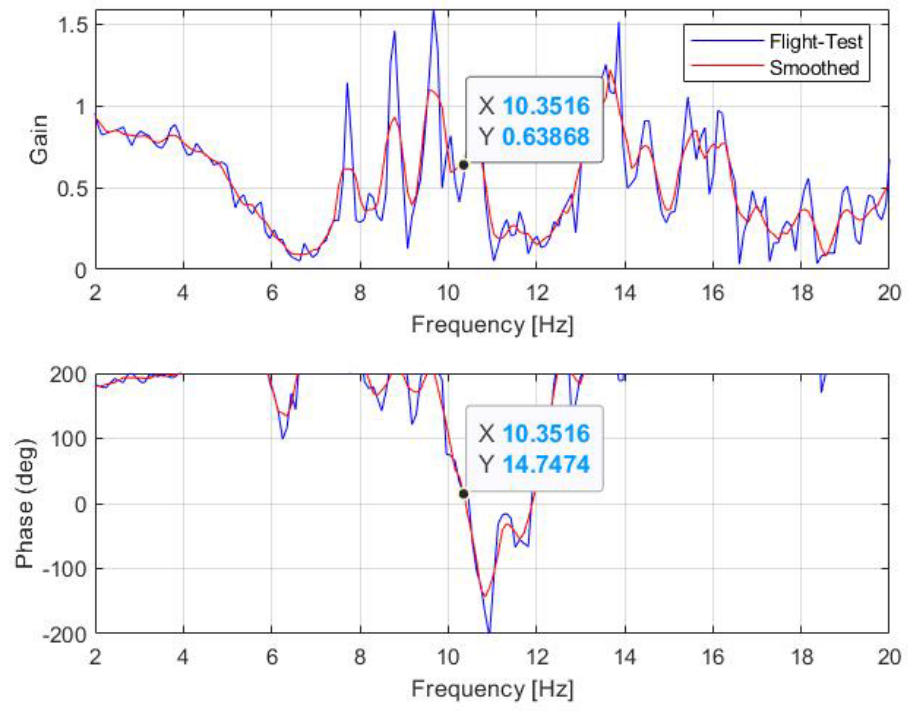


Figure D.20: Anti-symmetric FRF for $TAS = 38.8 \text{ ms}^{-1}$

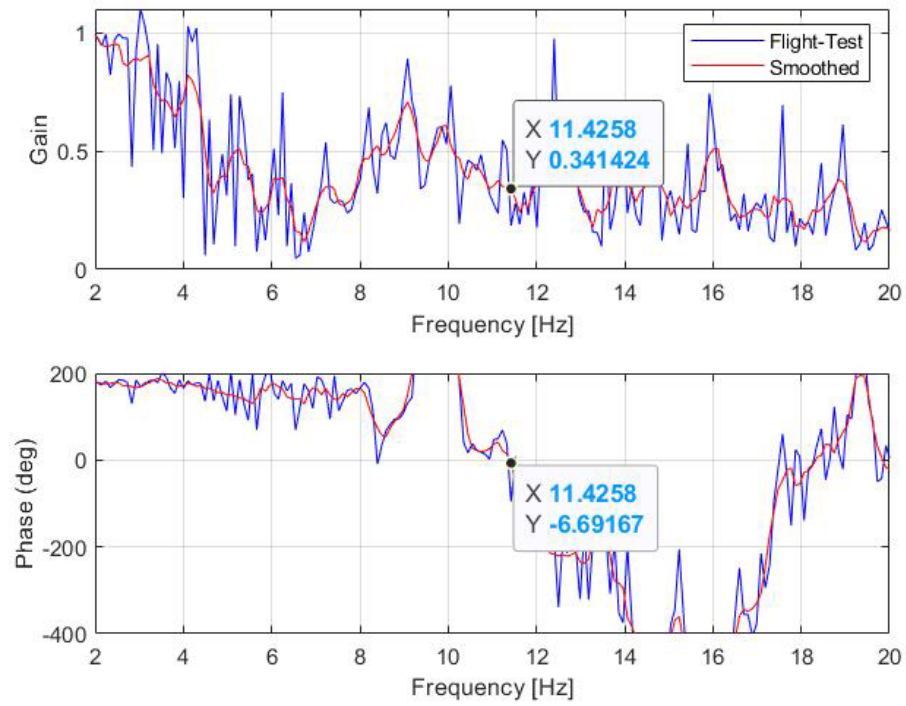


Figure D.21: Symmetric FRF for $TAS = 39.7 \text{ ms}^{-1}$

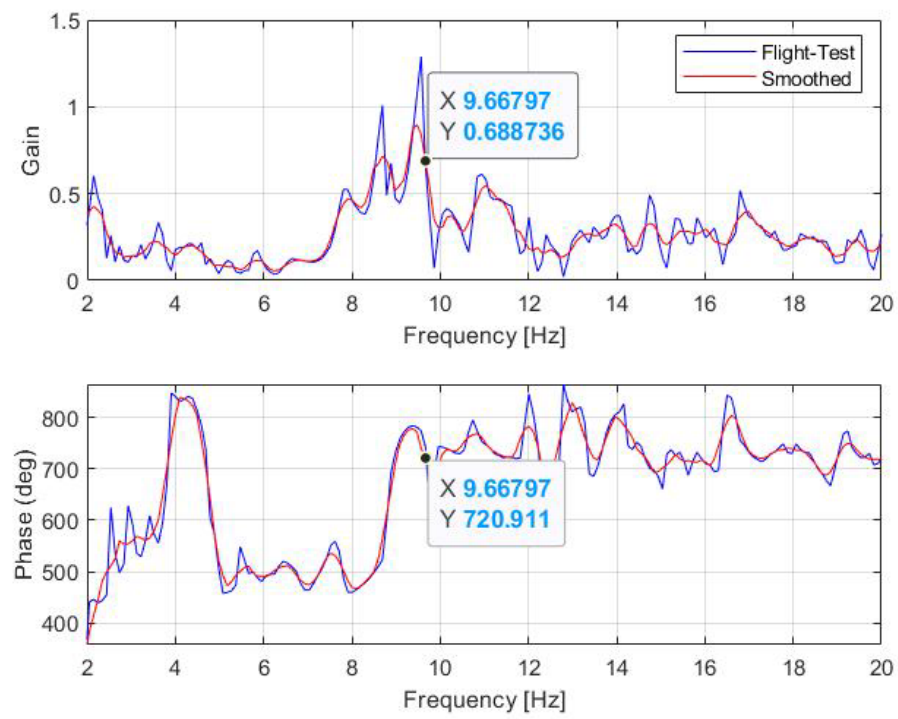


Figure D.22: Anti-symmetric FRF for $TAS = 39.7 \text{ ms}^{-1}$

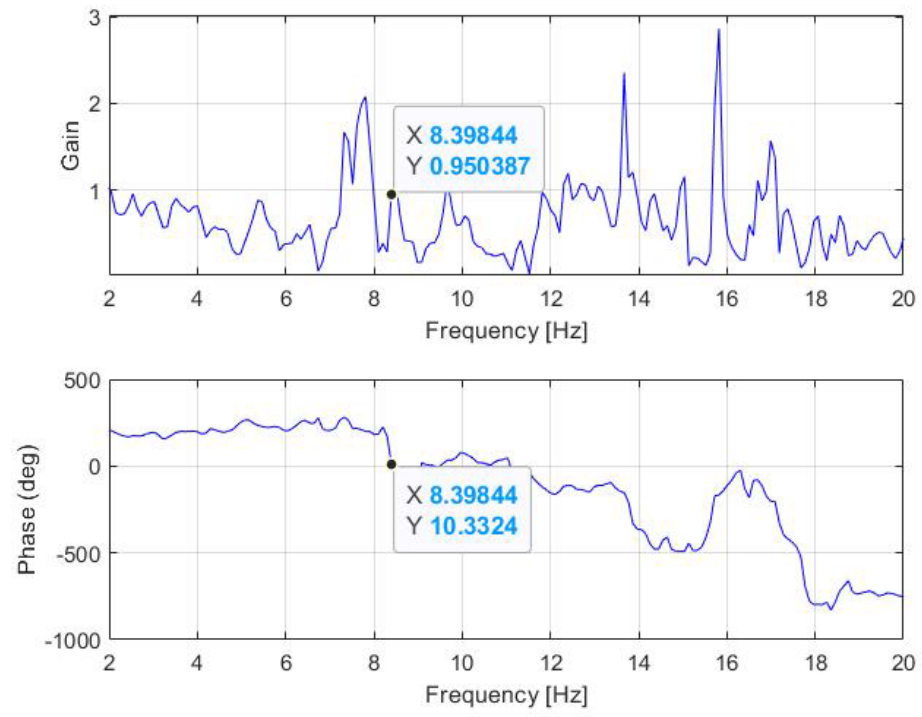


Figure D.23: Anti-symmetric FRF for $TAS = 42.5 \text{ ms}^{-1}$

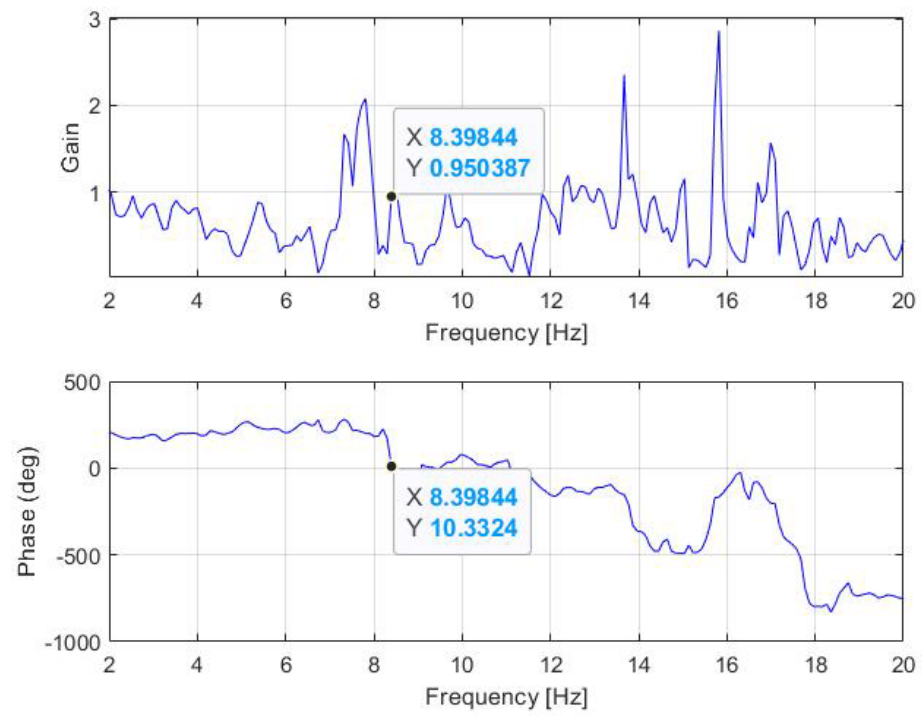


Figure D.24: Anti-symmetric FRF for $TAS = 42.5 \text{ ms}^{-1}$

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